MIL-C-60184(MU)
18 MARCH 1965

SUPERSEDING
FA(FC)-PD-109
2 JULY 1961

MILITARY SPECIFICATION

# COMPUTER SET, GUN DIRECTION: 10525735

## 1. SCOPE

**1.1** This specification covers one type of digital automatic computer set (see 3.2) known as Computer Set, Gun Direction: 10525735, used for field artillery applications.

## 2. APPLICABLE DOCUMENTS

**2.1** The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

### SPECIFICATIONS

MILITARY

MIL-F-13926 — Fire Control Materiel, General Specification Governing the Manufacture and Inspection of

MIL-P-14232 — Parts, Equipment and Tools for Ordnance Materiel: Packaging of

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of

MIL-C-46357 — Computer, Gun Direction: M18, Manufacturing Processes and Inspection of

MIL-P-60165 — Power Control A: 10526460

MIL-T-60171 — Tape Reader: 10526192

MIL-M-60172 —Manual Keyboard: 10526191

MIL-I-60173 — Input Display Matrix: 10526190

MIL-C-60174 — Connector, Receptacle, Electrical: 10526125

MIL-L-60175 — Logic Driver: 10526501

MIL-R-60176 — Rectifier - Transistor Assembly: 10525740

MIL-T-60177 — Transistor Assembly: 10525770

MIL-P-60178 — Power Supply Subassembly: 10525745

MIL-W-60179 — Write Switch: 10526581

MIL-M-60183 — Magnetic Disk, Data Storage: 10526600

MIL-G-60186 — Generator, Pulse: 10526464

MIL-C-60190 — Choke: 10525930

MIL-R-60191 — Reactor: 10525935

MIL-T-60192 — Transformer, Power, Step-Down: 10525905

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MIL–T–60193 — Transformer, Power, Step-Down: 10525915

MIL–C–60194 — Choke: 10525920

MIL–L–60196 — Logic Flip-Flop: 10526505

MIL–W–60197 — Write Amplifier: 10526585

MIL–R–60198 — Read Switch: 10526456

MIL–R–60199 — Read Amplifier: 10526589

MIL–T–60206 — Transformer, Power, Step-Down: 10525900

MIL–T–60207 — Transformer, Power, Step-Down: 10525910

MIL–C–60208 — Control Panel: 10526300

MIL–N–60209 — Network A: 10526472

MIL–N–60210 — Network B: 10526468

MIL–N–60211 — Network C: 10526476

MIL–P–60212 — Power Control B: 10526480

MIL–P–60213 — Power Control C: 10526593

MIL–V–60216 — Voltage Regulator Subassembly: 10525750

MIL–V–60217 — Voltage Regulator Subassembly: 10525755

MIL–V–60218 — Voltage Regulator Subassembly: 10525760

MIL–V–60219 — Voltage Regulator Subassembly: 10525765

## STANDARDS

### MILITARY

MIL–STD–105 — Sampling Procedure and Tables for Inspection by Attributes.

MIL–STD–109 — Quality Assurance Terms and Definitions

MIL–STD–202 — Test Methods for Electronic and Electrical Component Parts

## DRAWINGS

### U. S. ARMY MUNITIONS COMMAND

A10525735 — Computer Set, Gun Direction

F10526101 — Computer, Gun Direction: M18 (10526101)

F10526109 — Indicator Multilayer Panel: 10526109

C10526209 — Cable and Reel Assembly: 10526209

F10526355 — Table, Gun Direction Computer: 10526355

D10526771 — Adapter, Cable and Bracket: 10526771

IEL 10525735 — List of Inspection Documents for Computer Gun Direction: M18

## PACKAGING DATA SHEET

MIL–P–14232/P10525735 — Packaging of Computer Set, Gun Direction: 10525735

(Copies of specifications, standards, drawings, and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



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## 3.1 REQUIREMENTS

**3.1 Fabrication.** The computer set hall be manufactured in accordance with Drawing A10525735 and drawings pertaining thereto.

**3.2 Materials.** Materials shall be in accordance with the drawings, general specifications and material specifications forming a part of this specification.

**3.3 List of assemblies.** The computer set shall be comprised of the following assemblies which shall have been inspected and accepted in accordance with their respective drawings and specifications prior to incorporation as a set:

| *Item* | *Spec or Dwg No.* |
|---|---|
| a. Computer, Gun Direction: M18 (10526101) | F10526101 |
| (1) Connector, Receptacle, Electrical 10526125 | MIL-C-60174 |
| (2) Control Panel: 10526300 | MIL-C-60208 |
| (3) Generator, Pulse: 10526464 | MIL-G-60186 |
| (4) Input Display Matrix: 10526190 | MIL-I-60173 |
| (5) Logic Driver: 10526501 | MIL-L-60175 |
| (6) Logic Flip-Flop: 10526505 | MIL-L-60196 |
| (7) Magnetic Disk, Data Storage: 10526600 | MIL-M-60183 |
| (8) Manual Keyboard: 10526191 | MIL-M-60172 |
| (9) Network A: 10526472 | MIL-N-60209 |
| (10) Network B: 10526468 | MIL-N-60210 |
| (11) Network C: 10526476 | MIL-N-60211 |
| (12) Power Control A: 10526460 | MIL-P-60165 |
| (13) Power Control B: 10526480 | MIL-P-60212 |
| (14) Power Control C: 10526593 | MIL-P-60213 |
| (15) Power Supply Subassembly: 10525745 | MIL-P-60178 |
| (16) Choke: 10525920 | MIL-C-60194 |
| (17) Reactor: 10525925 | MIL-R-60191 |
| (18) Choke: 10525930 | MIL-C-60190 |
| (19) Read Amplifier: 10526589 | MIL-R-60199 |
| (20) Read Switch: 10526456 | MIL-R-60198 |
| (21) Rectifier-Transistor Assembly: 10525740 | MIL-R-60176 |
| (22) Tape Reader: 10526192 | MIL-T-60171 |
| (23) Transformer, Power, Step-Down: 10525900 | MIL-T-60206 |
| (24) Transformer, Power, Step-Down: 10525905 | MIL-T-60192 |
| (25) Transformer, Power, Step-Down: 10525910 | MIL-T-60207 |
| (26) Transformer, Power, Step-Down: 10525915 | MIL-T-60193 |
| (27) Transistor Assembly: 10525770 | MIL-T-60177 |
| (28) Voltage Regulator Subassembly: 10525750 | MIL-V-60216 |
| (29) Voltage Regulator Subassembly: 10525755 | MIL-V-60217 |
| (30) Voltage Regulator Subassembly: 10525760 | MIL-V-60218 |
| (31) Voltage Regulator Subassembly: 10525765 | MIL-V-60219 |
| (32) Writer Amplifier: 10526585 | MIL-W-60197 |
| (33) Write Switch: 10526581 | MIL-W-60179 |
| b. Table, Gun Direction Computer: 10526355 | F10526355 |
| c. Cable and Reel Assembly: 10526209 | C10526209 |
| d. Adapter, Cable and Bracket: 10526771 | D10526771 |
| e. Indicator Multilayer Panel: 10526109 | F10526109 |



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**3.4 General specifications.** The computer set shall meet the following requirements of MIL-F-13926 and MIL-C-46357.

3.4.1 *MIL-F-13926.*

(a) Order of precedence

(b) Dimensions and tolerances

(c) Inorganic protective surface finishes

(d) Part identification and marking

(e) Workmanship

(f) Soldering electrical connections

3.4.2 *MIL-C-46357.*

(a) Wiring

(b) Threaded fasteners

(c) Marking

(d) Finishes

**3.5 Environmental**

3.5.1 *Shock and vibration.* The computer in its transit case shall be capable of withstanding shock and vibration which would occur when the computer is placed on a vibrating table which is vibrated at a minimum of 273 cycles per minute in a 1.0 ± 0.1 inch diameter circular motion. The computer shall be constrained to a horizontal motion of a maximum of 3.25 inches. The computer shall not be damaged or adversely affected by the shock and vibration, applied a minimum of ½ hour on each of the 6 faces of the computer, and shall subsequently meet all the requirements of this specification.

3.5.2 *Sand and dust.* The computer shall meet the requirements of 3.7 at ambient temperature of 75 ± 15° F while subjected for 6 hours to an atmosphere of sand and dust. The density of the sand dust shall be 0.1 to 0.5 grams per cubic foot with a velocity of 200 ± 100 feet per minute. The relative humidity (RH) shall not exceed 30 percent. The computer shall also meet the requirements of 3.7 after exposure for 6 hours to the same sand and dust conditions at 160° F.

3.5.3 *Temperatures.*

3.4.3.1 *Operating.* The computer shall meet the requirements of 3.7.3 thru 3.7.9 in ambient temperatures of —40° and +125° F. The following characteristics shall also be exhibited:

(a) The computer memory shall deenergize at —25° ± 2.5° F.

(b) The temperature warning indicator lamp shall energize at internal temperatures of less than —20° For more than + 130° F.

(c) The computer shall deenergize at internal temperatures of less than —40° F or more than +135° F.

3.5.3.2 *Storage.* The computer shall be capable of withstanding temperature stabilization at +160° and —80° F. The computer shall not be damaged or adversely affected by the exposure and shall subsequently meet all requirements of this specification, at 75° ± 15° F.

3.5.4 *Humidity.*

3.5.4.1 *Operating.* The computer shall meet the requirements of 3.7.3 thru 3.7.9 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of 125° F.

3.5.4.2 *Non-operating.* The computer shall meet the requirements of 3.7 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.5.5 *Fungus resistance.* The computer shall show no evidence of damage and shall meet the requirements of 3.7, after being inoculated with the composite spore suspen-



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sion specified in MIL–F–13926 and incubated at an ambient temperature of between 77° F and 84° F with a RH of 97 ± 2%.

3.5.6 *Elevation.*

3.5.6.1 *Operating.* The computer shall be operated for a period of 8 ± 1 hours and meet the requirements of 3.7.9, while exposed to conditions encountered at an altitude of 10,000 feet above sea level.

3.5.6.2 *Non-operating.* The computer shall meet the requirements of 3.7, after exposure for 8 ±1 hours to conditions encountered at an altitude of 50,000 feet above sea level.

3.5.7 *Salt laden air.* The computer shall not be damaged or its performance impaired from exposure to salt laden air as encountered in coastal areas during operation or storage.

3.5.8 *Immersion.* The computer shall be capable of continuous performance (3.7.9) after being immersed, with all covers in place and pressure relief valve closed, in 3 feet of head water for 2 hours. The water temperature shall be 70° F.

3.5.9 *Rain and wind; equipment non-operating.* The computer shall be capable of continuous performance (3.7.9) after being exposed to simulated rainfall of 2 inches per hour, accompanied by winds at the rate of 20 miles per hour for a period of 4 hours. Water temperature shall be 60° F.

3.5.10 *Operating orientation.* The computer shall be capable of continuous performance (3.7.9) while orientated at 18 degrees from the normal in each of the mutually perpendicular directions.

**3.6 Computer characteristics and terminology.** Unless otherwise specified, computer functional characteristics shall be as defined in Appendix B. Terminology not defined in Sections 3 and 4 shall be as defiined in Appendix C.

**3.7 Performance.** The performance requirements contained herein shall be valid at all conditions of operation and environment specified in this specification except that the numerical values of 3.7.1 through 3.7.3 are accurate only at temperatures of 75 ± 15° F.

3.7.1 *Power supplies.* The computer power supplies when energized from a 120 ± 5 volts rms line to neutral, 400 ± 5 cycle, 3 phase, 4 wire ac source, shall possess the characteristics listed in Table I. Power indicator and control circuits shall operate as follows:

(a) Voltage transient lamp energizes at 132 volts rms and above (adjustable threshold).

(b) Voltage transient lamp energizes at 108 volts rms and below (adjustable threshold).

(c) Computer deenergizes at 140 volts rms and above (adjustable threshold).

(d) Computer deenergizes at 100 volts rms and below for periods exceeding 5 seconds; at 80 volts rms and below for periods not exceeding 5 seconds (both thresholds adjustable).

3.7.1.1 *Marginal test.* The marginal test switch shall set the power supplies to the values listed in Table II.

3.7.2 *Memory and timing signals.* The memory, previous to insertion into the computer, shall be in accordance with all the requirements of MIL–M–60183 and shall not have been damaged or mishandled during the insertion process. All memory channels shall retain and reproduce recorded information and shall possess output characteristics and signals compatible with the associated circuits. Signal amplitudes shall be at least 2



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times the threshold value of the input of the next circuit. Noise or spurious signals shall be a maximum of 0.5 times the threshold value of the input of the next circuit. Information pulses shall be phased with respect to the pulses in the clock channel. The signals at the outputs of the Pulse Generator: 10526464 (clock amplifier) shall possess rise and fall times of 0.15 microseconds maximum and 0.2 microseconds maximum at the "Clock" and "Strobe" outputs respectively.

3.7.3 *Automatic test circuits.* The automatic test diodes, located on each logic flip-flop subassembly, when connected to a minus 6 + 10 percent volts dc supply, shall be capable of setting each flip-flop circuit to each of its two states. The outputs of logic drivers, all "Primary and gates" which are not inputs to logic drivers, and leads from all regulated supplies shall be connected to external connectors, and shall display functional characteristics equivalent to those of the circuit at the source. The write amps and write switches shall be treated as flip-flops and drivers respectively except that the true condition of write switch outputs shall be +15 ± 1.5 Vdc.

3.7.4 *Control panel and operational commands.*

3.7.4.1 *Control panel switches.* When the computer is in the "Halt" mode the set-up switches shall insert an unconditional transfer to a specified memory location into the "I" register.

3.7.4.2 *Control panel keyboard.* The keyboard shall be capable of inserting binary bits into the "A" register; transferring the computer to "Fill", "Verify", "Halt", or "Compute" modes; setting up a memory location and entering information into the main memory.

3.7.4.3 *Control panel mechanical reader.* The mechanical reader shall be capable of inserting binary bits into the "A" register; transferring the computer to "Fill", "Verify", "Halt", or "Compute" modes; setting up a memory location and entering information into the main memory.

3.7.4.4 *Arithmetical commands.* The computer shall be capable of performing all commands listed in Table III.

3.7.4.5 *Special purpose commands.* The computer shall be capable of performing all commands listed in Table IV.

3.7.5 *Input-output.*

3.7.5.1 *Externally initiated inputs.*

3.7.5.1.1 *Teletype input in numeric mode.* The computer shall be capable of accepting externally initiated teletype codes.

3.7.5.1.2 *Field data inputs in numeric mode.* The computer shall be capable of accepting externally initiated field data codes.

3.7.5.1.3 *Error re-set.* The computer shall be capable of re-setting to the "Halt" mode from an external signal.

3.7.5.1.4 *Receive set-up.* The computer shall be capable of inserting into the I register an unconditional transfer command and a memory location when activated by an external signal.

3.7.5.1.5 *Output line.* The computer shall be capable of disabling the information output lines when activated by an external signal.

3.7.5.1.6 *"Verify" mode enable.* The computer shall be capable of entering the "Verify" mode when activately by an external signal.

3.7.5.2 *Internally initiated inputs.*

3.7.5.2.1 *Read numeric.* The computer shall



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be capable of performing the read numeric commands listed in Table V.

3.7.5.2.2 *Read alpha-numeric.* The computer shall be capable of performing the read alpha-numeric commands listed in Table VI.

3.7.5.2.3 *Discrete input to the accumulator.* The computer shall be capable of accepting discrete inputs into the accumulator when matrix selections are made on the control panel switch matrix.

3.7.5.2.4 *Discrete external inputs to the accumulator.* The computer shall be capable of accepting discrete inputs into the accumulator when signals are applied to the external input lines.

3.7.5.2.5 *Magnetic tape code conversion on input.* The computer shall be capable of converting magnetic tape code to paper tape code on inputs.

3.7.5.3 *Internally initiated outputs.*

3.7.5.3.1 *Write numeric.* The computer shall be capable of transmitting teletype or field data code by word or character.

3.7.5.3.2 *Write alpha-numeric.* The computer shall be capable of transmitting alpha numeric-5 and alpha-numeric-6 codes by word or character.

3.7.5.3.3 *Discrete output.* The computer shall be capable of initiating discrete outputs, energizing the "No-Solution" lamp to flash, and resetting all discrete outputs to the zero (0) state.

3.7.5.3.4 *Teletype output timing.* The computer shall be capable of using the teletype oscillator to transmit a signal equivalent to that shown in Figure 1.

3.7.5.3.5 *Fieldata signals.* The computer shall be capable of generating the fieldata timing signals equivalent to those shown in Figure 2.

3.7.5.3.6 *Magnetic tape code conversion on output.* The computer shall be capable of converting from paper tape code to magnetic tape code on output.

3.7.5.3.7 *Tape tub signals.* The computer shall be capable of generating an address signal and an information transfer signal.

3.7.6 *Matrix.* All matrix push-buttons shall be capable of being one-set (to the true position) when depressed, without mechanical interference or electrical failures.

3.7.7 *Display indicators.* All characters (including decimal points) of the display indicators shall be capable of being lighted with a minimum of delay. The delay shall be minimumized such that, on a rapid entry (maximum of 0.25 second time interval between key depressions) of any digit on the keyboard the indicator shall light before the next depression of the key.

3.7.8 *Problems.* The procuring activity will supply a program tape, a meteorological data tape, 5 problems, one for each battery, and input and output values to be used for verification of computer performance. The computer shall be capable of correctly solving these problems in each battery and in each of the 6 positions of the marginal test switch.

3.7.9 *Continuous performance.* The computer shall be capable of solving all the problems 240 times (40 times in each of the 6 positions of the marginal test switch).

3.8 **Computer run-in.** The computer shall be subjected to a 70 hour run-in period (excluding downtime) which includes operation at the extreme temperatures (3.8.2). The 70 hour run-in period shall start after the computer has satisfactorily met the performance requirements of 3.7.1 thru 3.7.9 and 3.8.1. The time required to conduct the vibration (3.8.1) shall not be included as part of the run-in period. The computer shall continuously meet the requirements of 3.7.3 thru 3.7.8 for the entire 70 hour run-in period.



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3.8.1 *Vibration.* The computer shall be capable of withstanding, without damage, a vibration of 273 cycles per minute (minimum) with a circular motion of $1 \pm 0.1$ inch diameter for a total period of one hour (30 minutes in a plane simulating its normal operating position and 30 minutes in a plane 180° from its normal operating position). The horizontal motion shall be constrained to a maximum of 3.25 inches total excursion.

3.8.2 *Temperature extremes.* The computer shall be capable of being operated, without damage, for 4 hours when stabilized at an ambient temperature of —22°F and for 4 hours when stabilized at +125°F. For two hours of operation at each temperature the computer shall be in a normal horizontal operating position and for two hours the computer shall be rotated 90° from normal (on its side).

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *General provisions.* The component inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

**4.2 Inspection provisions.**

4.2.1 *Submission of product.*

4.2.1.1 *Initial production test.* When specified in the contract, the first two production computers shall be presented to the Government Quality Assurance Representative for initial production testing (see 6.2). The Government representative shall witness all initial production testing required for the two computers. The samples shall be manufactured in the same manner, using the same materials, processes, and procedures as will be used in regular production. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary. Initial production testing shall be in accordance with 4.5 and the respective test requirements outlined below:

ENVIRONMENTAL INSPECTION

The computers shall have met the requirements and tests in Table A and shall then meet the following requirements and tests:

| | *Characteristics* | *Requirements* | *Test Procedures* |
|---|---|---|---|
| (a) | Shock and vibration | 3.5.1 | 4.5.1.1 |
| (b) | Sand and dust | 3.5.2 | 4.5.1.2 |
| (c) | Operating temperature | 3.5.3.1, 3.7.3 to 3.7.9 incl. | 4.5.1.3, 4.5.5 to 4.5.11 incl. |
| (d) | Storage temperature | 3.5.3.2 | 4.5.1.3.1 |
| (e) | Humidity-Operating | 3.5.4.1, 3.7.3 to 3.7.9 incl. | 4.5.1.4.2, 4.5.5 to 4.5.11 incl. |
| (f) | Humidity-Non-operating | 3.5.4.2 | 4.5.1.4.1 |
| (g) | Elevation-Operating | 3.5.6.1, 3.7.9 | 4.5.1.5.2, 4.5.9.9 |
| (h) | Elevation-Non-operating | 3.5.6.2 | 4.5.1.5.1 |
| (i) | Salt laden air | 3.5.7 | 4.5.1.6, 4.5.5 to 4.5.11 incl. |
| (j) | Immersion | 3.5.8 | 4.5.1.7 |
| (k) | Rain and wind, non-operating | 3.5.9 | 4.5.1.8 |
| (l) | Operating orientation | 3.5.10, 3.7.9 | 4.5.1.9, 4.5.9.9 |



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4.2.1.2 *Initial production sample failure.* Should one or both of the two computers subjected to the testing required by 4.2.1.1 fail to meet all requirements of this specification, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections thereto. When corrections in the process or controls have been made, two additional (new) production models shall be submitted in accordance with 4.2.1.1.

4.2.2 *Production.* Inspection of production computers shall be in accordance with all test requirements forming a part of 4.5.2 to 4.5.12 inclusive and Tables A and B. Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL–STD–105.

4.2.3 *Acceptance and rejection.* Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in acordance with "Acceptance and Rejection" as specified in MIL–STD–105.

4.2.4 *Functional tests.* The requirements and tests in Table A shall be inspected in a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE A
(100 percent inspection)

| | *Characteristics* | *Requirements* | *Test Procedures* |
|---|---|---|---|
| 101. | Power supplies | 3.7.1 | 4.5.3 |
| 102. | Marginal test | 3.7.1.1 | 4.5.3.1 |
| 103. | Memory and timing signals | 3.7.2 | 4.5.4 to 4.5.4.9 |
| 104. | Automatic test circuits | 3.7.3 | 4.5.5 to 4.5.5.6 |
| 105. | Control panel switches | 3.7.4.1 | 4.5.6.1 |
| 106. | Control panel keyboard | 3.7.4.2 | 4.5.6.2 |
| 107. | Control panel mechanical reader | 3.7.4.3 | 4.5.6.3 |
| 108. | Arithmetical commands | 3.7.4.4 | 4.5.6.4 |
| 109. | Special purpose commands | 3.7.4.5 | 4.5.6.5 |
| 110. | Teletype input in numeric codes | 3.7.5.1.1 | 4.5.7.1.1 |
| 111. | Field data inputs in numeric codes | 3.7.5.1.2 | 4.5.7.1.2 |
| 112. | Error re-set | 3.7.5.1.3 | 4.5.7.1.3 |
| 113. | Receive set-up | 3.7.5.1.4 | 4.5.7.1.4 |
| 114. | Output line | 3.7.5.1.5 | 4.5.7.1.5 |
| 115. | "Verify" mode enable | 3.7.5.1.6 | 4.5.7.1.6 |
| 116. | Read numeric | 3.7.5.2.1 | 4.5.8.1 |
| 117. | Read Alpa-numeric | 3.7.5.2.2 | 4.5.8.2 |
| 118. | Discrete input to the accumulator | 3.7.5.2.3 | 4.5.8.3 |
| 119. | Discrete external inputs to the accumulator | 3.7.5.2.4 | 4.5.8.4 |
| 120. | Magnetic tape code conversion on input | 3.7.5.2.5 | 4.5.8.5 |
| 121. | Write numeric | 3.7.5.3.1 | 4.5.9.1 |
| 122. | Write alpha-numeric | 3.7.5.3.2 | 4.5.9.2 |
| 123. | Discrete output | 3.7.5.3.3 | 4.5.9.3 |
| 124. | Teletype output timing | 3.7.5.3.4 | 4.5.9.4 |
| 125. | Field data signals | 3.7.5.3.5 | 4.5.9.5 |
| 126. | Magnetic tape code conversion on output | 3.7.5.3.6 | 4.5.9.6 |
| 127. | Tape tub signals | 3.7.5.3.7 | 4.5.9.7 |
| 128. | Matrix | 3.7.6 | 4.5.9.8 |
| 129. | Display indicators | 3.7.7 | 4.5.9.9 |
| 130. | Problems | 3.7.8 | 4.5.10 |
| 131. | Continuous performance | 3.7.9 | 4.5.11 |
| 132. | Computer run-in | 3.8 | 4.5.12 |
| 133. | Vibration (run-in) | 3.8.1 | 4.5.12.1 |
| 134. | Temperature extremes (run-in) | 3.8.2 | 4.5.12.2 |



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**4.3 Standard sampling.**

4.3.1 *General.* One unit shall be selected at random by a Government representative as a standard sample from each 100 produced. The sample shall meet the requirements and tests in Table B.

TABLE B

| Characteristic | Requirement | Test Procedures |
|---|---|---|
| 301. Fabrication | 3.1 | Applicable drawings-visual |
| 302. General specifications | 3.3 | MIL-F-13926, MIL-C-46377, visual |

4.3.2 *Failure of sample.* Should the sample fail to meet the specified requirements, acceptance of the represented production lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

**4.4 Inspection equipment.** Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 *Government furnished inspection equipment.* Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.4.2 *Contractor furnished inspection equipment.*

4.4.2.1 *Government design.* All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the requirements of 4.4 and the List of Inspection Documents, IEL-10525735.

4.4.2.2 *Contractor design.* The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration test equipment.

4.4.3 *Test equipment.* In conjunction with 4.4.2.2 the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

| Item No. | Description |
|---|---|
| Item 1 | Volt/OHM Meter; A D.C. range at least to 200 volts at no less than 10,000 ohms/volt. A.C. voltage at least to 300 volts (rms) at no less than 5000 ohms/volt. Capable of measuring 0 to 100 megohms. Accuracy to at least ± 3% of full scale (a-c voltage of 60 cps.) Ohmmeter accuracy of at least ± 3% of D.C. scale with full battery. |
| Item 2 | VOLTMETER, D.C. DIGITAL: A range of at least .001 to 999 volts D.C., or three significant figures. Accuracy of at least ± 0.2% of reading ± 1 count. Input impedance of at least 10 megohms on all ranges. |
| Item 3 | OSCILLOSCOPE: dual channel, rise time of scope plus measuring probe 50 milli-microseconds or less, sweep rate of 50 milli-microseconds per centimeter or greater, input resistance no less than 8 megohms, input capacitance less than 25 pico-farads. |



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**4.5 Test methods and procedures.**

4.5.1 *Environmental.*

4.5.1.1 *Shock and vibration.* For compliance with 3.5.1 the computer, in its transit case, shall be placed on a test table, but not fastened to it. Wooden fences shall be placed around the case and shall constrain its horizontal motion to a maximum of 3.25 inches total excursion. The equipment shall be vibrated at a minimum of 273 cycles per minute for a total of 3 hours. During each ½ hour period the transit case, containing the computer, shall rest on a different face. The motion of all points on the table shall be circular with a 1.0 ± 0.1 inch diameter. During the test the computer shall not be operated. It shall be subsequently inspected for physical damage, and shall be subjected to the tests defined in 4.5.2 to 4.5.11 inclusive, as specified in 3.5.1.

4.5.1.2 *Sand and dust.* This test shall be conducted in accordance with the sand and dust test defined in MIL–F–13926. The test shall not be repeated at 0 degrees Fahrenheit, and temperature changes shall not occur at a rate exceeding 2 degrees Fahrenheit per minute.

During the operating portion of the test, defined in 4.5.2 to 4.5.11 the computer air filters shall be replaced as required to obtain proper operation. Subsequent to the tests the computer shall be inspected for physical damage, and shall be subjected to the tests defined in 4.5.2 to 4.5.11 inclusive, as specified in 3.5.2.

4.5.1.3 *Operating temperatures.* This test shall be conducted in accordance with Procedure I, Conditions C and E of the temperature tests defined in MIL–F–13926 with the following exceptions and additions:

(a) The temperature shall never be changed at a rate exceeding 2 degrees Fahrenheit per minute.

(b) After temperature stabilization occurs at plus 125 degrees Fahrnheit, the computer shall remain non-operating at that temperature for 64 plus or minus 4 hours, and shall then be operated for 8 plus or minus 1 hours. Subsequent to operation, the computer shall be non-operating while the temperature is returned to room ambient.

(c) After temperature stabilization occurs at minus 40 degrees Fahrenheit, the computer shall be energized and, after the "Power Ready" lamp comes on, shall be operated for 8 plus or minus 1 hours. The computer shall be non-operating while the temperature is returned to room ambient.

(d) Throughout the operating portions of this test the computer shall be subjected to the tests defined in 4.5.5 to 4.5.11 inclusive, and shall meet the requirements specified in 3.5.3.1.

4.5.1.3.1 *Storage temperatures.* This test shall be conducted in accordance with the temperatures in 3.5.3.2 and test Procedure II of MIL–F–13926 with the following exceptoins and additions:

(a) The temperature shall not be changed at a rate exceeding 2 degrees Fahrenheit per minute.

(b) The computer covers shall be properly in place throughout the test, and shall be removed as required for performance of subsequent operating tests.

(c) After temperature stabilization at minus 80 degrees Fahrenheit occurs, the temperature shall be maintained constant for period of 24 plus or minus 2 hours. The temperature shall then be re-



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turned to room ambient and temperature stabilization shall occur before computer operation is initiated.

(d) After temperature stabilization at plus 160 degrees Fahrenheit occurs, the temperature shall be maintained constant for 4 hours. The temperature shall then be returned to room ambient and temperature stabilization shall occur before computer operation is initiated.

(e) Subsequent to the extreme temperature tests, the computer shall be inspected for physical damage, and shall be subjected to the tests defined in 4.5.5 to 4.5.11 inclusive as specified in 3.5.3.2.

4.5.1.4 *Humidity.*

4.5.1.4.1 *Non-operating.* This test shall be conducted in accordance with the humidity test defined in Procedure I of MIL–F–13926 except that the 4½ hour period shall be reduced to 2 hours, and the 6 hour period reduced to 2 hours and the test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and subjected to the tests defined in 4.5.2 to 4.5.11 inclusive, and shall meet the specified requirements of 3.5.4.2.

4.5.1.4.2 *Operating.* This test shall be conducted in accordance with the Humidity Test defined in Procedure III of MIL–F–13926 except that the upper temperature shall be modified to 125 degrees Fahrenheit, the humidity shall be maintained at 92 plus or minus 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL–STD–202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.5 to 4.5.11 inclusive, and shall meet the specified requirements of 3.5.4.1.

4.5.1.5 *Elevation.*

4.5.1.5.1 *Non-operating.* The computer with covers in place shall be placed within a test chamber and the absolute internal pressure of the chamber reduced to 3.44 inches of mercury (Corresponding to an altitude of 50,000 feet above sea level) and an ambient temperature of minus 65 degrees Fahrenheit. The rate of change of pressure shall not exceed 2.44 inches of mercury per minute, and the rate of change of temperature 2 degrees Fahrenheit per minute. The pressure and temperature shall be maintained constant for 8 plus or minus 1 hours and shall be returned to room nominal conditions at the same rates. Subsequent to exposure, the computer shall be inspected for physical damage, and shall be subjected to the tests defined in 4.5.2 to 4.5.11 inclusive, as specified in 3.5.6.2.

4.5.1.5.2 *Operating.* For compliance with 3.5.6.1 the computer shall be placed within a test chamber and the absolute internal pressure of the chamber reduced to 20.58 inches of mercury (Corresponding to an altitude of 10,000 feet). The rate of change of pressure shall not exceed 2.44 inches of mercury per minute. The temperature shall be maintained at 75 plus or minus 15 degrees Fahrenheit. The computer shall be operated for 8 plus or minus 1 hours in accordance with 4.5.11 of this specification. The return to room ambient pressure shall be at a maximum of 2.44 inches of mercury per minute.

4.5.1.6 *Salt spray.* The unit shall be subjected to the salt spray test defined in MIL–STD–202, Condition B. The computer shall be subsequently inspected for physical damage, and shall be operated for 4 hours at room ambient conditions. Operation subsequent to the test shall be in accordance with 4.5.5 to 4.5.11 inclusive to determine compliance with 3.5.7.

4.5.1.7 *Immersion.* For compliance with



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3.5.8, the computer with all covers in place, and pressure relief valve closed, shall be immersed under a 3 feet plus or minus 2 inch head of water for a period of 2 plus or minus 1/10 hours. The covers shall then be removed and the computer shall be operated in accordance with 4.5.11 of this specification. The water temperature shall be 70 plus or minus 10 degrees Fahrenheit.

4.5.1.8 *Rain and wind non-operating.* For compliance with 3.5.9, the computer with all covers in place, shall be exposed to simulated rainfall at the rate of 2 plus or minus 1/2 inches per hour, accompanied by a wind at a rate of 20 plus or minus 2 miles per hour for a period of 4 plus or minus 1/2 hours. The wind shall be directed at the front surface of the front cover of the computer and be inclined 45 plus or minus 20 degrees from the horizontal. The computer shall be mounted on the field table in its normal position. After exposure to this environment, covers shall be removed and the computer shall be subjected to the test defined in 4.5.11 of this specification. The water temperature shall be 60 plus or minus 25 degrees Fahrenheit.

4.5.1.9 *Operating orientation.* The computer shall be subjected to the test defined in 4.5.11 of this specification while being operated in the orientation of 18 plus or minus 2 degrees in each of 3 mutually perpendicular directions from the normal operating orientation as specified in 3.5.10.

4.5.1.10 *Fungus.* For compliance with 3.5.5 one of the first 5 computers produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.2 *Performance tests.* Unless otherwise specified, compliance with 3.7 shall be in accordance with the following test procedures, with all sub-assemblies and hardware properly installed and connected. The center transit cover of the computer shall be in place, except when access to components or test points is specifically required by the test procedure. The front and rear computer transit covers shall be removed. The computer shall be mounted on the Field Table and all cables connected. Unless stated otherwise, ac power shall be applied and the computer energized with the following conditions and procedures for testing.

(a) *Verification of information.* Throughout section 4 of this specification, the word "monitor" shall require a verification of information contained in computer storage locations by observation at external test points or equivalent comparison through performance of computer commands. External test points are listed, but such references, unless otherwise specified, shall not preclude verification through the use of internal computer operations.

(b) *Signal conventions.* Unless otherwise specified, the signal conventions defined herein shall be applicable throughout the performance of functional tests specified in section 4 of this specification. Nominal signal levels shall be implied and measurement of signal amplitude is not necessarily required.

(c) *Logic signals.* Binary terms shall have the following significance:

1. A "1" or "True" signal shall be represented by a potential between minus 4.5 and minus 6.5 volts (nominal minus 6 volts).
2. A "0" or "False" signal shall be represented by a potential between plus 0.5 and minus 1.0 volts (nominal 0.0 volts).

(d) *Fieldata, teletype, and alpha-numeric signals.* Binary terms shall have the following significance:



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1. A "1" or "True" signal shall be represented by a potential between minus 2.0 and minus 6.0 volts (nominal minus 3 volts).
2. A "0" or "False' 'signal shall be represented by a potential between minus 0.15 and plus 0.15 volts (nominal 0.0 volts).

(e) *Memory external output signals.* Binary terms shall have the following significance when monitored at the Memory External Output Test Points listed in Table VII.

1. A "1" or "True" signal shall be represented by a potential between plus 0.5 and minus 1.0 volts (nominal 0.0 volts).
2. A "0" or "False" signal shall be represented by a potential between 4.5 and minus 6.5 volts (nominal minus 6 volts).

4.5.3 *Power supplies.* With the marginal test switch in off position, the computer power supplies shall be tested with Magnetic Memory, and printed circuit sub-assemblies disconnected, except as indicated in (a). Tests, operations, and measurements shall be in accordance with the following procedures for compliance with 3.7.1.

(a) The following printed circuit subassemblies shall remain connected within the computer:

Regulator Subassembly, 10525750
Regulator Subassembly, 10525755
Regulator Subassembly, 10525760
Rectifier-Transistor Assembly
Transistor Assembly
Power Supply Subassembly
Power Control A
Power Control B
Power Control C
Regulator #4 Subassembly, 10525768
Network A Subassemblies, 10526475 (J401, J402, J404, J403)*
Network B Subassembly, 10526471 (J406)*
Network C Subassembly, 10526479*
Write Switch Subassemblies, 10526581*

* For tests of 4.5.3.1 (a) thru (b).

(b) Connect the resistors of Column C, Table VIII to the indicated supplies at the terminal block in the computer main frame. With the AC line adjusted to 102 ± 1 volts rms perform the measurements of Table I. Repeat the measurements with the line adjusted to 138 ± 1 volts rms.

(c) Replace the resistors at the terminal block with the resistors of Column B, Table VIII. Repeat the measurements of procedure (b) above.

(d) With the line adjusted to 120 ± 1 volts rms, connect a resistor from Column D, Table VIII across the supply and in parallel with the resistor of Column B. Observe that all regulated supplies have de-energized, and that the output voltages are 0 ± 1 volt, except pin a which shall have 60 volts when minimum load is connected and 23 volts when Ko load is connected. Re-apply computer power and sequentially repeat the procedure using each resistor of Column D across the indicated supply. As each resistor is connected, the associated pin J13 listed in Column E shall measure +.5 ± .5 volts concurrently. All other pins listed shall measure +6 ± 1 volts.



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4.5.3.1 *Marginal test.* For compliance with 3.7.7.1, replace the resistors of Column C, Table VIII. Set the "Marginal Test" switch to each position listed in Table II and perform the indicated measurements. At the conclusion of this test, set the "Marginal Test" switch to the OFF position.

(a) Insert network A and network C with the ac line adjusted to each value in Column A of Table IX observe that the condition described in Column B occurs. De-energize the computer between adjustments.

(b) Connect Power Control A to its connector through board extender or connector converter. With the computer energized, sequentially connect the resistors of Table X to the indicated terminals and observe that the effect described in Column D occurs. Re-apply computer power after each test, except after test 7. When the resistor of line 7 is disconnected pin 9 shall energize and the temperature lamp shall stop flashing. Twenty three (23) seconds later the power ready light shall light.

4.5.4 *Memory and timing signals.* Unless otherwise specified, memory and timing signal tests shall be performed with all circuit modules and sub-assemblies installed and operating to determine compliance with 3.2.2. The computer shall be opened for access to specified test point. Refer to appendices for memory loading procedures, word format, and computer characteristics.

4.5.4.1 *Clock channel.* Connect the pulse generator to its connector through a board extender or a connector converter. Energize the computer. The output of the clock channel at pin 56B on J432 shall have an amplitude not less than 1.0 volt peak-to-peak.

4.5.4.2 *Sector channel.* Observe the output of the Sector channel at the test point listed in Column C of Table XI. The signal shall have the following characteristics:

(a) The measured output shall consist of a train of pulses of amplitude between 4.0 and 8.3 volts peak-to-peak.

(b) Where information pulses do not occur, spurious signals shall be not greater than 0.45 volt, zero-to-peak.

4.5.4.3 *Read signal phasing.* The following test procedure shall be performed to verify that each channel output has the correct time relationship with respect to strobing pulses:

(a) Load a pattern of not less than 6 alternating ones and zeros into each addressable storage channel and recirculating loop.

(b) Memory channels of the non-recirculating type shall be measured at "MMN" test point listed in Column D, Table XI. Selection for monitoring purposes shall be accomplished by the application of input signals to the lines listed in Table XII. Input levels shall be those listed in 4.5.2(c). Measure Strobe Signal at board 301, Pin 1B.

(c) All measurements at test points listed in Column D of Table XI with the exception of the Sector channel and the output designate "MMP" shall be as specified in (f) below.

(d) Measurements at the output of the Sector channel at the test point in Column D of Table II shall be performed as in (f) below except that measurements shall be made at not less than 4 positions around the Sector Channel. Positions shall be chosen so that the bit pattern written in the



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sector channel provides an output waveform of not less than 3 adjacent alterations.

(e) An appropriate routine shall be generated and executed to cause the contents of not less than 1 of the non-recirculating storage channels to be displayed at "MMP" at the test point in Column D of Table XI.

(f) The time relationship of the strobing pulse to the memory output shall be as shown in Figure 3. Time shall be measured between the positive or negative going 0.5 volt levels of the leading edges of the memory output signals and the minus 5.0 volt level of the negative going strobe signal and shall be equal to (X ± 1)% of the strobe period. This strobe period shall be determined from leading edge of adjacent strobe pulses. X shall be 75 for all memory channels, recirculating loops and the sector track.

(g) The time measured shall be accurate to ± 5% of the strobe period.

(h) Variance in the time measured at "MMN" and that measured at "MMP" shall be not greater than 1% of the strobe period.

4.5.4.4 *Read outputs.* The outputs of all addressable storage channels and recirculating loops shall be measured by the following test procedure:

(a) Each storage channel shall be loaded with a bit pattern of alternating ones and zeros.

(b) A second pattern consisting of a reversed pattern of alternating zeros and ones shall be loaded into each storage channel.

(c) The outputs at "MMN" at the test point in Column C of Table XI shall be not less than 3.2 volts, peak-to-peak. Selection of storage channels shall be specified in Paragraph 4.5.4.3.(b).

(d) Each storage channel shall be loaded with a repetitive bit pattern consisting of 3 ones followed by 3 zeros. This bit pattern shall be reinforced by successive writing in each storage channel not less than 3 times.

(e) The outputs at "MMN" at the test point in Column C of Table XI, shall be not greater than 6.95 volts, peak-to-peak. Selection of storage channels shall be as specified in Paragraph 4.5.4.3(b).

(f) An appropriate routine shall be generated and executed to cause the contents of not less than 1 of the non-recirculating storage channels to be displayed at the test point "MMP" at Column C of Table XI for both test patterns specified. The measurements of (c) and (e) above shall be performed.

(g) Each recirculating loop shall be loaded with a bit pattern of alternating ones and zeros.

(h) The outputs at the test points listed in Column C of Table XI except "MMN", "MMP", and "S" shall be not less than 4.0 volts, peak-to-peak.

(i) Each recirculating loop shall be loaded with a repetitive bit pattern consisting of 3 ones followed by 3 zeros.

(j) The outputs at the test points listed in Column C of Table XI except "MMN", "MMP", and "S" shall be not greater than 8.3 volts, peakt-o-peak.



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4.5.4.5 *Adjacent channel interaction.* Each non-recirculating storage channel shall be tested for adjacent channel interaction using the following procedure:

(a) Storage channels physically adjacent to the channel under test shall be loaded with a bit pattern of alternating ones and zeros.

(b) The channel under test shall be loaded with a bit pattern of successive ones.

(c) The output of the channel to be tested shall be measured at "MMN" at the test point in Column C in Table XI. Selection of storage channels shall be as specified in 4.5.4.3(b).

(d) Noise in the channel under test shall be not greater than 0.45 volt, zero-to-peak.

(e) An appropriate routine shall be generated and executed to cause the contents of not less than 1 of the non-recirculating storage channels to be displayed at "MMP" at the test point in Column C of Table XI. The measurement of (d) above shall be performed.

4.5.4.6 *Recirculating loop interaction.* To insure the effectiveness of erase heads used to separate recirculating loops, and to test for interaction, each recirculating loop shall be tested using the following procedure:

(a) All recirculating loops except the loop under test shall be loaded with a bit pattern of alternating ones and zeros.

(b) The loop under test shall be loaded with a bit pattern of successive ones.

(c) The output of the loop under test shall be tested at the test point designated in Column C of Table XI.

(d) Noise in the loop under test shall be not greater than 0.45 volt, zero-to-peak.

4.5.4.7 *Clock pulse.* Connect the pulse generator to its connector through a board extender or connector converter. Refer to Figure 4. Observe the clock pulse at pins 5A and 13B of J432. The signal shall have the following characteristics:

(a) The rise time and fall time of the clock pulse, when measured at each test point specified, shall be not greater than 0.15 microseconds between the minus 0.6 and the minus 5.4 volt level.

(b) Pulse width at the minus 3.0 volt level, when measured at each test point specified, shall be between 0.4 and 0.6 microseconds.

4.5.4.8 *Marginal clock pulse variation.* The following procedure shall be used to observe marginal test variation of the clock pulse:

(a) Place the "Marginal Test" switch to the "1", "3", and "5" positions.

(b) Pulse width at the minus 3.0 volt level, when measured at either test point specified shall be increased by between 4 and 8 percent of the clock period for each switch position specified.

(c) Place the "Marginal Test" switch to the "2" and "4" positions.

(d) Pulse width at the minus 3.0 volt level, when measured at either test point specified shall be decreased by between 4 and 8 percent of the clock period for each switch position specified.

4.5.4.9 *Strobing pulse.* Connect the Network "C" sub-assembly to its connector through a board extender or connector converter. Refer to Figure 5. The strobing pulse,



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when measured at pin 62A of J323, shall have the following characteristics:

(a) pulse width at the minus 4.25 volt level of the strobing pulse shall be between 0.35 and 0.65 microseconds.

(b) The rise time and fall time of the strobing pulse shall be not greater than 0.2 microseconds when measured between the 0.35 and the minus 8.85 volt level of the pulse.

4.5.5 *Automatic test circuits.* Compliance with 3.7.3 shall be determined by applying a voltage of not less than minus 16.2 and not greater than minus 19.8 volts to pin NN of J12 (—18VF) and pin PP of J12 (—18VFT). Pin J of J13 shall be connected to system ground. Prior to performing individual tests of automatic test circuits, an initialization procedure shall be performed to insure that spurious responses do not invalidate the results of subsequent tests.

4.5.5.1 *Zero-set all.* Prior to making a one-set test on a given flip-flop, the following initialization procedure shall be performed to insure that all flip-flops are in the zero state:

(a) Apply an input level of between minus 6.6 and minus 5.4 volts to each of the selection lines listed in Table XIII.

(b) Apply an input level of between 2.7 and 3.3 volts to the One Inhibit (FD4) line on pin C of J13.

(c) Apply an input level of betweer minus 6.6 and minus 5.4 volts to the Zero Inhibit (FD5) line on pin D of J13.

(d) Apply an input level of between 2.7 and 3.3 volts to the One-set Read (1SR) line on pin n of J12.

(e) Momentarily apply an input level of between minus 6.6 and minus 5.4 volts to the Zero-set Read (OSR) line on pin p of J12.

(f) Apply an input pulse equivalent to that shown in Figure 6 of the Clock Trigger Input (CTI) line on pin m of J12.

4.5.5.2 *One-set all.* Prior to making a zero-set test on a given flip-flop, the following initialization procedure shall be performed to insure that all flip-flops are in the one state:

(a) Apply an input level of between minus 6.6 and minus 5.4 volts to each of the selection lines listed in Table XIII.

(b) Apply an input level of between 2.7 and 3.3 volts to the Zero Inhibit (FD5) line on pin D of J13.

(c) Apply an input level of between minus 6.6 and minus 5.4 volts to the One Inhibit (FD4) line on pin C of J13.

(d) Apply an input level of between 2.7 and 3.3 volts to the Zero-set Read (OSR) line on pin p of J12.

(e) Momentarily apply an input level of between minus 6.6 and minus 5.4 volts to the One-set Read (1SR) line on pin n of J12.

(f) Apply an input pulse equivalent to that shown in Figure 6 to the Clock Trigger Input (CTI) line on pin m of J12.

4.5.5.3 *One-set tests.* Each of the flip-flops shall be tested for one-set operation by energizing appropriate internal automatic test diodes. The Zero-set all initialization procedure specified in 4.5.5.1 shall be performed prior to any individual one-set test. The following test procedure shall be performed for each of the flip-flops listed in Table XIV.

(a) Apply an input level of between mi-



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nus 6.6 and minus 5.4 volts to the two selection lines associated with the flip-flop under test, as listed in Table XIV.

(b) Apply a level of between minus 0.6 and plus 0.6 volts to each of the DU and DL lines not associated with the flip-flop under test. Apply a level of between 2.7 and 3.3 volts to each of the DUR and DLR lines not associated with the flip-flop under test.

(c) If the selection lines associated with the flip-flop under test are designated DU and DL, perform the operations specified in Paragraph 4.5.5.1 (b), (c) and (f).

(d) If the selection lines associated with the flip-flop under test are designated DUR and DLR, perform the operations specified in Paragraph 4.5.5.1 (d) and (e).

(e) Output measured on the One Sense (FIT) line at pin i on J12 shall be between minus 6.5 and minus 4.5.

(f) Output measured on the Zero Sense (FOT) line at pin j on J12 shall be between minus 1.0 and plus 3.0 volts.

4.5.5.4 *Zero-set tests.* Each of the flip-flops shall be checked for zero-set operation by energizing appropriate internal automatic test diodes. The One-set all initialization procedure specified in 4.5.5.2 shall be performed prior to any individual zero-set test. The following test procedure shall be performed for each of the flip-flops listed in Table XIV.

(a) Apply an input level of between minus 6.6 and minus 5.4 volts to the two selection lines associated with the flip-flop under test, as listed in Table XIV.

(b) Apply a level of between minus 0.6 and plus 0.6 volts to each of the DU and DL lines not associated with the flip-flop under test. Apply a level of between 2.7 and 3.3 volts to each of the DUR and DLR lines not associated with the flip-flop under test.

(c) If the selection lines associated with the flip-flop under test are designated DU and DL, perform the operations specified in 4.5.5.2 (b), (c), and (f).

(d) If the selection lines associated with the flip-flop under test are designated DUR and DLR, perform the operations specified in 4.5.5.2 (d) and (e).

(e) Output measured on the Zero Sense (FOT) line at pin j on J12 shall be between minus 6.5 and minus 4.5 volts.

(f) Output measured on the One Sense (FIT) line at pin i on J12 shall be between minus 1.0 and plus 3.0 volts.

4.5.5.5 *Logic output tests.* The automatic test output lines listed in Table XV are energized by combinations of the outputs of computer flip-flops. Each of the outputs shall be tested by the following test procedure:

(a) Perform the initialization procedure of 4.5.5.1 or 4.5.5.2 specified in Column B of Table XV for the output line under test.

(b) Output measured on the line under test shall be between minus 1.0 and plus 0.5 volts.

(c) Using the procedures of 4.5.5.3 (a) through (d) and 4.5.5.4 (a) through (d), set up flip-flops associated with the output line under test as specified in Column C and D of Table XV.

(d) Output measured on the line under



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test shall be between minus 6.5 volts and minus 4.5 volts, except for the CTB and SS write switches on J13, which shall be +15 ± 1.5 Vdc.

4.5.5.6 *Voltage monitor lines.* Voltage monitor lines shall be tested by measurement of voltage levels at all pins specified in Column B of Table XVI. The output voltage level shall be within 5 percent of the nominal level specified in Column C of Table XVI.

4.5.6 *Control Panel and operational commands.* During the performance of each test specified herein, the computer shall be in the normal operating configuration as defined in 4.5.2 to insure conformance with 3.7.4.

4.5.6.1 *Control Panel switches.* Perform the following procedures to test the Control Panel switches for compliance with 3.7.4.1:

(a) Observe that the "Input-Output" and "Compute" indicator lamps are not energized.

(b) Apply a voltage between plus 5 and plus 7 volts to pin d of J17.

(c) Apply a voltage between minus 19 and minus 11 volts to pin e on J17.

(d) Momentarily depress the Control Panel switches listed in Column A of Table XVII in the order indicated.

(e) The computer word listed in Column B of Table XVII opposite the switch listed shall appear in the I register.

4.5.6.2 *Control Panel keyboard.* Perform the following procedures to test the operation of the manual keyboard on the computer Control Panel for compliance with 3.7.4.2:

(a) Apply a voltage between plus 5 and plus 7 volts to pin e on J17.

(b) Apply a voltage between minus 9 and minus 11 volts to pin d on J17.

(c) Momentarily depress the "Receive" switch on the Control Panel.

(d) Momentarily apply a voltage between minus 9.0 and minus 11.0 volts to pin f on J17.

(e) The "Keyboard" indicator lamp shall become energized.

(f) The voltage at pin GG on J17 shall be between plus 100.0 and plus 120.0 volts.

(g) The voltage at pin FF on J17 shall be between 0.0 and minus 6.0 volts.

(h) Clear the A register.

(i) Depress sequentially each key of the keyboard listed in Column A of Table XVIII.

(j) Information corresponding to bit patterns listed in Column A of Table XVIII shall appear in the three least significant positions of the A register.

(k) Momentarily depress the "—" key.

(l) The voltage at pin GG of J17 shall be between 0.0 and minus 6.0 volts.

(m) The voltage at pin FF at J17 shall be between plus 100.0 and 120.0 volts.

(n) Momentarily depress the "+" key.

(o) The voltage measured at pin FF of J17 shall be between 0.0 and minus 6.0 volts.

(p) Momentarily depress the "." key.

(q) Monitoring the A and L registers, observe that the contents of the A register have transferred into the register.



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(r) Monitoring the contents of the N register, momentarily depress the "Enter" key. Observe that the contents of the A register has transferred into the N register.

(s) Remove the voltages applied in (a) and (b).

(t) Momentarily depress the "9" key. The "Keyboard" and the "Input/Output" indicator lamps shall become de-energized and the "Compute" indicator lamp shall become energized.

(u) Momentarily depress the "Reset" button.

(v) The "Compute" indicator lamp shall become de-energized.

(w) Apply a voltage between plus 5 and plus 7 volts to pin e on J17. Apply a voltage between minus 9 and minus 11 volts to pin d on J17.

(x) Momentarily depress the "Receive" switch on the Control Panel.

(y) Momentarily apply a voltage between minus 9.0 and minus 11.0 volts to pin f on J17.

(z) The "Keyboard" indicator lamp shall become energized.

(aa) Repeat (s).

(bb) Momentarily depress the "8" key.

(cc) The "Keyboard" indicator lamp shall become de-energized.

4.5.6.3 *Control panel mechanical reader.* Perform the following test procedures to check operation of the Mechanical Reader on the computer Control Panel for compliance with 3.7.4.3.

(a) Apply a voltage between plus 5 and plus 7 volts to pin e on J17.

(b) Apply a voltage between 9 and minus 11 volts to pin d on J17.

(c) Momentarily depress the "Recall" switch on the Control Panel.

(d) Momentarily apply a voltage between minus 9.0 and minus 11.0 volts to pin f on J17. The "Input/Output" indicator lamp shall become energized.

(e) Insert prepared tapes containing the tape codes listed in Column A of Table XIX.

(f) Information corresponding to the bit patterns listed in Column B of Table XIX shall appear in the three least significant positions of the A register.

(g) Insert the tape code for "Verify".

(h) The voltage at pin FF on J17 shall be between plus 100.0 plus 120.0 volts.

(i) The voltage at pin GG on J17 shall be between 0.0 and minus 6.0 volts.

(j) Insert the tape code for "Fill".

(k) The voltage measured at pin FF on J17 shall be between 0.0 and minus 6.0 volts.

(l) Insert the tape code for "Location".

(m) Monitoring the A and L registers, observe that the contents of the A register have transferred into the L register.

(n) Remove the voltages applied in (a) and (b).

(o) Insert the prepared tape code for "Compute".

(p) The "Input-Output" indicator lamp shall become de-energized and the "Compute" light shall become energized.

(q) Momentarily depress the "Reset" switch.



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(r) The "Compute" light shall become de-energized.

(s) Repeat (a) and (b).

(t) Energize the Mechanical Reader.

(u) Insert the prepared tape code for "Halt".

(v) The "Input-Output" indicator lamp shall become de-energized.

(w) Insert prepared tape for illuminating all elements of nixie indicators on front panel.

4.5.6.4 *Arithmetical commands.* Perform the following procedure to test the arithmetical commands for compliance with 3.7.4.4.

(a) Generate and execute a routine to exercise each command listed in Table III.

(b) Monitor the appropriate memory locations to verify proper command execution.

4.5.6.5 *Special purpose commands.* Perform the following procedures to test the special purpose commands for compliance with 3.7.4.5:

(a) Generate and execute a routine to exercise each command listed in Table IV.

(b) Monitor the appropriate memory locations to verify proper command execution.

4.5.7 *Input-Output.*

4.5.7.1 *Externally initiated inputs.*

4.5.7.1.1 *Teletype code input in numeric mode.* Perform the following procedure to test externally initiated Numeric Teletype code inputs for determining compliance with 3.7.5.1.1:

(a) Apply an input voltage to each of the pins listed in Table XX.

(b) The voltage output at pin FF on J17 shall be between 0.0 and minus 0.6 volts.

(c) The voltage output at pin GG on J17 shall be between plus 100.0 and plus 120.0 volts.

(d) Apply appropriate Teletype input signals to the input pins such that proper computer words are written into main memory. The correspondence between the Teletype codes and the computer input lines are as shown in Figure 7.

(e) Simultaneously, apply to pin S on J17 an equivalent timing signal with respect to the information signal as shown in Figure 8.

(f) Simultaneously, apply a voltage between minus 9.0 and minus 11.0 volts to pin M on J17. This voltage shall be present during the time that signal inputs are applied to the input lines.

(g) After computer words composed of Teletype codes have been applied to the input lines, apply the Teletype code for "Halt".

(h) The voltage at pins FF on J17 shall be between plus 100.0 and plus 120.0 volts.

(i) Momentarily apply a voltage between minus 9 and minus 11 volts to pin f on J17.

(j) The voltage output at pin FF on J17 shall be between 0.0 and minus 0.6 volts.

(k) The voltage output at pin GG on J17 shall be between plus 100.0 and plus 120.0 volts.

(l) Apply the Teletype code for "Verify".

(m) The voltage at pin FF on J17 shall



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be between plus 100.0 and plus 120.0 volts.

(n) The voltage at pin GG on J17 shall be between 0.0 and minus 6.0 volts.

(o) Apply the same information with the same memory locations to the input lines as was applied in (d) of this paragraph.

(p) The "Error" lamp on the computer Control Panel shall be energized and not flashing during the time that the information signals are being re-entered.

4.5.7.1.2 *Fieldata code input in numeric mode.* Perform the following procedures to test externally initiated Numeric Fieldata code inputs for determining compliance with 3.7.5.1.2:

(a) Apply an input voltage to each of the pins listed in Table XXI.

(b) The voltage output at pin FF on J17 shall be between 0.0 and minus 6.0 volts.

(c) The voltage output at pin GG on J17 shall be between plus 100.0 and plus 120.0 volts.

(d) Apply Fieldata input signals to the input pins such that computer words are written into main memory. The correspondence between the Fieldata codes and the computer input lines are as shown in Figure 9.

(e) Simultaneously, apply to pin S on J17 a timing signal with respect to the information signal equivalent to that shown in Figure 8.

(f) Simultaneously, apply a voltage between minus 9.0 and minus 11.0 volts to pin M on J17. This voltage shall be present during the time that signal inputs are applied to the input lines.

(g) After computer words composed of Fieldata codes have been applied to the input lines, apply the Fieldata code for "Halt".

(h) The voltage at pins FF and GG on J17 shall be between plus 100.0 and plus 120.0 volts.

(i) Momentarily apply a voltage between minus 9 and minus 11 volts to pin f on J17.

(j) The voltage output at pin FF on J17 shall be between 0.0 and minus 6.0 volts.

(k) The voltage output at pin GG on J17 shall be between plus 100.0 and plus 120.0O volts.

(l) Apply the Fieldata code signal to "Verify".

(m) The voltage at pin FF on J17 shall be between plus 100.0 and plus 120.0 volts.

(n) The voltage at pin GG on J17 shall be between 0.0 and minus 6.0 volts.

(p) The "Error" lamp on the computer Control Panel shall be energized and not flashing during the time that the information signals are being re-entered.

4.5.7.1.3 *Error reset.* Perform the following procedures to test error reset for compliance with 3.7.5.1.3:

(a) Generate and execute a routine to give an overflow indication.

(b) The "Error" lamp shall be flashing.

(c) Momentarily apply a voltage between minus 5 and minus 7 volts, to pin b on J10.

(d) The "Error" lamp shall be energized and not flashing.

4.5.7.1.4 *"Receive" set-up.* Perform the following procedures to test the external



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"Receive" set-up for compliance with 3.7.5.1.4:

(a) Apply a voltage between minus 5 and minus 7 volts to pin c on J10.

(b) Monitor the appropriate memory location to insure that channel 0 sector 7 is the "Right" address in the "I" register.

4.5.7.1.5 *Output line disable.* Perform the following procedures to test the disabling of output lines for compliance with 3.7.5.1.5:

(a) Generate and execute a routine to output information on the output lines.

(b) Observing the output while information is being transmitted, apply a voltage between minus 1.0 and plus 1.0 volts to pin d on J10.

(c) The output lines shall be between minus 1.0 and plus 1.0 volts.

4.5.7.1.6 *"Verify" mode enable.* Perform the following procedures to test the "Verify" mode enable for compliance with 3.7.5.1.6:

(a) Apply a voltage between minus 1.0 and plus 1.0 volts to pin T on J17.

(b) Set the computer to the "Fill" mode.

(c) The voltage at pin FF on J17 shall be between plus 100.0 and plus 120.00 volts.

(d) The voltage at pin GG on J17 shall be between plus 0.0 and minus 6.0 volts.

4.5.8 *Internally initiated inputs under program control.*

4.5.8.1 *Read numeric.* Perform the following procedure to test the command operations listed in Table V for determining compliance with 3.7.5.2.1:

(a) Execute "Read Numeric" commands.

(b) Apply inputs for numeric codes.

(c) Apply a timing signal or signals having a nominal voltage swing between 0.0 and minus 4.0 volts and a frequency not greater than 600 cycles per second to the appropriate pins listed in Table XXII. The tolerance on the voltage is between minus 1.0 and plus 1.0 volts for the ground potential and between minus 2.0 and minus 6.0 for the negative potential.

(d) Monitor the appropriate memory locations to verify information acceptance and command operation.

4.5.8.2 *Read alpha-numeric.* Perform the following procedures to test the "Read Alpha-Numeric" commands listed in Table VI for determining compliance with 3.7.5.2.2:

(a) Execute "Read Alpha-numeric" command.

(b) Apply a timing signal or signals having a nominal voltage swing between 0.0 and minus 4.0 volts and a frequency not greater than 4250 cycles per second to the appropriate pins listed in Table XXII. The tolerance on the voltage is between minus 1.0 and plus 1.0 for the ground potential and between minus 2.0 and minus 6.0 volts for the negative potential.

(d) Monitor the appropriate memory locations to verify information acceptance and command operation.

4.5.8.3 *Discrete input to the accumulator.* Perform the following procedures to test the "Discrete Input to the Accumulator" com-



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mand for determining compliance with 3.7.5.2.3:

(a) Depress the switch buttons listed in Column A of Table XXIII in the order indicated.

(b) Execute "Discrete Input to Accumulator" command.

(c) Monitor the A register to determine that the bit patterns listed in Column B of Table XXIII, appear in the corresponding bit positions of the A register.

(d) Depress in sequence one (1) letter in left-hand vertical row and one (1) number in lower horizontal row of Matrix to illuminate each input display.

4.5.8.4 *Discrete external inputs to the A register.* Perform the following procedures to test the discrete external input to the A register for determining compliance with 3.7.5.2.4:

(a) Apply a voltage between minus 2 and minus 6 volts to the pins listed in Column A of Table XXIV.

(b) Execute the 'Discrete Input Command".

(c) Monitor the A register to determine that for each voltage applied the bit patterns listed in Column B of Table XXIV shall appear in the corresponding bit positions in the A register.

4.5.8.5 *Magnetic tape code conversion on input.* Perform the following procedures to test the Magnetic-Tape-Code conversion on input for determining compliance with 3.7.5.2.5:

(a) Apply Magnetic Tape code signal inputs, listed in Column A of Table XXV, to the input lines.

(b) Apply the timing signal inputs relative to the information signal inputs.

(c) Monitor the appropriate memory locations to determine the Magnetic-Tape-Code conversion for codes entered is in accordance with Column B of Table XXV.

4.5.9 *Internally initiated outputs.*

4.5.9.1 *Write-numeric.* Perform the following procedures to test the Write-Numeric commands listed in Table XXVI for determining compliance with 3.7.5.3.1:

(a) Execute "Write-Numeric" commands.

(b) Observe the signals on the output lines listed in Table XXVII to insure information transmittal in the proper code associated with the command executed.

(c) When applicable, apply a timing signal or signals having a nominal voltage swing between 0.0 and minus 4.0 volts and a frequency not greater than 600 cycles per second to the appropriate pins listed in Table XXVIII. The tolerance on the voltage is between minus 1.0 and plus 1.0 for the ground potential and between minus 2.0 and minus 6.0 volts for the negative potential.

4.5.9.2 *Write alpha-numeric.* Perform the following procedures to test the Write Alpha-Numeric commands listed in Table XXIX for determining compliance with 3.7.5.3.2:

(a) Execute the "Write Alpha-Numeric" command.

(b) Observe the signals on the output lines listed in Table XXVII to insure output transmittal in the proper code associated with the command executed.



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(c) When applicable, apply a timing signal or signals having a nominal voltage swing between 0.0 and 4.0 volts and a frequency not greater than 4250 cycles per second to the appropriate pins listed in Table XXVIII. The tolerance on the voltage is between minus 1.0 and plus 1.0 volts for the ground potential and between minus 2.0 and minus 6.0 for the negative potential.

4.5.9.3 *Discrete output.* Perform the following procedures to test the Discrete Output commands for determining compliance with 3.7.5.3.3:

(a) Execute the Discrete Output commands listed in Column A of Table XXX.

(b) Observe the signals on the output lines at the pins listed in Column B of Table XXX.

(c) The state of the term shall be as listed in Column C of Table XXX.

(d) Execute the Discrete Output command for "Discrete Output 7".

(e) The "No-Solution" lamp on the Control Panel shall be flashing.

(f) Execute the Discrete Output command for "Discrete Output O".

(g) The 'No-Solution" lamp shall be energized and not flashing.

(h) The state of the term listed in Column B of Table XXX shall be zero (0).

4.5.9.4 *Teletype output timing.* Perform the following procedures to test the teletype timing applicable to the 2-wire 5-level Teletype code format (7.42 system) for determining compliance with 3.7.5.3.4:

(a) Using the Teletype oscillator, generate and execute a routine to control a discrete output line.

(b) Observe the signal on the discrete output line to observe a waveform equivalent to that shown in Figure 1.

4.5.9.5 *Fieldata signals.* Perform the following procedures to test the Fieldata signals at the output pins for determining compliance with 3.7.5.3.5:

(a) Generate and execute a routine to output Fieldata code.

(b) The signal at pin W on J10 shall be equivalent to the signal shown in Figure 2 (c) when a signal equivalent to Figure 2 (a) is applied to pin Y on J17. The information signal for a binary "1" as shown in Figure 2 (b) shows the correspondence between the ready signal and the strobe signal.

4.5.9.6 *Magnetic tape code conversion on output.* Perform the following procedures to test the Magnetic Tape code conversion on output for determining compliance with 3.7.5.3.6:

(a) Apply a voltage between minus 2 and minus 6 volts to pin C on J10.

(b) Generate and execute a routine to output "Alpha-numeric 6" code signals.

(c) Observe the signal on the appropriate output lines to insure that the conversion from "Alpha-numeric 6" code to Magnetic Tape code is in accordance with the codes listed in Table XXV.

4.5.9.7 *Tape-tub signals.* Perform the following procedures to test the Tape-tub address signal and information signal for determining compliance with 3.7.5.3.7:

(a) Generate and execute a routine to output information in "Alpha-numeric 5" code.



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(b) Apply an applicable timing signal.

(c) The voltage at pin T on J10 shall be between minus 2 and minus 6 volts.

(d) Halt the "Alpha-numeric 5" output.

(e) Generate and execute a routine to output information in "Alpha-numeric 6" code.

(f) Apply timing signal.

(g) The voltage at pin U on J10 shall be between minus 2 and minus 6 volts.

**4.5.9.8** *Matrix.* For compliance with 3.7.6, each switch button on the matrix shall be free of any mechanical interference and shall operate without sluggishness.

**4.5.9.9** *Display (nixie) indicators.* Generate and execute a routine to provide for the display of each of the 0 to 9 digits and the decimal point of the keyboard on each of the 16 nixie indicators. Each digit shall then be entered at a minimum rate of 4 digits per second. The nixie indicator shall display each entered digit before the next depression of the key as specified in 3.7.7.

**4.5.10** *Problems.* For compliance with 3.7.8, perform the following procedures to test the computer operation with the Field Artillery Program. All solutions and target recalls shall be in accordance with data supplied in Tables XXXI through XXXIV inclusive.

(a) Set the computer's hot channel selector switch to 12. Using the SDR AN/GSQ–64 or equivalent and tape 8213315–17, fill the pro gram into the computer. Verify the program then turn off the computer and disconnect the SDR.

(b) Turn on the computer and depress the "Program Test" button. Depress "1" key on the keyboard. The reading of the nixie display shall be 9 00000 00000 05055.

(c) Depress the "Program Test" button. Depress the "2" key on the keyboard. The "Quadrant" nixies shall count from 070 and come to a stop at 136.

(d) Without turning off the computer, repeat "b" and "c" in all marginal switch positions.

**4.5.10.1** The following switches and buttons shall be set as specified herein:

| | |
|---|---|
| Marginal Test switch | to 5 |
| Battery button | to A |
| Matrix | to F–5 |
| Weapon select | to 1 |

(a) Depress set-up button. Set marginal test switch to 4. Set weapon select to 2 and depress battery B button.

(b) Depress set-up button. Set marginal test switch to 3. Set weapon select to 1 and depress battery C button.

(c) Depress set-up button. Set marginal test switch to 2. Set weapon select to 2 and depress battery D button.

(d) Depress set-up button. Set the marginal test switch to 1. Set weapon select to 1 and depress battery E button.

(e) Depress set-up button. Using targets 1 through 5 of Table XXXI enter problems of Table XXXII and obtain appropriate solutions of Table XXXIII.

| Enter: | | |
|---|---|---|
| Battery A | in | Marginal 5 |
| Battery B | in | Marginal 4 |
| Battery C | in | Marginal 3 |
| Battery D | in | Marginal 2 |
| Battery E | in | Marginal 1 |



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**4.5.10.2** *Select Matrix position G–5.* Insert Meteorological Data tape (MET tape) into the mechanical tape reader. Set marginal test switch to the Off position. Depress the sample matrix (SM) button. Depress "1" key on the keyboard. Depress "Enter" key. The mechanical reader shall energize and the tape message shall be read into the computer.

(a) Depress "compute" button to obtain the appropriate solution of Table XXXIV in each battery. Battery to be selected by the battery button. High angle and low angle solutions shall be obtained for each battery in each marginal position.

**4.5.10.3** *Target store.* Depress the A battery button. Select Matrix position E4. Depress SM button. Depress "1" and "Enter" keys. Nixie display should indicate target information ofr battery A problem.

(a) Depress the B battery button. Depress the SM button. Depress "2" and "Enter" keys. Nixie display should indicate the target information for battery B problem.

(b) Depress the C battery button. Depress the SM button. Depress "3" and "Enter" keys. Nixie display should indicate target information for battery C problem.

(c) Depress the battery D button. Depress the SM button and "4" and "Enter" keys. Nixie display should indicate target information for battery D problem.

(d) Depress the battery E button. Depress the SM button and "5" and "Enter" keys. Nixie display should indicate target information for battery E problem.

**4.5.10.3.1** *Target recall.* Depress the A battery button. Select matrix position A–1. Depress the SM button and "1" and "Enter" keys. Target information stored in 4.5.10.3 shall be displayed on the nixies.

(a) Depress the battery B button. Depress the SM button and "2" and "Enter" keys. Target information stored in 4.5.10.3 (a) should be displayed on the nixies.

(b) Depress the battery C button. Depress the SM button and "3" and "Enter" keys. Target information stored in 4.5.10.3 (b) should be displayed on the nixies.

(c) Depress the battery D button. Depress SM button and "4" and "Enter" keys. Target information stored in 4.5.10.3 (c) should be displayed on the nixies.

(d) Depress the battery E button. Depress SM button and "5" and "Enter" keys. Target information stored in 4.5.10.3 (d) should be displayed on the nixies.

(e) Repeat 4.5.10.3 through 4.5.10.3.1 (d) in all marginal positions.

**4.5.10.4** *Failures.* The blinking of the parity or error light at any time other than when the marginal test switch is being switched or imediately after it has been switched, is considered a failure. A parity error in the sum check in 4.5.10 (c) shall not be considered as a failure, if it can be cleared by the program tape hot storage routine and subsequent problem solutions do not cause recurrence of parity error.

(a) *Incorrect displays*—Any incorrect display shall be considered a failure.

(b) If a failure occurs, repeating the failed operation successfully is not acceptable unless either a positive corrective action for the failure has been accomplished or the failed operation is repeated 50 consecutive times successfully without failure.



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**4.5.11** *Continuous performance.* As specified in 3.7.9, the solution of each battery problem shall be repeated 240 times–40 times in each marginal position. This test shall be performed utilizing the Cable; Table, Gun Direction and the Computer with the front and back covers removed.

**4.5.12** *Run-in.* As specified in 3.8 and after each computer has successfully passed all the performance tests, each computer shall be run for a total of 70 hours excluding down-time. The tests shall be run for at least 4 hours at —22°F, 4 hours at +125°F and at least 50 hours after the —22°F, +125°F and vibration tests are completed. The tests at the high and low temperatures shall be with the table and cable.

**4.5.12.1** *Vibration.* To determine compliance with 3.8.1, the computer in its full transit case shall be vibrated at 271 ± 2 cycles per minute with a circular motion of 1.0 ± 0.1 inch diameter and constrained to a maximum of 3.25 inches total excursion horizontally for a 1 hour total. ½ hour in its normal operating position and ½ hour resting on its opposite side. After vibration, the computer shall be operated for at least 4 hours, excluding down-time. During this operation, the following tests shall be successfully performed with the computer's front and back covers removed.

(a) The memory shall be cleared by writing a 1 in a field of zeros (00000100000) into all memory locations.

(b) Repeat 4.5.10 through 4.5.10.3.1 (e).

**4.5.12.2** *High temperature run-in.* Each computer, after successfully passing the vibration test, shall be placed in a chamber and the temperature shall be raised to +125°F at a rate not exceeding 2° per minute. The computer shall be exposed at the +125°F for at least 4 hours. After the exposure, the computer shall be operated at the +125°F for at least 4 hours excluding down time. The following tests shall be performed at least once:

(a) Automatic logic test specified in 4.5.5 through 4.5.5.6.

(b) Control panel tests specified in 4.5.6 through 4.5.6.3.

(c) Commands specified in 4.5.6.4 and 4.5.6.5.

(d) Input-Output specified in 4.5.7, 4.5.8, and 4.5.9.

(e) Matrix and Nixie indicators specified in 4.5.9.8 and 4.5.9.9.

(f) All connector caps shall be assembled to connector and then removed without the use of special tools.

**4.5.12.2.1** The following tests shall be performed at least 3 times:

(a) The memory shall be cleared by writing a 1 in a field of zeros (00000100000) into all memory locations.

(b) Repeat 4.5.10 through 4.5.10.3.1 (e).

(c) Two hours of computer operation shall be in its normal mode specified in 4.5.2 and for two hours of operation the computer shall be on its side (rotated 90°).

**4.5.12.3** *Low temperature run-in.* Repeat 4.5.12.2, except the computer shall be exposed to the —22°F temperature.

**4.5.12.4** The remainder of the 70 hours, the computer shall be operating at temperatures between 55° to 95°F. The tests specified in 4.5.12.2 shall be performed, except the test specified in 4.5.10 through 4.5.10.3.1 (e) shall be run at least 50 times and the tests specified in 4.5.12.2 (a) to (f) and 4.5.12.2.1 shall be performed during the last 4 hours of operation.

**4.5.12.5** *Run-in failure.* Any failure may be corrected and tests continued without redoing previous acceptable tests provided:



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(a) Any part, module, component or subassembly that is used as a replacement shall have previously passed the environmental tests.

(b) Any replacement part shall have been exposed for at least 1 hour at the temperature being run.

(c) After a correction has been accomplished, at least one complete test sequence shall be performed at the temperature at which failure occurred.

**4.5.12.6** The following shall also be considered as failures during run-in:

(a) Failures as defined in 4.5.10.4 through 4.5.10.4 (b).

(b) Computer losing "SYNC", except if loss of "SYNC" is caused by the switching of the marginal test switch.

## 5. PREPARATION FOR DELIVERY

**5.1 Packaging, packing and marking.** Packaging, packing and marking shall be in accordance with MIL–P–14232 and Packaging Data Sheet MIL–P–14232/P10525735. The level of protection shall be as specified in the procurement document.

## 6. NOTES

**6.1 Intended use.** The Computer Set, Gun Direction performs computations required in the solution of ballistic trajectories, meteorological data reduction, and geodetic surveys.

**6.2 Ordering data.** Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Selection of applicable level of preservation, packaging and packing.

(c) Applicable packaging data sheet number (see 5.1).

(d) Applicable stock number.

(e) A block of serial numbers to identify each computer set.

(f) Provisions for initial production samples including dates of delivery (see 4.2.1.1).

**Custodian:**
**Army—MU**

**Preparing activity:**
**Army—MU**

**Project No. 1220–A211Y**



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TABLE I

| | A | B** | C |
|---|---|---|---|
| Supply | Voltage average DC to ground | Ripple P-P MV maximum to ground | Terminal |
| +6R | +6 ± .12 | 50 | 139E–17 |
| +3 R | +3 ± .12 | 50 | 139E–24 |
| —3R | —3 ± .12 | 50 | 139E–20 |
| —10R | —10 ± .2 | 50 | 139E–10 |
| +150R | +150R ± 2.32 | 500 | 103E–1 |
| +15 R | +15 ± .45 | 50 | 103E–18 |
| —1.5R | —1.5 ± .045 | 20 | 105E–19 |
| +1.25R | +1.25 ± .038 | 100 | 135E–11 |
| —18R | —18 ± .54 | 500 | 107E–1 |
| +35R | +35 ± .7 | 50 | 105E–13 |
| —6R | —6 ± .18 | 300 | 107E–20 |
| +6Z | +6 ± .6 | 200 | 105E–11 |
| —12Z | —12 ± 1.2 | 300 | 105E–4 |
| —27U | —25.0 ± 1.08* | 6000 | 105E–1 |
| —32U | —30.5 + 1.28* | 6000 | 105E–6 |
| +66Z | +58 to + 72 | No measurement | 139E–7 |
| +73Z | +68 to + 80 | No measurement | 139E–6 |
| +138Z | +124 to + 152 | No measurement | 139E–3 |

R = Regulated
Z = Zener controlled
U = Unregulated

* Valid only at line = 120 volts RMS. At maximum line voltage multiply sum of nominal and tolerance by 1.15 for limits and at minimum line voltage by 0.85 for limits.

** Ripple is defined as the peak to peak value of any alternating component of the output voltage. The ripple voltage measurement shall exclude noise of less than 10 us duration. The noise level measurement of less than 10 us duration shall be no greater than ± 0.75V off the dc level.

TABLE II

| A | B | | | | | | |
|---|---|---|---|---|---|---|---|
| Switch Position | DC Voltage ±2% at Poower Supply | | | AC Source = 120 ± 1 Volt RMS, 400 ± 2CPS | | | |
| Off | —18 | —10 | —6 | +1.25 | +6 | +15 | +35 |
| 1 | —17.1 | —9.5 | —5.7 | +1.188 | +6.3 | +15.75 | +33.25 |
| 2 | —18.9 | —10.5 | —6.3 | +1.312 | +5.7 | +14.2 | +36.75 |
| 3 | —17.1 | —9.5 | —6.3 | +1.312 | +5.7 | +15.75 | +33.25 |
| 4 | —17.1 | —9.5 | —5.7 | +1.188 | +5.7 | +14.25 | +36.75 |
| 5 | —18.9 | —10.5 | —6.3 | +1.188 | +5.7 | +15.75 | +33.25 |



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TABLE III

| A | B | C | | | |
|---|---|---|---|---|---|
| | | Quasi-Octal | | | |
| Commands | Mnemonic | Spare | OP | Right Channel | Right Sector |
| Clear and Add | CLA | 0<br>2 | 24 | | |
| Clear and Subtract | CLS | 0<br>2 | 26 | | |
| Add | ADD | 0<br>2 | 00 | | |
| Subtract | SUB | 0<br>2 | 02 | | |
| Multiply | MPY | 0 | 20 | | |
| Divide | DIV | 0<br>2 | 30 | | |
| Store Accumulator | STO | 0<br>2 | 50 | | |
| Store N — Register | STN | 0 | 40 | | |
| Store D — Register | STD | 0<br>2 | 42 | | |
| Store L — Register | STL | 0<br>2 | 52 | | |
| Store Address | STA | 0 | 70 | | |
| Store Program Add | STP | 0 | 60 | | |
| Load R — Loop | LDR | 0<br>2 | 72 | | |
| Load Q — Loop | LDQ | 0 | 72 | | |
| Store R — Loop | STR | 0 | 62 | | |
| Transfer on Zero | TZE | 0<br>2 | 12 | | |
| Transfer on Plus | TPL | 0<br>2 | 10 | | |
| Transfer on Overflow | TOV | 0 | 16 | | |
| Transfer | TRA | 0<br>2 | 14 | | |
| "And" to Accumulator | ANA | 0 | 34 | | |
| Equal Search | EQS | 0<br>2 | 64 | | |
| Greater than or Equal Search | QES | 0<br>2 | 66 | | |
| Halt Program | HLT | | 36 | | 120 |



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TABLE IV

| A | B | C | | | |
|---|---|---|---|---|---|
| | | Quasi-Octal | | | |
| Commands | Mnemonic | Spare | OP | Right Channel | Right Sector |
| "A" Right Shift | ARS | 0 | 76 | 002 | |
| "A" Left Shift | ALS | 0 | 76 | 006 | |
| | | 2 | | | |
| Long Right Shift | LRS | 0 | 76 | 022 | |
| Long Left Shift | LLS | 0 | 76 | 026 | |
| | | 2 | | | |
| "A" Right Cycle | ARC | 0 | 76 | 000 | |
| "A" Left Cycle | ALC | 0 | 76 | 004 | |
| Long Right Cycle | LRC | 0 | 76 | 020 | |
| Long Left Cycle | LLC | 0 | 76 | 024 | |
| Replace A by Minus A | RMA | 0 | 36 | 170 | |
| | | 2 | | | |
| Replace A by Minus L | RML | 0 | 36 | 172 | |
| | | 2 | | | |
| Replace A by Minus N | RMN | 0 | 36 | 174 | |
| | | 2 | | | |
| Zero L | ZEL | 0 | 36 | 162 | |
| Halt Display Mode | HDM | 0 | 36 | 166 | |
| Initiate Display Mode | IDM | 0 | 36 | 164 | |

TABLE V

| A | B | C | | | |
|---|---|---|---|---|---|
| | | Quasi-Octal | | | |
| Commands | Mnemonic | Spare * | OP | Right Channel | Right Sector |
| *Numerical — Octal* | | | | | |
| Keyboard | RDN | 0 | 54 | 000 | |
| Mechanical Reader | RDN | 0 | 54 | 020 | |
| Other External | RDN | 0 | 54 | 030 | |
| | | | | Binary | |
| *Numeric — BCD* | | | | | |
| Keyboard | RDN | 0 | 54 | 100 | Amount |
| | | 2 | | | |
| Mechanical Reader | RDN | 0 | 54 | 101 | Amount |
| | | 2 | | | |
| Other External | RDN | 0 | 54 | 111 | Amount |
| | | 2 | | | |

* Note spare bit.



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TABLE VI

| A | B | C | | | |
|---|---|---|---|---|---|
| | | Quasi-Octal | | | |
| Commands | Mnemonic | Spare * | OP | Right Channel | Right Sector |
| | | | | Binary | |
| *Alpha-Numeric 5* | | | | | |
| Keyboard | RDA | 0 2 | 56 | 111 | Amount |
| Mechanical Reader | RDA | 0 2 | 56 | 101 | Amount |
| Other External | RDA | 0 2 | 56 | 100 | Amount |
| *Alpha-Numeric 6* | | | | | |
| Keyboard | RDA | 0 2 | 56 | 011 | Amount |
| Mechanical Reader | RDA | 0 2 | 56 | 001 | Amount |
| Other External | RDA | 0 2 | 56 | 000 | Amount |
| Magnetic Tape | RDA | 0 2 | 56 | 010 | Amount |

TABLE VII

| A | B | C | D |
|---|---|---|---|
| Register | Term (Prime) | Location pin | Connector |
| Accumulator | Ax- | b | J 16 |
| Lower Accumulator (Also Location) | Lx- | g | J 16 |
| Number | Nx- | h | J 16 |
| Instruction | Ix- | j | J 16 |
| Buffer Instruction | Xo- | k | J 16 |
| Display | Dx- | l | J 16 |
| 16 word Rapid Access | Rx- | m | J 16 |
| 16 word Rapid Access | Qx- | n | J 16 |
| Main Memory | Mmn- | p | J 16 |



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TABLE VIII

| Item | A<br>Supply | B<br>Max. load—ohms ± 5% to ground* | C<br>Min. load—ohms ± 5% to ground* | D<br>Kickout load—ohms ± 5% to ground | E<br>Connector J13 pin |
|---|---|---|---|---|---|
| 1 | +6R | 13.3, 5 Watt | 24, 2W | 2, 25W | W |
| 2 | +3R | 3.1, 5W | 10.3, 2W | ..... | — |
| 3 | —3R | 4.8, 5W | 20, 1W | ..... | — |
| 4 | —10R | 7.7, 25W | 27, 5W | 5, 25W | Y |
| 5 | +150R | 1019, 5W | 2980, 2W | 330, 100W | a |
| 6 | +15R | 12.1, 25W | 15.3, 25W | 7, 50W | V |
| 7 | —1.5R | 13.75, 25W* | 75, 5W* | ..... | — |
| 8 | +1.25R | 0.67,5W | 1, 5W | ..... | — |
| 9 | —18R | 4.1, 150W | 6.0, 75W | 1.8, 150W | Z |
| 10 | +35R | 34.3, 50W | 89.7, 25W | 23, 50W | U |
| 11 | —6R | 0.89, 50W | 1.58, 50W | 0.5, 100W | X |
| 12 | +6Z | 75, 5W | 200, 5W | ..... | — |
| 13 | —12Z | 17, 5W | 46, 2W | ..... | — |
| 14 | —27V | 67.5, 15W | No Resistor | ..... | — |
| 15 | —32V | 22.8, 50W | No Resistor | ..... | — |
| 16 | +66Z | 15.5K, 1W | 22K, 1W | ..... | — |
| 17 | +73Z | 78K, 1W | 78K, 1W | ..... | — |
| 18 | +138Z | 47K, 1W | 47K, 1W | ..... | — |

R = Regulated
Z = Zener, controlled
U = Unregulated
* Return —1.5R loads to —18R, not to ground.

TABLE IX

| | A<br>Line voltage ± 1 volt RMS, AC | B<br>Condition |
|---|---|---|
| 1 | 141 ..... | Computer deenergizes |
| 2 | 99 longer than 5 sec ..... | Computer deenergizes |
| 3 | 82 for 3 ± 1 sec ..... | Transient indicator lamp energizes |
| 4 | 107 ..... | Transient indicator lamp flashes |
| 5 | 133 ..... | Transient indicator lamp flashes |



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TABLE X

| | A | B | C | D |
|---|---|---|---|---|
| | Resistor ohms ± 5% ½ W | From connector 204, pin | From connector 304, pin | Effect |
| 1 | 1500 | 14B | 15B | The temperature warning lamp flashes |
| 2 | 4700 | 20B | 23B | " |
| 3 | 470 | 2A | 2B | " |
| 4 | 1200 | 49B | 54B | The computer deenergizes |
| 5 | 2700 | 43B | 46B | " |
| 6 | 470 | 64B | 63B | " |
| 7 | 560 | 29B | 28B | P009 Pin 9 (was 120 volts AC to ground) deenergizes and temperature warning lamp flashes and power ready light shall go out. |

TABLE XI

| A | B | C | D |
|---|---|---|---|
| Memory channel | Connector | Read signal test point (TP) | Read signal phasing test point (TP) |
| Main Memory | | | |
| MMP (000 – 336) | J301 | TP14 | TP16 |
| MMN (000 – 366) | J302 | TP14 | TP16 |
| Sector Channel | | | |
| S | J303 | TP34 | TP43 |
| Recirculating Loops | | | |
| X | J304 | TP34 | TP43 |
| D | J305 | TP34 | TP43 |
| R | J306 | TP34 | TP43 |
| Rm | J307 | TP34 | TP43 |
| Q | J308 | TP34 | TP43 |
| Qm | J309 | TP34 | TP43 |
| N | J310 | TP34 | TP43 |
| L | J311 | TP34 | TP43 |
| I | J312 | TP34 | TP43 |
| A | J313 | TP34 | TP43 |



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TABLE XII

| Storage channel No. | Selection line state | | | | | | |
|---|---|---|---|---|---|---|---|
| | RFS | $R_7$ | $R_6$ | $R_5$ | $R_4$ | $R_3$ | $R_2$ |
| 000 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 002 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 004 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| 006 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 010 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 012 | 0 | 0 | 0 | 0 | 1 | 0 | 1 |
| 014 | 0 | 0 | 0 | 0 | 1 | 1 | 0 |
| 016 | 0 | 0 | 0 | 0 | 1 | 1 | 1 |
| 020 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 022 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |
| 024 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |
| 026 | 0 | 0 | 0 | 1 | 0 | 1 | 1 |
| 030 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |
| 032 | 0 | 0 | 0 | 1 | 1 | 0 | 1 |
| 034 | 0 | 0 | 0 | 1 | 1 | 1 | 0 |
| 036 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |
| 040 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 042 | 0 | 0 | 1 | 0 | 0 | 0 | 1 |
| 044 | 0 | 0 | 1 | 0 | 0 | 1 | 0 |
| 046 | 0 | 0 | 1 | 0 | 0 | 1 | 1 |
| 050 | 0 | 0 | 1 | 0 | 1 | 0 | 0 |
| 052 | 0 | 0 | 1 | 0 | 1 | 0 | 1 |
| 054 | 0 | 0 | 1 | 0 | 1 | 1 | 0 |
| 056 | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
| 060 | 0 | 0 | 1 | 1 | 0 | 0 | 0 |
| 062 | 0 | 0 | 1 | 1 | 0 | 0 | 1 |
| 064 | 0 | 0 | 1 | 1 | 0 | 1 | 0 |
| 066 | 0 | 0 | 1 | 1 | 0 | 1 | 1 |
| 070 | 0 | 0 | 1 | 1 | 1 | 0 | 0 |
| 072 | 0 | 0 | 1 | 1 | 1 | 0 | 1 |
| 074 | 0 | 0 | 1 | 1 | 1 | 1 | 0 |
| 076 | 0 | 0 | 1 | 1 | 1 | 1 | 1 |
| 100 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 102 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| 104 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| 106 | 0 | 1 | 0 | 0 | 0 | 1 | 1 |
| 110 | 0 | 1 | 0 | 0 | 1 | 0 | 0 |
| 112 | 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| 114 | 0 | 1 | 0 | 0 | 1 | 1 | 0 |
| 116 | 0 | 1 | 0 | 0 | 1 | 1 | 1 |
| 120 | 0 | 1 | 0 | 1 | 0 | 0 | 0 |
| 122 | 0 | 1 | 0 | 1 | 0 | 0 | 1 |
| 124 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| 126 | 0 | 1 | 0 | 1 | 0 | 1 | 1 |
| 130 | 0 | 1 | 0 | 1 | 1 | 0 | 0 |
| 132 | 0 | 1 | 0 | 1 | 1 | 0 | 1 |
| 134 | 0 | 1 | 0 | 1 | 1 | 1 | 0 |
| 136 | 0 | 1 | 0 | 1 | 1 | 1 | 1 |
| 300 | 1 | 1 | 0 | 0 | 0 | 0 | 0 |
| 302 | 1 | 1 | 0 | 0 | 0 | 0 | 1 |
| 304 | 1 | 1 | 0 | 0 | 0 | 1 | 0 |



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TABLE XII (continued)

| Storage channel No. | Selection line state | | | | | | |
|---|---|---|---|---|---|---|---|
| | RFS | $R_7$ | $R_6$ | | $R_4$ | $R_3$ | $R_2$ |
| 306 | 1 | 1 | 0 | 0 | 0 | 1 | 1 |
| 310 | 1 | 1 | 0 | 0 | 1 | 0 | 0 |
| 312 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |
| 314 | 1 | 1 | 0 | 0 | 1 | 1 | 0 |
| 316 | 1 | 1 | 0 | 0 | 1 | 1 | 1 |
| 320 | 1 | 1 | 0 | 1 | 0 | 0 | 0 |
| 322 | 1 | 1 | 0 | 1 | 0 | 0 | 1 |
| 324 | 1 | 1 | 0 | 1 | 0 | 1 | 0 |
| 326 | 1 | 1 | 0 | 1 | 0 | 1 | 1 |
| 330 | 1 | 1 | 0 | 1 | 1 | 0 | 0 |
| 332 | 1 | 1 | 0 | 1 | 1 | 0 | 1 |
| 334 | 1 | 1 | 0 | 1 | 1 | 1 | 0 |
| 336 | 1 | 1 | 0 | 1 | 1 | 1 | 1 |

TABLE XIII

| A | B |
|---|---|
| Selection line | * Pin |
| DU02 | A |
| DU03 | B |
| DU04 | C |
| DU05 | D |
| DU06 | E |
| DU07 | F |
| DU11 | G |
| DU12 | H |
| DU13 | J |
| DU14 | K |
| DU15 | L |
| DU16 | M |
| DL20 | N |
| DL21 | P |
| DL22 | R |
| DL23 | S |
| DL24 | T |
| DL25 | U |
| DL26 | V |
| DL27 | W |
| DL30 | X |
| DL31 | Y |
| DL32 | Z |
| DUR02 | a |
| DUR03 | b |
| DUR04 | c |
| DUR05 | d |
| DLR33 | e |
| DLR34 | f |
| DLR35 | g |
| DLR36 | h |

* All pins are located on Connector J12.



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TABLE XIV

| A | B | C | A | B | C |
|---|---|---|---|---|---|
| Flip-Flop | Selection lines | | Flip-Flop | Selection lines | |
| | DU | DL | | DU | DL |
| AC | 05 | 20 | DX | 05 | 23 |
| AK | 06 | 22 | DSH | 07 | 23 |
| AP | 15 | 27 | DSP | 14 | 23 |
| AX | 14 | 30 | D8 | 16 | 23 |
| B6 | 05 | 22 | D7 | 02 | 20 |
| B5 | 02 | 31 | D6 | 14 | 20 |
| B4 | 04 | 31 | D5 | 07 | 22 |
| B3 | 11 | 24 | D4 | 11 | 22 |
| B2 | 04 | 25 | D3 | 16 | 27 |
| B1 | 05 | 26 | D2 | 16 | 30 |
| CL7 | 16 | 22 | D1 | 11 | 30 |
| CL6 | 07 | 30 | DA8 | 15 | 25 |
| CL5 | 15 | 26 | DA7 | 12 | 22 |
| CL4 | 07 | 25 | DA6 | 22 | 321 |
| CL3 | 15 | 30 | DA5 | 11 | 21 |
| CL2 | 02 | 21 | DA4 | 12 | 21 |
| CP7 | 13 | 25 | DA3 | 03 | 24 |
| CP6 | 15 | 20 | DA2 | 05 | 25 |
| CP5 | 13 | 23 | 02 | 14 | 22 |
| CP4 | 13 | 27 | PE | 06 | 24 |
| CP3 | 02 | 30 | PN | 02 | 27 |
| CP2 | 16 | 20 | PP | 13 | 20 |
| CP76 | 03 | 30 | QC | 14 | 26 |
| CR7 | 07 | 26 | QX | 04 | 26 |
| DR6 | 03 | 22 | RC | 11 | 27 |
| CR5 | 03 | 21 | RX | 03 | 27 |
| CR4 | 03 | 23 | K | 06 | 23 |
| CR3 | 16 | 24 | LC | 15 | 22 |
| CR2 | 05 | 27 | LP | 04 | 20 |
| CN7 | 02 | 36 | LX | 02 | 23 |
| CN6 | 03 | 20 | MC | 04 | 23 |
| DA1 | 05 | 31 | MN | 04 | 24 |
| D | 06 | 21 | MP | 02 | 25 |
| E | 13 | 20 | NC | 07 | 27 |
| FS | 03 | 31 | ND | 04 | 27 |
| G | 02 | 24 | NX | 06 | 27 |
| IC | 12 | 24 | OB6 | 15 | 27 |
| ID | 14 | 21 | OB5 | 04 | 27 |
| IP | 06 | 30 | OB4 | 04 | 21 |
| CN5 | 07 | 20 | OB3 | 11 | 23 |
| CN4 | 11 | 20 | OB2 | 07 | 24 |
| CN3 | 12 | 21 | OFLO | 03 | 25 |
| CN2 | 06 | 26 | OP3 | 05 | 26 |
| C7 | 14 | 24 | OP2 | 12 | 22 |
| C6 | 16 | 26 | OP1 | 05 | 30 |
| C5 | 16 | 21 | O6 | 13 | 22 |
| C4 | 13 | 21 | 05 | 13 | 26 |
| C3 | 02 | 22 | O4 | 03 | 26 |
| C2 | 16 | 25 | 03 | 04 | 30 |
| CFS | 06 | 25 | L32* | 03 | 32 |
| CPFS | 12 | 25 | DP * | 02 | 32 |
| DC | 15 | 23 | | | |



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TABLE XIV (continued)

| A | B | C |
|---|---|---|
| Flip-Flop | Selection lines | |
| | DU | DL |
| NP * | 16 | 31 |
| XP * | 15 | 31 |
| RP * | 13 | 31 |
| I32 * | 12 | 31 |
| A32 * | 11 | 31 |
| QP * | 07 | 31 |
| MWO ** | 06 | 31 |
| MW1 ** | 14 | 31 |
| TSO | 11 | 26 |
| TP | 05 | 24 |
| TX | 13 | 24 |
| TO | 06 | 25 |
| XC | 12 | 30 |
| X2 | 14 | 25 |
| X1 | 12 | 23 |
| XD | 07 | 26 |
| XR | 15 | 24 |
| CFSW | 11 | 25 |

| A | B | C |
|---|---|---|
| Flip-Flop | Selection lines | |
| | DU | DL |
| CLC | 12 | 25 |
| | DUR | DIR |
| S | 02 | 33 |
| Xo | 02 | 33 |
| Do | 02 | 35 |
| Ro | 02 | 36 |
| Rm | 03 | 33 |
| Qo | 03 | 34 |
| Qm | 03 | 35 |
| No | 03 | 36 |
| Lo | 04 | 33 |
| Ix | 04 | 34 |
| Ao | 04 | 35 |
| MMn | 04 | 36 |
| MMp | 05 | 33 |
| Td/Wi | 05 | 34 |

* To obtain proper readings at FIT and FOT for para. 4.5.5.3 & 4.5.5.4. Appropriate write coil or equiv. resistance shall be across memory plug terminals of above write amps.

** Same as * except appropriate write switches must be turned on.

TABLE XV

| A | B | C | D | E | F |
|---|---|---|---|---|---|
| Term | Initialization | Setup | Flip-Flops | Connector | Pin |
| (BDX) | 1 (One-set all) | 0 (Zero-set following) | DSH, KD, Wi/Td | J14 | A |
| (BL1) | 0 (Zero-set all) | 1 (One-set following) | B1 | ↑ | B |
| (BL2) | 0 | 1 | B2 | | C |
| (BL3) | 1 | 0 | B3 | | C |
| (BL5) | 1 | 0 | B2 | | E |
| (BL6) | 1 | 0 | B1 | | F |
| (BL7) | 0 | 1 | B3, B2, B1 | | G |
| (B5L4) | 0 | 1 | B5, B3 | | H |
| (CNU6) | 1 | 0 | C5 | | J |
| (CPS4) | 1 | 0 | CP6, CP5 | | K |
| (CPS5) | 1 | 0 | CP6 | | L |
| (CRT5) | 1 | 0 | CR5 | | M |
| (CR65) | 1 | 0 | CR6, CR3 | | N |
| (CL34 | 0 | 1 | CL3, 04 | | P |
| (D21–) | 1 | 0 | D2, D1 | | R |
| (D43) | 0 | 1 | D4, D3 | | S |
| (D43–) | 0 | 1 | D4 | | T |
| (D51) | 1 | 0 | D5, D4, D3, D2, D1 | | U |
| (DOP) | 1 | 0 | D, B4 | | V |
| (DSPH) | 0 | 1 | DSP, DSH | | W |
| (EAKO) | 1 | 0 | E, AK, PE | | X |
| (ERR–) | 1 | 0 | OFLO, PE | | Y |
| (FST) | 1 | 0 | CL7, CL6, B1 | | Z |
| (1CP) | 1 | 0 | D | | a |
| (KAKI) | 1 | 0 | K | J14 | b |



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TABLE XV (continued)

| A | B | C | D | E | Pin |
|---|---|---|---|---|---|
| Term | Initialization | Setup | Flip-Flops | Connector | F |
| (KB5) | 0 | 1 | K, B5 | J14 | c |
| (KD3M) | 1 | 0 | NC | ↑ | d |
| (KED3) | 1 | 0 | K | ↑ | e |
| (KEO) | 1 | 0 | K, E | ↑ | f |
| (KHRS) | 0 | 1 | B4, D | ↑ | g |
| (K1OC) | 1 | 0 | K, D | ↑ | h |
| (K10P) | 1 | 0 | K, D | ↑ | i |
| K10X) | 1 | 0 | O5 | ↑ | j |
| (KLC-) | 1 | 0 | K | ↑ | k |
| (KSUO) | 0 | 1 | E | ↑ | m |
| (KSU1) | 0 | 1 | E, AK | ↑ | n |
| (KFST) | 1 | 0 | FS | ↑ | p |
| (KW) | 0 | 1 | AK, MC | ↑ | g |
| (MAXO) | 1 | 0 | MN, AX | ↑ | r |
| (MAX1) | 1 | 0 | MN | ↑ | s |
| (MAX2) | 1 | 0 | AX | ↑ | t |
| (MAX3) | 0 | 1 | MN, AX | ↑ | u |
| (MMPT) | 1 (One-set all) | 0 (Zero-set following) | MC | ↑ | v |
| (NDX-) | 0 (Zero-set all) | 1 (One-set following) | TX | ↑ | w |
| (OA) | 0 | 1 | K, E, AC | ↑ | x |
| (OBN) | 0 | 1 | OB5, OB3, OB4, OB2 | ↑ | y |
| (OBW) | 1 | 0 | OB5, OB3 | ↑ | z |
| (OCL) | 0 | 1 | AK, OB4, 04 | ↑ | AA |
| (ODO) | 1 | 0 | 06, 03 | ↑ | BB |
| (OEL) | 1 | 0 | 03 | ↑ | CC |
| (OF) | 1 | 0 | 06, 04, TP | ↑ | DD |
| (OGO) | 0 | 1 | AK, OB3, OB2 | ↑ | EE |
| (OJ) | 1 | 0 | 06, 05 | ↑ | FF |
| (OLP) | 1 | 0 | 06, 02, TP | ↑ | GG |
| (OLTP) | 0 | 1 | 05, OB3, LC, TP | ↑ | HH |
| (OLO) | 1 | 0 | 04, 03, 02 | ↑ | JJ |
| (OL4) | 0 | 1 | 04 | ↑ | KK |
| (OMDL) | 0 | 1 | E, K, 05 | ↑ | LL |
| (OMDM) | 1 | 0 | 06, 03, NC | ↑ | MM |
| (OMF) | 1 | 0 | 06, 04 | ↑ | NN |
| (OMNF) | 0 | 1 | K, E, 05 | J14 | PP |
| (OOP) | 0 | 1 | PP | J15 | A |
| (OSA) | 0 | 1 | AK, OB3, 04 | ↑ | B |
| (OSHB) | 0 | 1 | OB2, OB4, AK | ↑ | C |
| (OTR) | 1 | 0 | E, K, D, OB4 | ↑ | D |
| (OTR4) | 1 | 0 | K, E, D, OB4, X1 | ↑ | E |
| (OW) | 1 | 0 | 05, 03 | ↑ | F |
| (OWT) | 0 | 1 | AK, OB4, OB3 | ↑ | G |
| (OYAC) | 0 | 1 | E, K, 06, 05, 04, AC, 03, 02 | ↑ | H |
| (OYB) | 0 | 1 | K, OB6, OB5, OB3, OB4, OB2 | ↑ | J |
| (OYC5) | 0 | 1 | C5, E, K, 06, 05, 04, 03, 02 | ↑ | K |
| (OYR) | 0 | 1 | K, E, 06, 04, 05, 03, 02, CR5, CR3 | ↑ | L |
| (OTL) | 1 | 0 | 04 | ↑ | N |
| (QT1) | 0 | 1 | B3 | ↑ | P |
| (TXP-) | 1 | 0 | TP, TX | J15 | R |



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TABLE XV (continued)

| A | B | C | D | E | F |
|---|---|---|---|---|---|
| Term | Initialization | Setup | Flip-Flops | Connector | Pin |
| (TO1C) | 1 | 0 | B6, B4, TP, TX, C3 | J15 | T |
| (T15) | 0 | 1 | B6 | | U |
| (T19) | 0 | 1 | B6, B3 | | V |
| (VER) | 0 | 1 | AK, O4 | | W |
| (XCT) | 1 | 0 | S, B2 | | X |
| /C430/ | 1 | 0 | C4, C3, Td/Wi | | Y |
| /C431/ | 0 | 1 | C3 | | Z |
| /C432/ | 0 | 1 | C4 | | a |
| /C433/ | 1 (One-set all) | 0 (Zero-set following) | Td/Wi | | b |
| /C5A/ | 0 (Zero-set all) | 1 (One-set following) | C5 | | c |
| /C760/ | 1 | 0 | C7, C6 | | d |
| /C761/ | 1 | 0 | C7 | | e |
| /CFSU/ | 0 | 1 | C7 | | f |
| /CFSS/ | 1 | 0 | C6 | | g |
| /DA-/ | 1 | 0 | D | | h |
| /1Pa/ | 0 | 1 | 1P | | i |
| /1PA-/ | 1 | 0 | 1P | | j |
| /1XA/ | 0 | 1 | 1X | | k |
| /KA-/ | 1 | 0 | K | | m |
| /KED2/ | 0 | 1 | E | | n |
| /KD1/ | 0 | 1 | D | | p |
| /K1OA/ | 0 | 1 | AK, OB4 | | q |
| /KOPC/ | 0 | 1 | AK | | r |
| /KTP/ | 0 | 1 | K, TP | | s |
| /MNA/ | 0 | 1 | MN | | t |
| /MNA-/ | 1 | 0 | MN | | u |
| /OCT/ | 1 | 0 | K, E, D, OB3, B2 | | v |
| /OM/ | 1 | 0 | 06, 04, 03 | | w |
| /OMD/ | 1 | 0 | 06, 03 | | x |
| /O2A/ | 0 | 1 | 02 | | y |
| /O4A-/ | 1 | 0 | 04 | | z |
| /OX/ | 1 | 0 | 06, TX, TP | | AA |
| /SA/ | 0 | 1 | S | | BB |
| /TOA/ | 0 | 1 | TO | | CC |
| TOA-/ | 1 | 0 | TO | | DD |
| /TOB/ | 0 | 1 | TO | | EE |
| /TOX-/ | 1 | 0 | TX, TO | | FF |
| /TPA/ | 0 | 1 | TP | | GG |
| /TSH/ | 0 | 1 | AK, OB2 | | HH |
| /TXA-/ | 1 | 0 | TX | | JJ |
| /TXC/ | 0 | 1 | TX | | KK |
| /TRC/ | 0 | 1 | B5 | | LL |
| /TRS/ | 0 | 1 | B4 | | MM |
| /TLS/ | 1 | 0 | B5 | | NN |
| /T32/ | 0 | 1 | B6, TSO | J15 | PP |



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TABLE XV—Continued

| A | B | C | D | E | F |
|---|---|---|---|---|---|
| Term | Initialization | Setup | Flip-Flops | Connector | Pin |
| CTB1 | 1 (one set all) | 0 (zero set following) | C7, C6, C5 | J013 | q |
| CTB2 | 0 (zero set all) | 1 (one set following) | C5 | | r |
| CTB3 | 0 | 1 | C6 | | s |
| CTB4 | 1 | 0 | C7 | | t |
| CTB5 | 0 | 1 | C7 | | m |
| CTB6 | 0 | 1 | C7, C5 | | v |
| CTB7 | 1 | 0 | C6, C5 | | w |
| CTB8 | 1 | 0 | C6 | J013 | x |
| SS1 * | 1 | 0 | C7, C6, C5, C4, C3, C2, td/Wi | J013 | y |
| SS2 * | 1 | 0 | C7, C6, C5, C4 C3, C2, td/Wi | | z |
| SS3 * | 0 | 1 | C2 | | AA |
| SS4 * | 0 | 1 | C2 | | BB |
| SS5 * | 0 | 1 | C3 | | CC |
| SS6 * | 0 | 1 | C3 | | DD |
| SS7 * | 0 | 1 | C3, C2 | | EE |
| SS8 * | 0 | 1 | C3, C2 | | FF |
| SS9 * | 0 | 1 | C4 | | GG |
| SS10 * | 0 | 1 | C4 | | HH |
| SS11 * | 0 | 1 | C4, C2 | | JJ |
| SS12 * | 0 | 1 | C4, C2 | | KK |
| SS13 * | 0 | 1 | C4, C3 | | LL |
| SS14 * | 0 | 1 | C4, C3 | | MM |
| SS15 * | 0 | 1 | C4, C3, C2 | | NN |
| SS16 * | 0 | 1 | C4, C3, C2 | J013 | PP |

* To obtain true output appropriate write coils or equiv. resistance shall be across memory plug terminals assoc. with above write switches.



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TABLE XVI

| A | B | C |
|---|---|---|
| Voltage term | * Pin | Normal DC volts to ground |
| —18 V | M | —18.0 |
| +35 | P | +35.0 |
| +6 V Z | b | +6.0 |
| +20 V | *c* | +20.0 |
| —27 V | d | —25.0 |
| —32 V | e | —30.5 |
| —10 V | f | —10.0 |
| —12 V | *g* | —12.0 |
| —3 V | h | —3.0 |
| —1.5 V | i | —1.50 |
| +1.25 V | j | +1.25 |
| +3 V | *k* | +3.0 |
| +6 V | m | +6.0 |
| +15 V | n | +15.0 |
| +150 | *p* | +150.0 |
| —6 V | B | —6.0 |

* All pins are located on Connector J13.

TABLE XVII

| A | B |
|---|---|
| Switches | Computer word quasi-octal format |
| Sm | XXXXX140000 * |
| Test | XXXXX140001 |
| Set-Up | XXXXX140002 |
| Recall | XXXXX140003 |
| Send | XXXXX140004 |
| Compute | XXXXX140005 |
| Trig | XXXXX140006 |
| Receive | XXXXX140007 |

* X denotes "Non-applicable".

TABLE XVIII

| A | B | | |
|---|---|---|---|
| "Key" | Bit pattern in 3 least significant bit positions | | |
| | $t_3$ | $t_2$ | $t_1$ |
| 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 |
| 2 | 0 | 1 | 0 |
| 3 | 0 | 1 | 1 |
| 4 | 1 | 0 | 0 |
| 5 | 1 | 0 | 1 |
| 6 | 1 | 1 | 0 |
| 7 | 1 | 1 | 1 |

TABLE XIX

| A | | B | | |
|---|---|---|---|---|
| Teletype tape code | | Bit patterns in 3 least significant bit positions | | |
| Octal | Code format | $t_3$ | $t_2$ | $t_1$ |
| 0 | 01.101 | 0 | 0 | 0 |
| 1 | 11.101 | 0 | 0 | 1 |
| 2 | 11.001 | 0 | 1 | 0 |
| 3 | 10.000 | 0 | 1 | 1 |
| 4 | 01.010 | 1 | 0 | 0 |
| 5 | 00.001 | 1 | 0 | 1 |
| 6 | 10.101 | 1 | 1 | 0 |
| 7 | 11.100 | 1 | 1 | 1 |
| Halt | 01.100 | ..... * | | |
| Compute | 00.011 | ..... * | | |
| Fill | 10.001 | ..... * | | |
| Verify | 11.000 | ..... * | | |
| Location | 00.111 | ..... * | | |
| Clear | 10.111 | ..... * | | |
| Enter | 00.010 | ..... * | | |
| Blank | 00.000 | ..... * | | |

* Not-applicable.

TABLE XX

| A | B | | C |
|---|---|---|---|
| Term | Location | | Voltage average D.C. to ground |
| | Pin | Connector | |
| GPHC' | e | J17 | 6.0 ± 1.0 V |
| GPRC' | d | J17 | —12.0 ± 1.0 V |
| CTTI | NN | J17 | —4.0 + 2.0 V |
| FBPR | f | J17 | —12.0 ± 1.0 V (momentary) |



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TABLE XXI

| A | B | | C |
|---|---|---|---|
| Term | Location | | Voltage average D.C. to ground |
| | Pin | | |
| GPHC′ | e | J17 | 6.0 ± 1.0 V |
| GPRC′ | d | J17 | —12.0 ± 1.0 V |
| IFS | U | J17 | 0.0 ± 0.5 V |
| FBPR | f | J17 | —12.0 ± 1.0 V (momentary) |

TABLE XXII

| A | B | |
|---|---|---|
| Term | Location | |
| | Pin | |
| $T_G$ | M | J17 |
| $T_G'$ | N | J17 |
| $T_{GO}'$ | P | J17 |
| TEIP | R | J17 |
| $T_R'$ | S | J17 |
| PSIA (Strobe Stretch) | DD | J17 |
| FBIO | V | J17 |

TABLE XXIII

| A | | | | B | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 13 least significant bits in the "A" register | | | | | | | | | | | |
| | | | | Bit times | | | | | | | | | | | |
| Vertical row | Horizontal row | Battery | Weapon type | $t_{13}$ | $t_{12}$ | $t_{11}$ | $t_{10}$ | $t_9$ | $t_8$ | $t_7$ | $t_6$ | $t_5$ | $t_4$ | $t_3$ | $t_2$ | $t_1$ |
| A | 1 | A | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| A | 2 | A | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| A | 3 | A | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 1 |
| A | 4 | A | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 |
| A | 5 | A | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |
| A | 6 | A | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 1 |
| A | 7 | A | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 1 |
| A | 8 | A | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |
| B | 1 | B | 2 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 |
| B | 2 | B | 2 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 |
| B | 3 | B | 2 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 |
| B | 4 | B | 2 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
| B | 5 | B | 2 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 |
| B | 6 | B | 2 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 |
| B | 7 | B | 2 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 1 |
| B | 8 | B | 2 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 |



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TABLE XXIII.—(Continued)

| A | | | | B | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 13 least significant bits in the "A" register | | | | | | | | | | | |
| | | | | Bit times | | | | | | | | | | | |
| Vertical row | Horizontal row | Battery | Weapon type | $t_{13}$ | $t_{12}$ | $t_{11}$ | $t_{10}$ | $t_9$ | $t_8$ | $t_7$ | $t_6$ | $t_5$ | $t_4$ | $t_3$ | $t_2$ | $t_1$ |
| C | 1 | C | | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| C | 2 | C | | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 |
| C | 3 | C | | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| C | 4 | C | | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 1 |
| C | 5 | C | | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 |
| C | 6 | C | | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 1 |
| C | 7 | C | | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 1 |
| C | 8 | C | | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 1 |
| D | 1 | D | | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 |
| D | 2 | D | | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 1 |
| D | 3 | D | | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 0 | 1 |
| D | 4 | D | | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 1 |
| D | 5 | D | | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 |
| D | 6 | D | | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 |
| D | 7 | D | | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 1 |
| D | 8 | D | | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 |
| E | 1 | E | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| E | 2 | E | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 |
| E | 3 | E | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 1 |
| E | 4 | E | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 1 |
| E | 5 | E | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| E | 6 | E | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 1 |
| E | 7 | E | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 1 |
| E | 8 | E | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 1 |
| F | 1 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 1 |
| F | 2 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 |
| F | 3 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| F | 4 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 1 |
| F | 5 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 1 |
| F | 6 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | 1 |
| F | 7 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 0 | 1 |
| F | 8 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 |
| G | 1 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 |
| G | 2 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 |
| G | 3 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |
| G | 4 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 1 |
| G | 5 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 |
| G | 6 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 0 | 1 | 1 |
| G | 7 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 |
| G | 8 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 1 | 1 |
| H | 1 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 |
| H | 2 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 1 |
| H | 3 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 0 | 1 |
| H | 4 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 |
| H | 5 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 1 |
| H | 6 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | 1 |
| H | 7 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 |
| H | 8 | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |



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TABLE XXIV

| A | B | | C | D | E |
|---|---|---|---|---|---|
| "F" line | Location | | Voltage average DC to ground | Bit time in the "A" register | Binary designation at specified bit time |
| | Pin | Connector | | | |
| F14 | HH | J 17 | —6 ± 1.0 V | t14 | 1 |
| F1 | JJ | J 17 | —6 ± 1.0 V | t15 | 1 |
| F16 | KK | J 17 | —6 ± 1.0 V | t16 | 1 |
| F17 | LL | J 17 | —6 ± 1.0 V | t17 | 1 |
| F18 | MM | J 10 | —6 ± 1.0 V | t18 | 1 |
| F19 | *k* | J 10 | —4.0 ± 2.0 V | t19 | 1 |
| F20 | m | J 10 | —4.0 ± 2.0 V | t20 | 1 |
| F21 | n | J 10 | —4.0 ± 2.0 V | t21 | 1 |
| F22 | *p* | J 10 | —4.0 ± 2.0 V | t22 | 1 |
| F23 | q | J 10 | —4.0 ± 2.0 V | t23 | 1 |
| F24 | r | J 10 | —4.0 ± 2.0 V | t24 | 1 |
| F25 | g | J 17 | —4.0 ± 2.0 V | t25 | 1 |
| F26 | h | J 17 | —4.0 ± 2.0 V | t26 | 1 |
| F27 | i | J 17 | —4.0 ± 2.0 V | t27 | 1 |
| F28 | j | J 17 | —4.0 ± 2.0 V | t28 | 1 |
| F29 | *k* | J 17 | —4.0 ± 2.0 V | t29 | 1 |
| F30 | m | J 17 | —4.0 ± 2.0 V | t30 | 1 |

TABLE XXV

| Character | A | | | | | | | B | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Magnetic tape code | | | | | | | FADAC binary code (paper tape) | | | | |
| MASTER SPACE | 0 | 1 | 00 | 0 | 0 | 0 | 0 | 00 | 0 | 0 | 0 | 0 |
| UPPER CASE | 1 | 1 | 00 | 0 | 0 | 0 | 1 | 00 | 0 | 0 | 0 | 1 |
| LOWER CASE | 1 | 1 | 00 | 0 | 0 | 1 | 0 | 00 | 0 | 0 | 1 | 0 |
| TAB | 0 | 1 | 00 | 0 | 0 | 1 | 1 | 00 | 0 | 0 | 1 | 1 |
| CAR. RET. | 1 | 1 | 00 | 0 | 1 | 0 | 0 | 00 | 0 | 1 | 0 | 0 |
| SPACE | 0 | 1 | 00 | 0 | 1 | 0 | 1 | 00 | 0 | 1 | 0 | 1 |
| Aa | 0 | 1 | 00 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 |
| Bb | 1 | 1 | 00 | 0 | 1 | 1 | 1 | 00 | 0 | 1 | 1 | 1 |
| Cc | 1 | 1 | 00 | 1 | 0 | 0 | 0 | 00 | 1 | 0 | 0 | 0 |
| Dd | 0 | 1 | 00 | 1 | 0 | 0 | 1 | 00 | 1 | 0 | 0 | 1 |
| Ee | 0 | 1 | 00 | 1 | 0 | 1 | 0 | 00 | 1 | 0 | 1 | 0 |
| Ff | 1 | 1 | 00 | 1 | 0 | 1 | 1 | 00 | 1 | 0 | 1 | 1 |
| Gg | 0 | 1 | 00 | 1 | 1 | 0 | 0 | 00 | 1 | 1 | 0 | 0 |
| Hh | 1 | 1 | 00 | 1 | 1 | 0 | 1 | 00 | 1 | 1 | 0 | 1 |
| Ii | 1 | 1 | 00 | 1 | 1 | 1 | 0 | 00 | 1 | 1 | 1 | 0 |
| J | 0 | 1 | 00 | 1 | 1 | 1 | 1 | 00 | 1 | 1 | 1 | 1 |
| Kk | 1 | 1 | 01 | 0 | 0 | 0 | 0 | 01 | 0 | 0 | 0 | 0 |
| Ll | 0 | 1 | 01 | 0 | 0 | 0 | 1 | 01 | 0 | 0 | 0 | 1 |
| Mm | 0 | 1 | 01 | 0 | 0 | 1 | 0 | 01 | 0 | 0 | 1 | 0 |
| Nn | 1 | 1 | 01 | 0 | 0 | 1 | 1 | 01 | 0 | 0 | 1 | 1 |
| Oo | 0 | 1 | 01 | 0 | 1 | 0 | 0 | 01 | 0 | 1 | 0 | 0 |
| Pp | 1 | 1 | 01 | 0 | 1 | 0 | 1 | 01 | 0 | 1 | 0 | 1 |
| Qq | 1 | 1 | 01 | 0 | 1 | 1 | 0 | 01 | 0 | 1 | 1 | 0 |
| Rr | 0 | 1 | 01 | 0 | 1 | 1 | 1 | 01 | 0 | 1 | 1 | 1 |
| Ss | 0 | 1 | 01 | 1 | 0 | 0 | 0 | 01 | 1 | 0 | 0 | 0 |



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TABLE XXV— (Continued)

| Character | A | | | | | | | B | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Magnetic tape code | | | | | | | FADAC binary code (paper tape) | | | | |
| Tt | 1 | 1 | 01 | 1 | 0 | 0 | 1 | 01 | 1 | 0 | 0 | 1 |
| Uu | 1 | 1 | 01 | 1 | 0 | 1 | 0 | 01 | 1 | 0 | 1 | 0 |
| Vv | 0 | 1 | 01 | 1 | 0 | 1 | 1 | 01 | 1 | 0 | 1 | 1 |
| Ww | 1 | 1 | 01 | 1 | 1 | 0 | 0 | 01 | 1 | 1 | 0 | 0 |
| Xx | 0 | 1 | 01 | 1 | 1 | 0 | 1 | 01 | 1 | 1 | 0 | 1 |
| Yy | 0 | 1 | 01 | 1 | 1 | 1 | 0 | 01 | 1 | 1 | 1 | 0 |
| Zz | 1 | 1 | 01 | 1 | 1 | 1 | 1 | 01 | 1 | 1 | 1 | 1 |
| ) | 1 | 1 | 10 | 0 | 0 | 0 | 0 | 10 | 0 | 0 | 0 | 0 |
| - | 0 | 1 | 10 | 0 | 0 | 0 | 1 | 10 | 0 | 0 | 0 | 1 |
| + | 0 | 1 | 10 | 0 | 0 | 1 | 0 | 10 | 0 | 0 | 1 | 0 |
| < | 1 | 1 | 10 | 0 | 0 | 1 | 1 | 10 | 0 | 0 | 1 | 1 |
| = | 0 | 1 | 10 | 0 | 1 | 0 | 0 | 10 | 0 | 1 | 0 | 0 |
| < | 1 | 1 | 10 | 0 | 1 | 0 | 1 | 10 | 0 | 1 | 0 | 1 |
| — | 1 | 1 | 10 | 0 | 1 | 1 | 0 | 10 | 0 | 1 | 1 | 0 |
| $ | 0 | 1 | 10 | 0 | 1 | 1 | 1 | 10 | 0 | 1 | 1 | 1 |
| " | 0 | 1 | 10 | 1 | 0 | 0 | 0 | 10 | 1 | 0 | 0 | 0 |
| ( | 1 | 1 | 10 | 1 | 0 | 0 | 1 | 10 | 1 | 0 | 0 | 1 |
| " | 1 | 1 | 10 | 1 | 0 | 1 | 0 | 10 | 1 | 0 | 1 | 0 |
| : | 0 | 1 | 10 | 1 | 0 | 1 | 1 | 10 | 1 | 0 | 1 | 1 |
| ? | 1 | 1 | 10 | 1 | 1 | 0 | 0 | 10 | 1 | 1 | 0 | 0 |
| ! | 0 | 1 | 10 | 1 | 1 | 0 | 1 | 10 | 1 | 1 | 0 | 1 |
| , | 0 | 1 | 10 | 1 | 1 | 1 | 0 | 10 | 1 | 1 | 1 | 0 |
| STOP | 1 | 1 | 10 | 1 | 1 | 1 | 1 | 10 | 1 | 1 | 1 | 1 |
| 0 | 0 | 1 | 11 | 0 | 0 | 0 | 0 | 11 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 11 | 0 | 0 | 0 | 1 | 11 | 0 | 0 | 0 | 1 |
| 2 | 1 | 1 | 11 | 0 | 0 | 1 | 0 | 11 | 0 | 0 | 1 | 0 |
| 3 | 0 | 1 | 11 | 0 | 0 | 1 | 1 | 11 | 0 | 0 | 1 | 1 |
| 4 | 1 | 1 | 11 | 0 | 1 | 0 | 0 | 11 | 0 | 1 | 0 | 0 |
| 5 | 0 | 1 | 11 | 0 | 1 | 0 | 1 | 11 | 0 | 1 | 0 | 1 |
| 6 | 0 | 1 | 11 | 0 | 1 | 1 | 0 | 11 | 0 | 1 | 1 | 0 |
| 7 | 1 | 1 | 11 | 0 | 1 | 1 | 1 | 11 | 0 | 1 | 1 | 1 |
| 8 | 1 | 1 | 11 | 1 | 0 | 0 | 0 | 11 | 1 | 0 | 0 | 0 |
| 9 | 0 | 1 | 11 | 1 | 0 | 0 | 1 | 11 | 1 | 0 | 0 | 1 |
| ' | 0 | 1 | 11 | 1 | 0 | 1 | 0 | 11 | 1 | 0 | 1 | 0 |
| ; | 1 | 1 | 11 | 1 | 0 | 1 | 1 | 11 | 1 | 0 | 1 | 1 |
| / | 0 | 1 | 11 | 1 | 1 | 0 | 0 | 11 | 1 | 1 | 0 | 0 |
| . | 1 | 1 | 11 | 1 | 1 | 0 | 1 | 11 | 1 | 1 | 0 | 1 |
| SPECIAL | 1 | 1 | 11 | 1 | 1 | 1 | 0 | 11 | 1 | 1 | 1 | 0 |
| BACKSPACE | 1 | 1 | 11 | 1 | 1 | 1 | 1 | 11 | 1 | 1 | 1 | 1 |



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TABLE XXVI

| A | B | C | | | |
|---|---|---|---|---|---|
| | | Quasi-Octal | | | |
| Commands | Mnemonic | Spare * | OP | Right Channel | Right Selector |
| *Numeric — Octal* | | | | | |
| Teletype | WRN | 0<br>2 | 44 | 000 | |
| Fieldata | WRN | 0<br>2 | 44 | 010 | |
| *Numeric — BCD* | | | | | |
| Teletype | WRN | 0<br>2 | 44 | 100 | |
| Fieldata | WRN | 0<br>2 | 44 | 110 | |

* Note spare bit.

TABLE XXVII

| A | B | |
|---|---|---|
| | Location | |
| Output line | Pin | Connector |
| D100 | J | J 10 |
| D200 | K | J 10 |
| D300 | L | J 10 |
| D400 | M | J 10 |
| D500 | N | J 10 |
| D600 | O | J 10 |
| D700 | P | J 10 |
| D800 | R | J 10 |

TABLE XXVIII

| Term | Location | |
|---|---|---|
| | Pin | Connector |
| TFBO | h | J10 |
| TFBO' | i | J10 |
| TEOP | j | J10 |
| TFAO' | g | J10 |
| TFAI' | f | J10 |
| PSIB (strobe stretch) | EE | J17 |
| FBOO | V | J10 |

TABLE XXIX

| A | B | C | | | |
|---|---|---|---|---|---|
| | | Quasi-Octal | | | |
| Commands | Mnemonic | Spare * | OP | Right Channel | Right Sector |
| | | | | Binary | |
| Alpha Numeric 5 | WRA | 0<br>2 | 46 | 0 | Amount |
| Alpha Numeric 6 | WRA | 0<br>2 | 46 | 1 | Amount |

* Note spare bit.



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TABLE XXX

| A | B | | C | | | | | |
|---|---|---|---|---|---|---|---|---|
| Discrete output selected | Location | | Binary state of each discrete output line | | | | | |
| | Pin | Connector | $O_{P6}$ | $O_{P5}$ | $O_{P4}$ | $O_{P3}$ | $O_{P2}$ | $O_{P1}$ |
| OPL1 | X | J10 | 0 | 0 | 0 | 0 | 0 | 1 |
| OPL2 | Y | J10 | 0 | 0 | 0 | 0 | 1 | 0 |
| OPL3 | Z | J10 | 0 | 0 | 0 | 1 | 0 | 0 |
| OPL4 | a | J17 | 0 | 0 | 1 | 0 | 0 | 0 |
| OPL5 | b | J17 | 0 | 1 | 0 | 0 | 0 | 0 |
| OPL6 | c | J17 | 1 | 0 | 0 | 0 | 0 | 0 |

TABLE XXXI

TARGET INFORMATION

| *Bat.* | *Target* | *Target East* | *Target North* | *Target Alt* |
|---|---|---|---|---|
| A | 1 | 33125 | 00768 | 428 |
| A | 6 | 36500 | 01100 | 450 |
| B | 2 | 84989 | 09563 | 27 |
| B | 7 | 82000 | 08500 | 75 |
| C | 3 | 11230 | 07178 | 105 |
| C | 8 | 09800 | 07000 | 140 |
| D | 4 | 99157 | 57724 | 105 |
| D | 9 | 97000 | 58000 | 130 |
| E | 5 | 50889 | 31330 | 389 |
| E | 10 | 53000 | 31500 | 410 |

TABLE XXXII

Lat N 50
GRID DECL +50
PROBLEM INFORMATION

| Battery | A–105 | B–155 | C–105 | D–155 | E–105 |
|---|---|---|---|---|---|
| Bat. East | 33485 | 86240 | 12340 | 00294 | 50000 |
| Bat North | 99567 | 10070 | 08560 | 56767 | 30000 |
| Altitide | 422 | 20 | 1000 | 105 | 500 |
| Bat. AZ | 1600 | 4800 | 3200 | 1600 | 1600 |
| Bat. Defl. | 2800 | 2600 | 2800 | 2600 | 2800 |
| * Defl. Corr. | L1 | R10 | R20 | R10 | R15 |
| * Time Corr. | —.2 | +.5 | +.5 | +.5 | +.6 |
| * Range Corr. | —50 | +50 | +50 | +100 | —100 |
| Charge | 1 | 2 | 1 | 1 | 1 |
| Proj. Type | 3 | 2 | 4 | 1 | 1 |
| Fuse Type | 2 | 2 | 2 | 1 | 2 |

* Charge Associated.



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TABLE XXXIII

PROBLEM SOLUTIONS
MET. STANDARD

| Bat. | Target | Deflection | | Fuse time | | Quad. elev. | |
|---|---|---|---|---|---|---|---|
| | | Low | High | Low | High | Low | High |
| A | 1 | 1500 | 1717 | 4.8 | 38.7 | 171 | 1412 |
| A | 6 | 95 | 115 | 23.8 | 26.4 | 615 | 955 |
| B | 2 | 2986 | 3141 | 6.7 | 46.5 | 156 | 1450 |
| B | 7 | 2973 | 2979 | 27.0 | 37.9 | 744 | 848 |
| C | 3 | 2094 | 2194 | 11.3 | 37.0 | 221 | 1271 |
| C | 8 | 1761 | 1763 | 22.5 | 31.0 | 735 | 772 |
| D | 4 | 1882 | 1982 | 8.1 | 40.7 | 205 | 1381 |
| D | 9 | 2248 | 2251 | 22.2 | 34.7 | 770 | 827 |
| E | 5 | 587 | 756 | 9.5 | 42.0 | 136 | 1374 |
| E | 10 | 68 | 103 | 23.9 | 32.1 | 491 | 1048 |

TABLE XXXIV

PROBLEM SOLUTIONS
MET "H" TAPE

| Bat. | Target | Deflection | | Fuse time | | Quad. elev. | |
|---|---|---|---|---|---|---|---|
| | | Low | High | Low | High | Low | High |
| A | 1 | 1499 | 1705 | 4.8 | 38.7 | 171 | 1408 |
| A | 6 | Out of range | Out of range | Out of range | Out of range | Out of range | Out of range |
| B | 2 | 2985 | 3123 | 617 | 46.5 | 156 | 1451 |
| B | 7 | 2967 | 2975 | 26.5 | 38.2 | 706 | 882 |
| C | 3 | 2093 | 2181 | 11.2 | 371 | 217 | 1273 |
| C | 8 | 1754 | 1758 | 22.1 | 31.2 | 690 | 821 |
| D | 4 | 1881 | 1969 | 8.1 | 40.6 | 205 | 1378 |
| D | 9 | 2242 | 2243 | 22.2 | 34.7 | 754 | 832 |
| E | 5 | 588 | 767 | 9.6 | 42.0 | 137 | 1372 |
| E | 10 | 73 | 110 | 24.3 | 32.1 | 498 | 1049 |

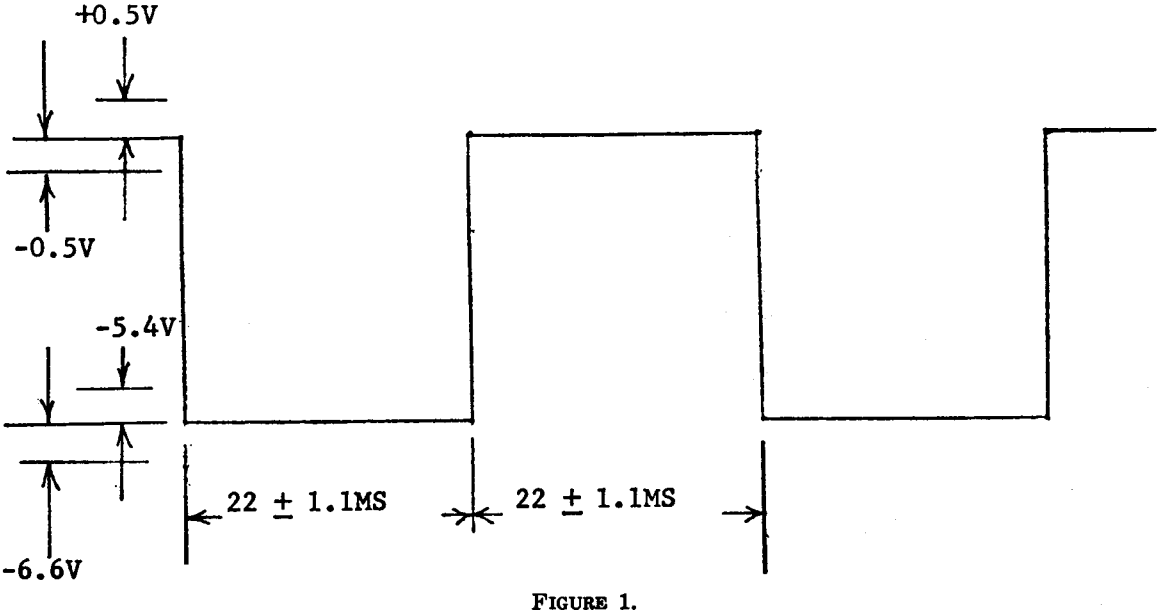
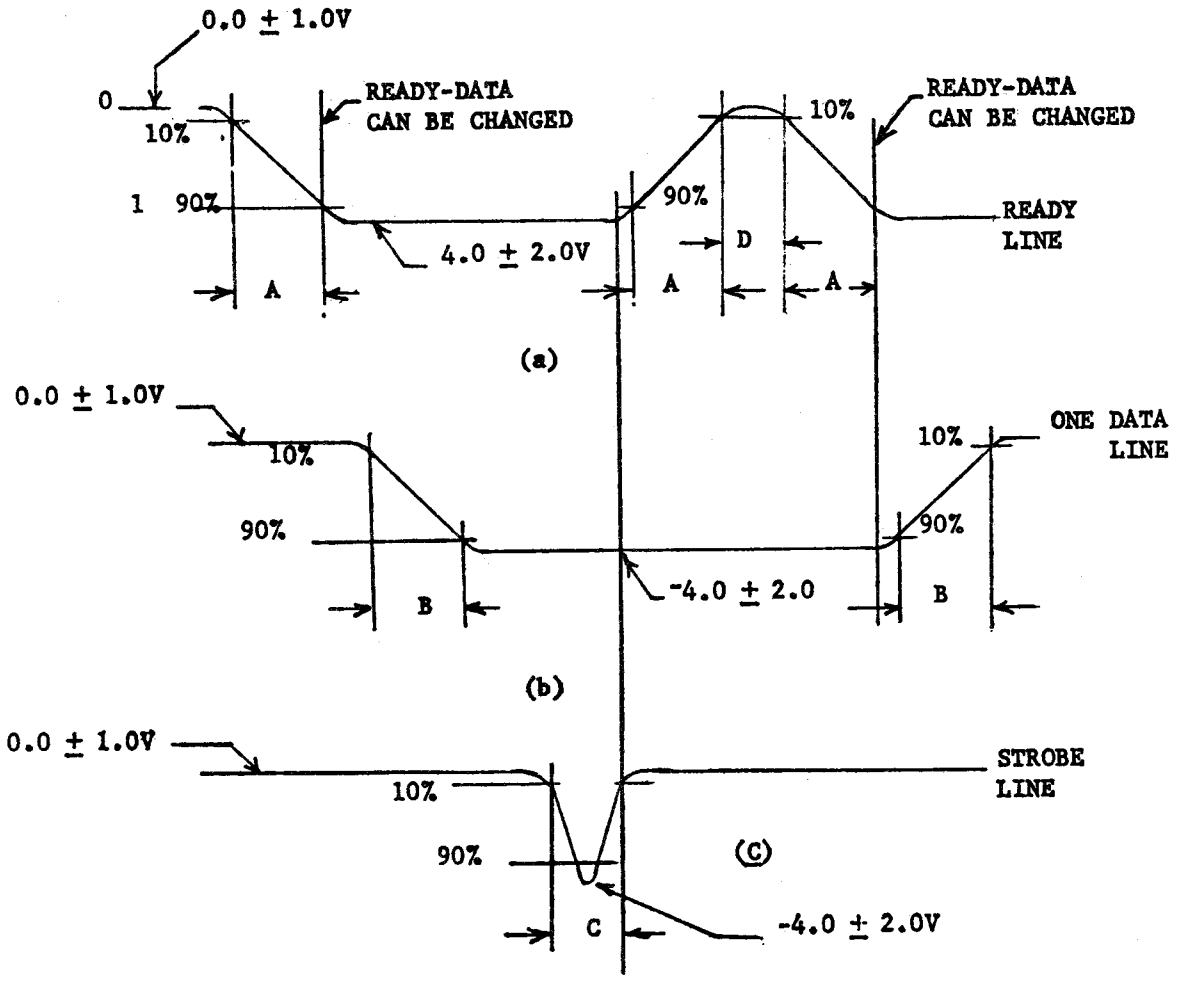



FIGURE 1.



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A = not less than 2 microsec no more than 4 microsec.

B = not less than 2 microsec. Maximum time compatible with character rate.

C — not less than 1 microsec, nor more than 2 microsec at 10% points. Rise and decay times no more than those attainable through generation of a half sinusoid to represent the strobe.

D = not less than 2 microsec. Maximum time compatible with character rate.

FIGURE 2.



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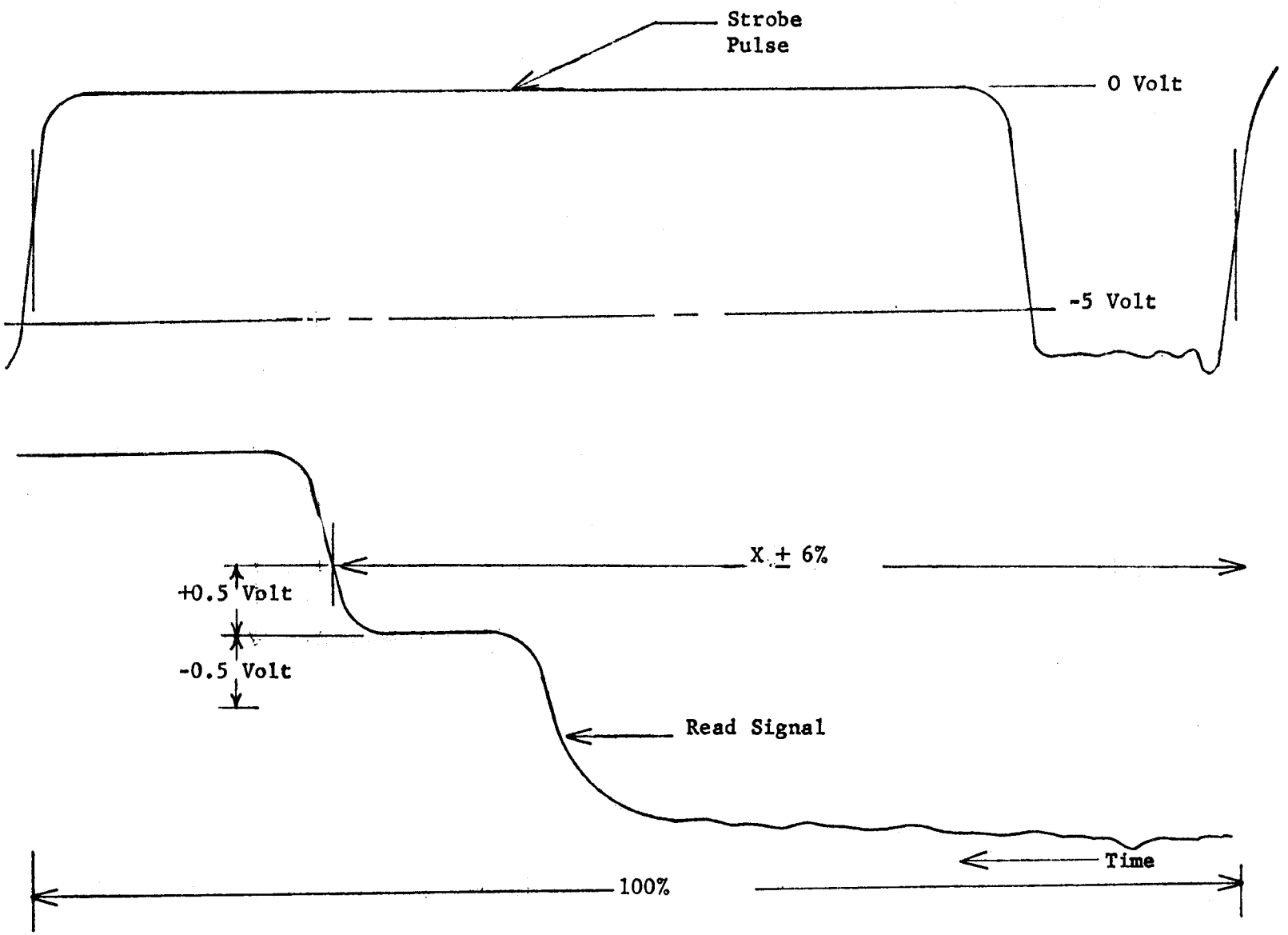



FIGURE 3. *Position of strobing pulse relative to memory signals.*





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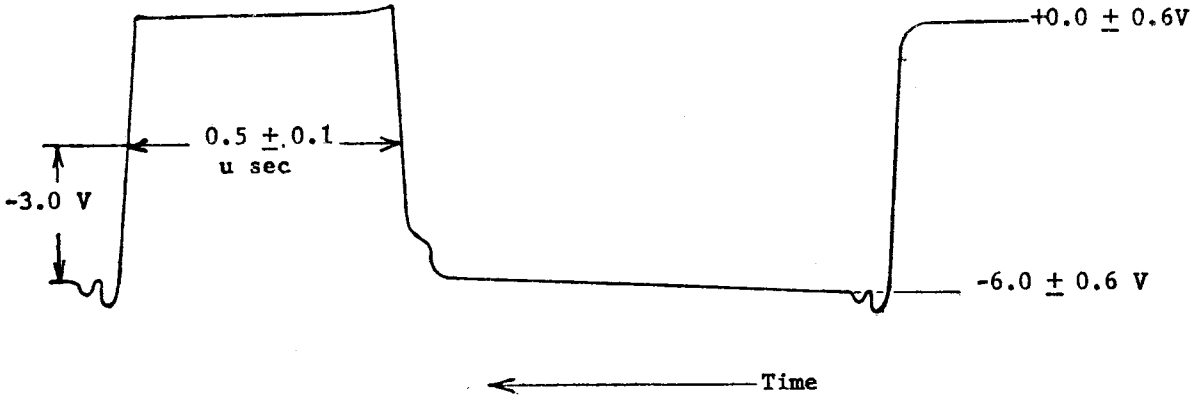



FIGURE 4. *Clock Pulse waveform*

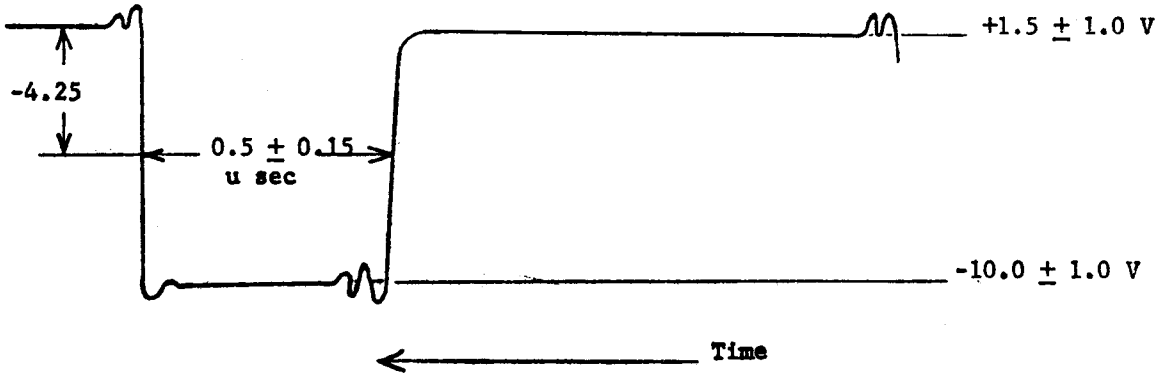
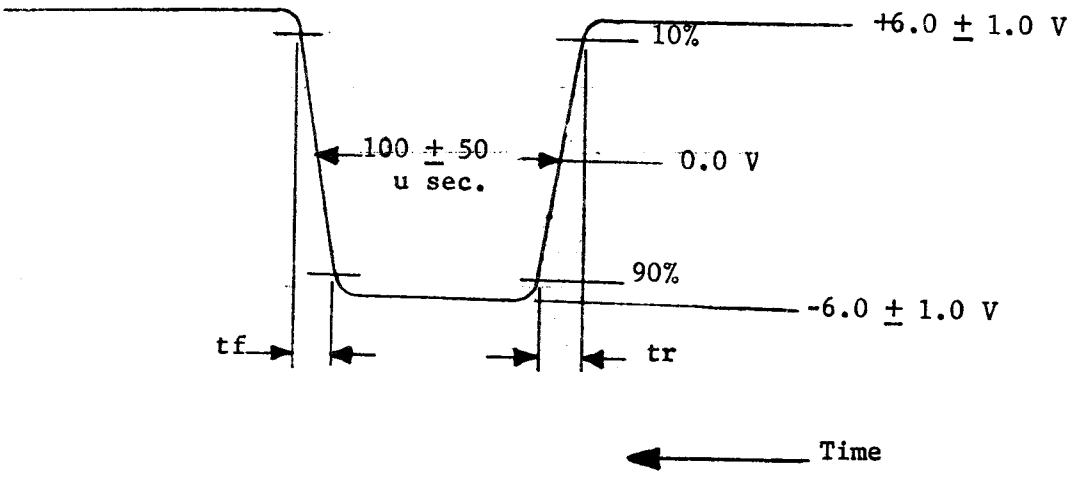



FIGURE 5. *Strobing pulse waveform.*



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tr - Less than 1.0 u sec.

tf - Less than 1.0 u sec.

FIGURE 6. *Clock trigger waveform.*

| Input Line Designation | $I_{5G}$ | $I_{4G}$ | | $I_{3G}$ | $I_{2G}$ | $I_{1G}$ |
|---|---|---|---|---|---|---|
| Teletype Channel Designation | 5 | 4 | • | 3 | 2 | 1 |
| Teletype Code For "0" | 0 | 1 | • | 1 | 0 | 1 |

No hole - Binary 0

Hole - Binary I

Sprocket Hole

FIGURE 7.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
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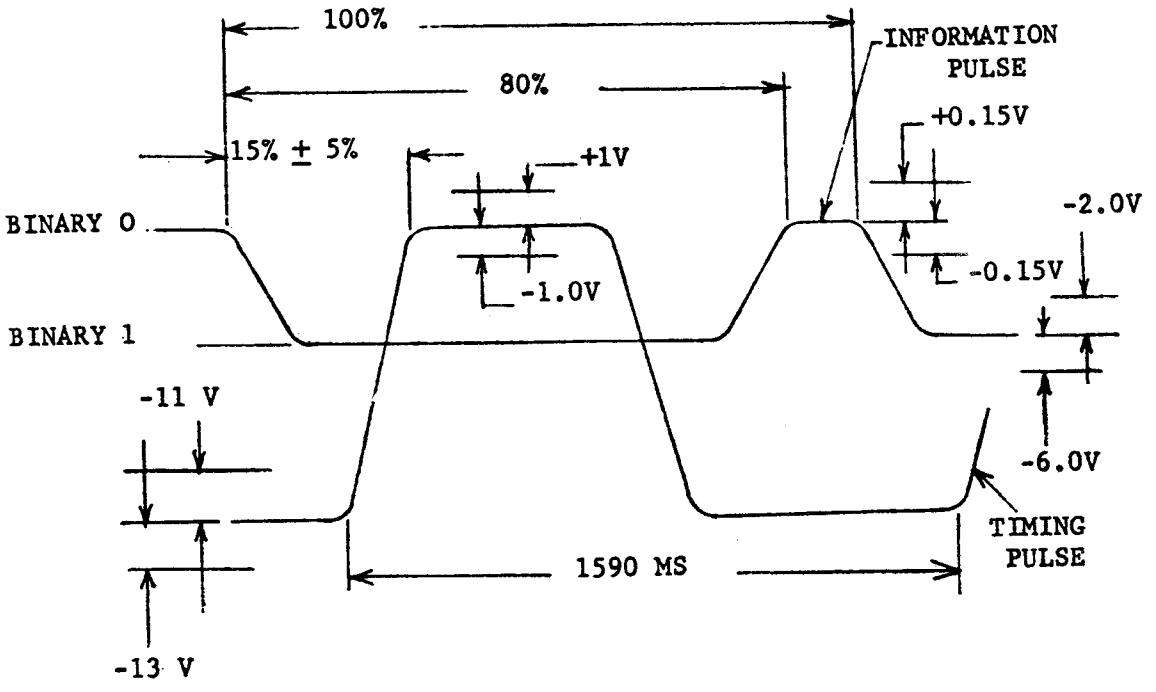



FIGURE 8.

| Input Line Designation | $I_{8G}$ | $I_{7G}$ | $I_{6G}$ | $I_{5G}$ | $I_{4G}$ | | $I_{3G}$ | $I_{2G}$ | $I_{1G}$ |
|---|---|---|---|---|---|---|---|---|---|
| Fieldata Channel Designation | 8 | 7 | 6 | 5 | 4 | • | 3 | 2 | 1 |
| Fieldata Code for "0" | 0 | 1 | 1 | 0 | 0 | • | 0 | 0 | 0 |

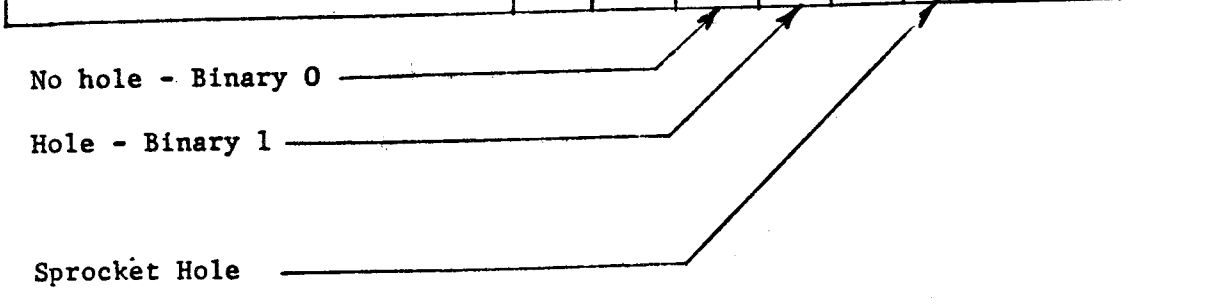



FIGURE 9.



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# APPENDIX A

# COMPUTER OPERATIONS

## 10. SCOPE

10.1 This appendix to MIL-C-60184(MU), Computer Set, Gun Direction: 10525735, provides instructions and commands for the operation of the computer.

## 20. APPLICABLE DOCUMENTS

None.



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## 30. INSTRUCTIONS

*Operation Commands*

HALT, HLT:
Execution word times:
Execution:

36, C = 120
1
Computation is halted. Flag bit and operand address are ignored by this command.

INITIATE DISPLAY MODE, IDM:
Execution word times:
Execution:

36, C = 164
1
Visual display activated until a HALT DISPLAY command is received. Sixteen binary-coded decimal (BCD) digits at a time are displayed from the 2-word D register. Computations can proceed while in this display mode. Flag bit and operand address are ignored by this command.

HALT DISPLAY MODE, HDM:
Execution word times:
Execution:

36, C = 166
1
Visual display mode halted. Flag bit and operand address are ignored.

DISCRETE INPUT TO ACCUMULATOR, DIA:
Execution word times:
Execution:

36, C = 040
1
Information on 32 discrete input lines are input into the A register, enabling program to sample various input signals in A register. Currently these are as follows: (flag bit and operand address are ignored).

| *A Register Bits* | *Contents* |
|---|---|
| 1 | Signal that both column and row matrix buttons have been depressed |
| 2 - 7 | Matrix code |
| 8 | Control Panel button 1 or 2 depressed |
| 9 - 13 | Control Panel button A, B, C, D, or E depressed |
| 14 - 30 | Spare bits |
| 32 | Two-wire Teletype timing signal for programming |

DISCRETE OUTPUT:

DO 0: 36, C = 100
DO 1: 36, C = 102
DO 2: 36, C = 104
DO 3: 36, C = 106
DO 4: 36, C = 110
DO 5: 36, C = 112
DO 6: 36, C = 114
DO 7: 36, C = 116

GREATER THAN OR EQUAL SEARCH, GES:
Execution word times:
Execution

66
2 minimum
Starting with i = 0, the contents of the A register and memory location (m + i) are compared in bit positions as directed by a mask appearing in the L register. This mask must have been stored in the L registerer by the program. It informs the computer of the bits of the A register and (m + i) that should be compared. Whenever a 1 appears in the L register, the corresponding bits in A and (m + i) will be checked. If in the designated bit positions (m + i) (A), then the ensuing word (m + i + 1) is put into the A register. At this time if the flag bit of the command is a 1, the contents of the A register are transferred to the L register. If the flag bit is 0, the contents of the L register remain unaltered. If in the designated bit positions the contests of the A register are not less than or equal to the contents of (m + i), then i is increased by one and the comparison continues if the end of the channel has not been reached.



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If the end of the channel has been reached, the operation is terminated.

*Arithmetic Commands*

ADD, ADD m:
00
Execution word times:
1
Execution
The word in location m is added to the contents of the A register and the sum is stored in the A register. Overflow can occur and, unless followed by a TRANSFER ON OVERFLOW command, the computer will halt with an error indication. A flag bit of 1 will cause the contents of location m to be put into the N register. A flag bit of 0 leaves the contents of the N register unaltered.

Execution word times:
1
Execution
The eight DO commands can set a 3-bit register to any one of its eight possible counts. This register, in turn, controls several output lines and lights. Currently, the eight counts are used as follows:

| *Count* | *To* | *Function* |
|---|---|---|
| OPL0 | ..... | Idle |
| OPL1 | Output plug | |
| OPL2 | Output plug | |
| OPL3 | Output plug | |
| OPL4 | Input plug | |
| OPL5 | Input plug | |
| OPL6 | Input plug | |
| OPL7 | | No Solution Light |

*Control Commands*

TABLE SEARCH
EQUALITY, EQS:
64
Execution word times:
2 minimum
Execution
Starting with $i = 0$, the contents of the A register are compared with those in memory location $(m + i)$ in bit positions as indicated by a mask appearing in the L register. This mask must have been stored in the L register by the program. It directs which bits of the A register and memory should be compared. Whenever a 1 appears in the L register, the corresponding bits in A and $(m + i)$ will be checked. If in the designated bit positions the contents of the A register are found to be equal to the contents of $(m + i)$, the ensuing word in memory $(m + i + 1)$ is put into the A register. At this time, if the flag bit of the command is a 1, the contents of the A register will be transferred to the L register. If the flag bit is a 0, the contents of the L register remain unaltered. If the contents of the A register, in the designated bit positions, are not equal to the contents of $(m + i)$, then i is increased by one and the comparison continues if the end of a channel has not been reached. If the end of the channel has been reached, the operation is terminated.

SUBTRACT, SUB m:
02
Execution word times:
1
Execution
The word in location m is subtracted from the A register, the difference appearing in the A register. Overflow can occur and, unless followed by a TRANSFER ON OVERFLOW command, will cause the computer to halt and indicate an error. A flag bit of 1 will cause contents of m to be put into the N register. A flag bit of 0 leaves contents of N register unaltered.



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MULTIPLY, MPY m: 20
Execution word times: 18
Execution

The contents of memory location m are multiplied by the contents of the A register. The product appears in the A and L registers unrounded. The most significant 31 bits and sign are in the A register. The least significant 31 bits are in the L register. Because the number in the L register is the least significant part of the total product, it does not have a sign associated with it. The most significant bit of the 31 bits of the least significant part of the product appears in the sign position of the L register. The least significant bit position of the L register always contains a 0 after multiplication which is not considered a part of the product. The contents of location m are stored in the N register. The flag bit is ignored by this instruction.

DIVIDE, DIV m: 30
Execution word times: 18
Execution

The 62-bit number appearing in the A and L registers is divided by the contents of location m. The most significant bit of the portion of the dividend that appears in the L register is in the sign position. If the flag bit is 0, then the A register will contain the rounded quotient and the L register will contain a pseudo-remainder. If the flag bit is 1, then the A register will contain the unrounded quotient and the L register will contain a remainder with the same sign as the divisor. This remainder satisfies the relation: Divisor $\times$ Quotient $+$ Remainder $=$ Dividend. The contents of location m are put in the N register.

Overflow can occur if the absolute value of the contents of the A register is equal to or exceeds the absolute value of the contents of m. In case of overflow, the computer will halt with an error signal unless the command is followed by a TRANSFER ON OVERFLOW command.

*Semiarithmetic Commands*

ACCUMULATOR RIGHT CYCLE, ARC S: 76, C = 00

Execution word times: $\frac{1}{2}\ (1 + S + \cos\frac{2S}{2})$

Execution

The sector address portion of this command does not refer to a location in memory, but indicates the number of binary positions the word in the A register is to be cycled. On execution of this command, the contents of the A register are shifted right the number of binary positions designated by S. The bits shifted off the right end of the A register appear at the left end of the register in the same sequence being shifted in through the sign bit. The flag bit must be a 0.

ACCUMULATOR RIGHT SHIFT, ARS S: 76, C = 02

Execution word times: $\frac{1}{2}\ (1 + S + \cos\frac{2S}{2})$

Execution:

The sector address portion of this command does not refer to a location in memory, but indicates the number of binary positions the word in the A register is to be shifted. On execution of this command, the contents of the A register are shifted right the number of binary places designated by S. The sign is spread in the left-hand bits, while bits shifted beyond the right-hand limits of the A register are lost. The flag bit must be a 0.



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ACCUMULATOR LEFT CYCLE, ALC S:

76, C = 04

Execution word times:

$$\tfrac{1}{2}\ (1 + S + \cos\frac{2\ S}{2})$$

Execution:

The sector address portion of this command does not refer to a location in memory, but indicates the number of binary positions the word in the A register is to be cycled. On execution of this command, the contents of the A register are shifted left, through the sign bit, the number of binary places indicated by S. The bits shifted off the left end of the register are shifted in at the right in the same sequence. The flag bit of 1 indicates that the computer should check for overflow.

ACCUMULATOR LEFT SHIFT, ALS S:

76, C = 06

Execution word times:

$$\tfrac{1}{2}\ (1 + S + \cos\frac{2\ S}{2})$$

Execution:

The sector address portion of this command does not refer to a location memory, but indicates the number of binary positions the word in the A register is to be shifted. On execution of this command, the contents of the A register are shifted left, through the sign bit, the number of binary places designated by S. Positions left vacant, as the contents in the A register are shifted left, are filled with zeros. Any bits shifted beyond the sign position of the A register are lost. It is possible to have an overflow with this command.

A flag bit of 1 indicates that the computer should check for overflow. When the flag bit is 0, overflow is ignored.

LONG RIGHT CYCLE LRC S:

76, C = 20

Execution word times:

$$\tfrac{1}{2}\ (1 + S + \cos\frac{2\ S}{2})$$

Execution:

The sector address portion of this command designates the number of binary places the information is to be cycled. Upon execution of this command, the contents of the A and L registers are shifted right S binary places. The bits shifted off the right end of the A register are shifted into the L register through its sign bit position. The bits shifted off the right end of the L register are shifted into the A register through its sign bit position. The flag bit must be 0.

LONG RIGHT SHIFT, LRS S:

76, C = 22

Execution word times:

$$\tfrac{1}{2}\ (1 + S + \cos\frac{2\ S}{2})$$

Execution:

The sector address portion of this command designates the number of binary places the information is to be shifted. Upon execution of this command, the contents of both the A and L registers are shifted right the number of binary places designated by S. The bits shifted off the right end of the A register are shifted into the L register through its sign bit position. The bits shifted off the right end of the L register are lost. The sign of the A register is spread. The flag bit must be a 0.



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LONG LEFT CYCLE, LLC S: 76, C = 24

Execution word times: $\frac{1}{2}\ (1 + S + \cos\frac{2\ S}{2})$

Execution: The sector address portion of this command designates the number of binary places the information is to be cycled. Upon execution of this command, the contents of the A and L registers are shifted left the number of binary places indicated by S. The bits shifted left through the sign of the A register are shifted into the right end of the L register. Bits shifted through the sign bit of the L register are shifted into the right end of the A register. The flag bit is ignored for this command.

LONG LEFT SHIFT, LLS S: 76, C = 26

Execution word times: $\frac{1}{2}\ (1 + S + \cos\frac{2\ S}{2})$

Execution: The sector address portion of this command designates the number of binary places the information is to be shifted. Upon execution of this command, the contents of both the A and L registers are shifted left the number of binary places designated by S. The bits shifted through the sign bit of the L register are shifted into the right-hand end of the A register. The right-hand bits of the L register vacated by the shifting process are filling with zeros. Bits shifted off the left-hand end of the A register are lost. It is possible to have overflow with this command. A flag bit of 1 indicates the computer should check for overflow.

*Control Transfer*

TRANSFER ON PLUS, TPL: 10

Execution word times: 1

Execution: Control is transferred to the instruction found in location m if the contents of the A register are greater than or equal to zero. If the contents of the A register are negative, the address of the next instruction to be executed is obtained from the NEXT INSTRUCTION address of the command. If the transfer is executed, a flag bit of 1 will put the instruction into the N register. A flag bit of 0 leaves the contents of the N register unaltered.

TRANSFER ON OVERFLOW, TOV m: 16

Execution word times: 1

Execution: Control is transferred to the instruction found in location m when an overflow has occurred as a result of the previous instruction. If no overflow has occurred, control will proceed to the next instruction. If the transfer is executed, a flag bit of 1 will put the instruction into the N register. A flag bit of 0 leaves the N register unaltered.

TRANSFER UNCONDITIONALLY, TRA m: 14

Execution word times: 1

Execution: Control is transferred to the instruction found in location m. A flag bit of 1 will put the transfer instruction into the N register. A flag bit of 0 leaves the N register unaltered.

TRANSFER ON ZERO: TZE m: 12

Execution word times: 1

Execution: Control is transferred to the instruction found in location m if the contents of the A register are equal to zero. If the contents of the A



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register are not equal to zero, the transfer will not be executed, and the next instruction will come from the location specified by the NEXT INSTRUCTION address portion of the command. If the transfer is executed, a flag bit of 1 will put the instruction into the N register. A flag bit of 0 leaves the N register unaltered.

*Information Transfer*

CLEAR AND ADD, CLA m: 24
Execution word times: 1
Execution: The contents of location m are transferred to the A register. Location m can refer to any addressable location. A flag bit of 1 transfers the previous contents of the A register into the L register. If the flag bit is 0, the contents of the L register remain unaltered.

CLEAR AND SUBTRACT, CLS m: 26
Execution word times: 1
Execution: The negative of the contents of memory location m are transferred to the A register. Location m can refer to any addressable location. A flag bit of 1 transfers the previous contents of the A register into the L register. If the flag bit is 0, the contents of the L register remain unaltered.

STORE A, STO m: 50
Execution word times: 1
Execution: The contents of the A register are put (stored) in location m. If the flag bit is 1, this word will be put into the N register as well as into location m. If the flag bit is 0, the previous contents of the N register remain unaltered. Location m can refer to any addressable location except when m = A, L, or N registers, then no storage takes place.

STORE L, STL m: 25
Execution word times: 1
Execution: The contents of the L register are put (stored )in location m. If the flag bit is 1, the word will be put into the N register as well as into location m. If the flag bit is 0, the previous contents of the N register remain unaltered. Location m can refer to any addressable location except when m + A, L, or N registers, then no storage takes place.

STORE N, STN m: 40
Execution word times: 1
Execution: The contents of the N-register are put (stored) in location m. Location m can refer to any addressable location except when m = A, L, or N registers, then no storage takes place. The bit is ignored.

STORE D, STD m: 42
Execution word times: 1
Execution: The contents of sector i of the 2-word D register are put into location m, where i = o if m is even and i = 1 if m is odd. If the flag bit is 1, this word will be put into the N register as well as into location m. If the flag bit is 0, the previous contents of the N register remain unaltered. Location m can refer to any addressable location except when m = A, L, or N, then no storage takes place.

STORE R, STR m: 62
Execution word times: 18
Execution: The contents of the 16 words of the R loop are transferred into main memory starting at location m. The last four bits of m will determine the sector of R from which the first word is being stored. The contents of the D register are altered. The flag bit is ignored.



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CONDITIONAL REPLACE NEGATIVE, CRN:
36, C = 170 = A
C = 172 = L
C = 174 = L

Execution word times:
1

Execution:
If the contents of the A register are negative, then the negative of the contents of the A, L, or N register, respectively, are transferred to the A register. Note than when C = 170, this amounts to replacing the contents of the A register with its absolute value. If the contents of the A register are negative, and the flag bit is 1, then the old contents of the A register are transferred to the L register. If the flag bit is 0, the contents of the L register remain unaltered.

ZERO L, ZEL:
36, C = 162

Execution word times:
1

Execution:
The L register is cleared to zero. The flag bit and the operand sector are ignored by this command.

LOGICAL AND TO ACCUMULATOR, ANA m:
34

Execution:
The contents of the A register are replaced by the logical AND of the contents of A and the contents of m. This command works in the following manner: when both the bit in memory and the bit in the A register are 1, a 1 will appear in the same bit position in the A register at the end of the execution. When either the bit in memory or the bit in the A register is 0, a 0 will appear in that bit position of the A register upon execution of this command. The flag bit is ignorer by this command.

STORE ADDRESS, STA m:
70

Execution word times:
1 if m refers to R or Q; otherwise 3.

Execution:
The right address of the word in location m is replaced by the right address of the word in the A register (13 bits). The modified contents of m are put in the N register. The m can refer to any addressable location except A and L. The flag bit is ignored for this operation.

STORAGE PROGRAM ADDRESS, STP m:
60

Execution word times:
1 of m refers to R or Q; otherwise 3.

Execution:
The left address of the word in location m is replaced by the left address of the word in the A register (13 bits). The modified contents of m are put in the N register. The m can refer to any addressable location except A and L. The flag bit is ignored for this operation.

LOAD Q or LOAD R
LDQ m or LDR m:
72

Execution word times:
16

Execution:
The contents of 16 consecutive words of memory starting with location m are transfered to one of the rapid access loops. The specific loop, i.e., R or Q, is determined by the flag bit. A flag bit of 1 indicates the R loop, while a flag bit of 0 indicates the Q loop. Each word is transferred into R or Q in the sector in which the last four bits agree with the sector of these loops.

*Input-Output Commands*

READ NUMERIC, RDN:
54

Execution:
The program is allowed to select the keyboard, mechanical reader, or an external device. The form of input (octal or BCD) is designated by the most significant bit of the right channel number. The input



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device is selected by the next two bits of the channel number.

In Numeric Input mode, the computer will accept certain Fieldata or Teletype characters, convert these characters to a 4-bit code and then, depending on the mode (Octal or BCD) and perhaps the character (if in Octal mode), will process these four bits as shown in the following descriptions.

In Octal Input mode, the acceptable characters are the octal digits 0 through 7 and the control characters for Location, Enter, Fill, Verify, Halt, and Compute. All other characters are ignored. The computer will interpret a VERIFY code and initiate the Verify mode, as well as interpret a FILL code and initiate the Fill mode. Upon receipt of an octal digit, the computer will shift left the corresponding three bits into the least significant end of the A register. Upon receipt of a location code, the contents of the A register are transferred to the L register, thus setting up the initial location where the following information is to be stored or where computation is to start. Upon receipt of an ENTER code, information in the A register will be transferred to (if in FILL) or compared with the contents of (if in VERIFY) the memory location specified by the contents of the L register. The location designation will then be increased by one. A HALT code will halt the operation. A COMPUTE code will send the computer into the Program Halt mode.

In the BCD Input mode the acceptable characters are the decimal digits 0 through 9 and the symbols (+), (—), ( . ), (clear), (enter), and (blank). All other characters are ignored.

Upon receipt of an acceptable character, the computer will shift left the corresponding four bits into the least significant end of the A register.

The amount of data to be accepted is determined by a choice of block size and the number of blocks. A flag bit of 0 indicates the block is a character. A flag bit of 1 indicates the block is a word. In the BCD mode, a word is made up of eight characters. The number of blocks is indicated by the 10 least significant bits of the command.

If the computer is still in the Input mode after a word is assembled, the resulting word is transferred to the N register.

In addition, if the flag bit is 1, then the word just transferred to the N register is also transferred to the location specified by the 13 least significant bits of the L register, and this location designation will then be increased by one.

After the specified number of blocks have been accepted (and stored in memory if the flag bit is 1), the computer will return to COMPUTE and execute the next command in its program sequence.

READ ALPHA-NUMERIC.

RDA: 56

Execution:

The program is allowed to select the keyboard, mechanical reader, or an external device. The form of the input (either five or six characters per word) is designated by the most significant bit of the right channel number. The input device is selected by the next two bits of the channel number.

In the Alpha-numeric Input mode, the computer will accept all Fieldata and Teletype characters. The computer will shift left these characters directly (without conversion) into the six least significant bit positions of the A register. The amount of data to be accepted is determined by a choice of block size and the number of clocks. A flag bit of 0 indicates the block is a character. A flag bit of 1 indicates the block is a word. In the Alpha-numeric-Five mode, a word is made up to five characters.

In the Alpha-numeric-Six mode, a word is made up of six characters.



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The number of blocks to be accepted is indicated by the 10 least significant bits of the command.

If the computer is still in the Input mode after a word is assembled, the resulting word is transferred into the N register. In addition, if the flag bit is 1, the word just transferred to the N register is also transferred to the location specified by the 13 least significant bits of the L register, and the location designation will then be increaesd by one.

After the specified number of blocks has been accepted (and stored in memory if the flag bit is 1), the computer will return to COMPUTE and execute the next command in its program sequence.

WRITE NUMERIC, WRN: 44

Execution:

The most significant channel bit of this command designates whether the form is octal or BCD. The next two channel bits designate whether the output should be in Teletype or Fieldata code.

In Numeric Output mode the computer will shift left an octal or BCD character out of the most significant end of the A register, convert this character to the proper Fieldata or Teletype code, and then put it on the output lines.

The amount of data to be transmitted is determined by a choice of block size and the number of blocks. A flag bit of 0 indicates the block is a character. A flag bit of 1 indicates the block is a word. In the Octal mode, a word is made up of 11 octal characters plus 1 "enter" character. In the BCD mode, a word is made up of eight BCD characters. The number of blocks to be transmitted is indicated by the 10 least significant bits of the command. If the computer is still in the Output mode after a word has been transmitted, then the word in the N register is transferred to the A register. In addition, if the flag bit is 1, the contents of the location specified by the 13 least significant bits of the L register are transferred into the N register, and the location designation will then be increased by one.

At the start of the operation, if the flag bit is 1, the computer will simulate two words transmitted, thus initializing the A and N registers with the first two words to be transmited.

Afer the specified number of blocks have been transmitted, the computer will return to COMPUTE and execute the next command in its program squence.

WRITE ALPHANUMERIC, WRA: 46

Execution:

The most significant channel bit of this command designates whether the form of the output is five or six characters per word.

In Alpha-numeric Output mode, the computer will shift left a 6-bit Fieldata or Teletype character out of the most significant end of the A register and put it directly on the output lines (without any conversion).

The amount of data to be transmitted is determined by a choice of block size and the number of blocks. A flag bit of 0 indicates the block is a character. A flag bit of 1 indicates the block is a word. In the Alpha-numeric-Five mode, a word is made up of five characters. In the Alpha-numeric-Six mode a word is made up of six characters. The number of blocks to be transmitted is indicated by the least significant bits of the command.

If the computer is still in the Output mode after a word has been transmitted, then the word in the N register is transferred to the A register. In addition, if the flag bit is 1, the contents of the location specified by the 13 least significant bits of the L register are trans-



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ferred into the N register, and the location designation will then be increased by one.

At the start of the operation, if the flag bit is 1, the computer will simulate two words transmitted, thus initializing the A and H registers with the first two words to be transmitted.

After the specified number of blocks has been transmitted, the computer will return to COMPUTE and execute the next command in its program sequence.



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# APPENDIX B

## COMPUTER CHARACTERISTICS

### 10. SCOPE

10.1 This appendix to MIL–C–60184(MU), Computer Set, Gun Direction: 10525735, provides the operator with the computer characteristics required for computer operation.

### 20. APPLICABLE DOCUMENTS

None.

**30.1 Computer Word Formats and Storage Capacity.** Information stored in memory contains no inherent distinction between numerical data and computer instructions. Memory outputs are interpreted in accordance with the state of computer logic elements classified as mode controls. Various word formats have been developed to relieve the operator of as many time-consuming number conversions as possible.

30.1.1 *Memory Capacity.* Information is stored in the computer memory in binary code —32 bits per information word. In addition, 3 bits are used for synchronization and one bit for parity information, giving a total of 36 bits per machine word. The capacity of the memory is given in Table I, this appendix, including the storage capacity of registers normally required to perform program functions. The nominal clock rate is 460.8 kilocycles per second.

30.1.2 *Number Formats.* In addition to the normal binary representation of numbers, other number formats are available for use by external communication to the computer Control Panel or other devices.

30.1.2.1 *Binary Representation.* Information used by the computer as an operand is interpreted as a signed, 31-bit, binary coded word with a fixed binary point located between the sign bit and most significant digit. Positive numbers have a zero sign bit and negative numbers a one sign bit. Negative umbers are stored and interpreted in two's complement form. Binary representation is shown in Figure 10 (b).

30.1.2.2 *Octal Number Representation.* Each group of three binary digits, beginning with the three least significant digits of the word, are interpreted as a number to the base eight. The sign and most significant digit are grouped together and interpreted as a number to the base four. Octal representation is shown in Figure 10 (b).

30.1.2.3 *Alpha-numeric Representation.* Certain input and output operations require recognition of mixed alphabetic and numeric symbols. Alpha-numeric information requires sufficient bits for non-ambiguous representation of each symbol, and is shown in Figure 10 (c).

30.1.3 *Instruction Word Format*

30.1.3.1 *Instruction Words.* Computer words containing program information are interpreted by the computer as instructions when properly sequenced. If required by the program, certain portions of instruction words may be treated as operands. Instruction words contain an operation code, an operand address, and the address of the next sequential instruction. Instruction word composition is shown in Figure 1 (a).

30.1.3.2 *Octal Instruction Representation.* Information contained in an instruction word may be interpreted as a sequence of numbers to the base eight as defined in paragraph 1.2.2. Octal instruction representation is shown in Figure 10 (a).

30.1.3.3 *Quasi-Octal Representation.* In the instruction word the octal digits overlap



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logical portions of the word; which makes it inconvenient to write an instruction word in true octal form. For this reason, instructions are usually written in quasi-octal fom. Octal digits that overlap instruction word subunits are split into two quasi-octal digits. After the program is written, adjacent quasi-octal digits are summed for the final form of the program. Quasi-octal representation is shown in Figure 10 (a).

### 30.2 Memory Addressing

**30.2.1** *Memory Disk.* The memory disk consists of functional bands, or channels, each with read and write heads.

**30.2.2** *Sector Track.* The Sector track is permanently recorded, and has external connections only to the read head. Its purpose is to notify the control unit which sector is about to pass under the read heads of the other channels. The main memory read heads are all at the same effective sector location.

**30.2.3** *Main Memory.* There are 64 main memory channels and are numbered 000 through $136_8$ and $300_8$ through $336_8$. Words or sectors of each channel are numbered in consecutive octal numbers from 000 to 177 inclusive.

**30.2.4** *Short Loops.* A, L, N, D, R, and Q are short recirculating loops. Each bit is sensed by the read head and rewritten by the write head. Each word is shifted, bit by bit, through the loop. The loop information remains the same unless changed by some arithmetic or control operation. A, L, and N are 1-word loops whose contents are rewritten each word time. D is a 2-word loop, and R and Q are 16-word storage loops. R and Q each have a secondary read head located in the middle of the loop. This head is not involved in the recirculation process.

**30.2.5** *Short Loop Addressing.* The rapid access loops R and Q may have sector address 000 to 177, but only the last four bits are recognized. The channel address of R is 142; Q is 152. Registers A, L, and N are all addressable. The program may store into and read out of them or they may be used as temporary storage for operands and instructions when not required by the program operations. The sector number has no significance when addressing A, L, or N, because they are one word loops. The display register is a 2-word loop in channel 160. The right word is addressed by odd sector numbers and even sector numbers refer to the left word; where the relative position of each word is defined with respect to the Control Panel display.

### 30.3 Input

**30.3.1** *Initial Program Fill.* The most useful form of input for loading a program into memory, is the octal mode. Information is loaded in octal Teletype code. The basic form of the program is as follows:

(a) The memory location of incoming words is specified with five octal characters (channel and sector) followed by a "LOCATION" code. The channel and sector are read into A. The "LOCATION" code transfers these bits to the L register. A new location may be specified at any point in the program.

(b) The program and constants are entered in octal digits; 11 characters per word; and each word followed by an "ENTER" code. The "ENTER" code stores the contents of A in the location contained in L, and then increases L by one. In this way, incoming words are stored in sequential location, unless a new location is entered.

(c) During octal fill, control is dependent upon the input lines and not the computer itself. If it is desired to proceed directly from in-



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put to computing, the last characters should be five characters specifying the start address, fo'lowed by a "LOCATION" code and a "COMPUTE" code. The five characters load into A; the "LOCATION" code transfers A to L; then the "COMPUTE" code begins computation at the address specified in L.

30.3.2 *Computer Initiated Inputs*

30.3.2.1 *Octal Fill.* Using the input commands, the computer will:

(a) Accept Fieldata or Teletype characters as listed in Table II.

(b) Convert Fieldata and Teletype codes listed in Table II, and additionally the blank tape code, to a four-bit code. This conversion is accomplished by the code conversion information written on the Sector channel.

(c) Interpret the four-bit code derived from (b) as one of the following:

(1) An octal character, and insert it into the A register.

(2) A "FILL" code, and initiate "Fill" mode.

(3) A "VERIFY" code, and initiate "Verify" mode.

(4) A "LOCATION" and transfer the contents of A to the L register.

(5) An "ENTER" code, and will transfer the contents of A to memory if in the "Fill" mode or will compare the contents of A to memory if in the "Verify" mode.

(6) A "HALT" code, and will go to the "Manual Halt" mode.

(7) A "COMPUTE" code, and will initiate the "Compute" mode if the computer is in the "Program Halt" mode.

30.3.2.2 *Decimal Fill.* Using the input commands, the computer will:

(a) Accept and convert up to 16 binary-coded decimal characters, and will store them by character by A and N on input.

(b) While in "Decimal Fill by Word", convert and store 8 characters per word in A and store the contens of A in memory after each 8 characters.

30.3.2.3 *Alpha-numeric Fill.* Using the input commands and the codes listed in Tables III and IV, this appendix, for "Alpha-numeric-5" and "Alpha-numeric-6" respectively, the computer will:

(a) Accept up to 10 characters and will store them by character in A and N while in "Alpha-numeric-5" Fill".

(b) Store 5 characters per word in A and store the contents of A in memory after each 5 characters while in "Alpha-numeric-5 Fill."

(c) Accept up to 12 characters and will store them by character in A and N while in "Alpha-numeric-5" Fill".

(d) Store 6 characters per word in A and store the contents of A in memory by word, after each 6 characters while in "Alpha-numeric-6 Fill".

**30.4 Output**

**4.1 Nixie Display.** The Nixie tubes on the Control Panel are activated by the command "ENTER DISPLAY".

30.4.1.1 The contents of $D_0$ and $D_1$ are displayed as binary-coded decimal numbers. In order to display binary numbers, the program must convert them to decimal, store the eight most significant decimal digits in



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$D_1$, and the seven least significant decimal digits in $D_0$. The sign must be examine, and a + or — code (four bits) store in the four least significant bits of $D_0$.

**30.4.1.2** Once the "Display" mode is entered it will continue until the order is given to "HALT DISPLAY". Computation may proceed during display.

**30.4.2** *Computer Initiated Outputs*

**30.4.2.1** *Octal Output.* Using the output commands the computer will:

(a) Convert the octal characters in A and N to their Fieldata or Teletype equivalent via the Sector track conversion table, and generate a carriage return after every eleventh character while in "Octal Output by Character".

(b) Continually load A and N with the words from main memory to be transmitted and proceed as per "Octal Output by Character" while in "Octal Output by Word".

**30.4.2.2** *Decimal Output.* Using the output commands the computer will:

(a) Convert the decimal characters in A and N to their Fieldata or Teletype equivalents via the Sector track conversion table, while in "Decimal Output by Character".

(b) Continually load A and N with the words from main memory to be transmitted and proceed as per "Decimal Output by Character", while in "Decimal Output by Word".

**30.4.2.3** *Alpha-numeric Output.* Using the output commands the computer will:

(a) Send up to four 8-bit characters from A while in "8-bit Character Output".

(b) Continually load A and N with the words from main memory to be transmitted, and send 5 characters per word while in "Alpha-numeric-5" output by word.

(c) Send 6 characters per words from A and N while in "Alpha-numeric-6 output by character".

(d) Continually load A and N with the words in main memory to be transmitted and proceed as per "Alpha-numeric-6" output by character while in "Alpha-numeric-6 Output by Word".

**30.4.3** *Two-wire Teletype Output.* The Teletype oscillator provides a precision time reference for computer 2-wire, 5-level (serial) Teletype output. Machine logic is such that a character of information may be transmitted in the following manner:

(a) Assuming that information already exists in an applicable register of the computer, such information will be in binary code.

(b) Under direction of the control unit, the arithmetic of the computer converts the information into 5-level Teletype code, which is sent out serially on a 2-wire line.

**30.4.3.1** *Format.* A start pulse is defined as minus 3 volts and a stop pulse as 0 volts. A start pulse of 1 unit time precedes information pulses. The start pulse is followed by 5 units of information, and terminated by a stop pulse. The stop pulse must be at least 1.42 units. A unit of time is defined as 22 milliseconds ± 5 percent. The stop pulse cannot be less than 1.42 units; and, since a unit cannot be divided, two units of time are used as the stop pulse (44 milliseconds).

**30.4.3.2** *Operation.* The output signal from the Teletype oscillator is called the F32 line (or term). F32 is referred to as the sam-



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pling line. Sampling occurs by the X-special command "READ SWITCH MATRIX". The X-special command causes the 32nd bit in the A register to become one or zero, depending on the state of the sampling line. Upon a command such as "Transfer On Positive", if the F32 line has been setting true for 11 milliseconds, the A register, has been receiving a one in the sign position (32nd bit) indicating a negative number. Assume next that F32 goes false and a zero replaces the one in the sign bit of the word. With the advent of the zero, the "Transfer On Positive" command causes the output driver to transmit a true signal (—3 volts), which is the start pulse. During output of the start pulse, F32 again goes true. A delay occurs until the F32 line again goes false, at which time the second bit of the character is sent out. The process continues through the remaining bits of the character, after which the stop pulse is sent out, permitting a new character to be transmitted.

**30.5 Input-Output Terms.** The terms associated with the Input-Output plugs are described in Table V, this appendix.



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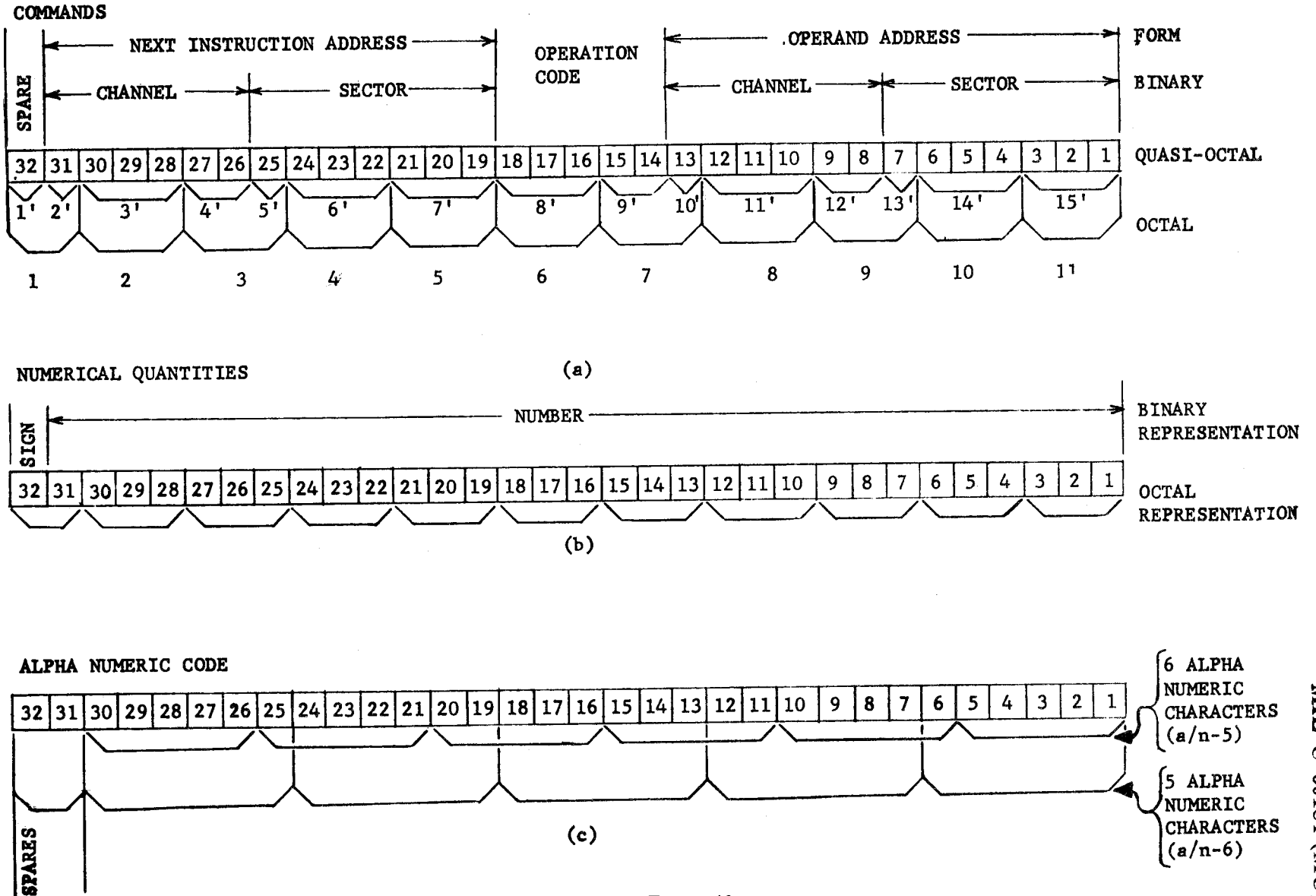



FIGURE 10.



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TABLE I. *Memory Contents*

| Register | | Capacity | |
|---|---|---|---|
| R | : | 16 words | |
| Q | : | 16 words | |
| A | : | 1 word | |
| L | : | 1 word | may be stored into |
| N | : | 1 word | by program |
| $D_0$, $D_1$ | : | 2 words | |
| Main Memory | : | 8192 words | |
| I | : | 1 word | may not be stored |
| X | : | 1 word | into by pragram |

TABLE II. *Numeric Input-Output Codes*

| Teletype meaning | | FADAC meaning | | | Fieldata meaning | |
|---|---|---|---|---|---|---|
| Character | Tape code | Otcal mode | BCD mode | Binary conversion | Tape code | Character |
| 0, P | 01101 | 0 | 0 | 0000 | 00110000 | 0 |
| 1, Q | 11101 | 1 | 1 | 0001 | 10110001 | 1 |
| 2, W | 11001 | 2 | 2 | 0010 | 10110010 | 2 |
| 3, E | 10000 | 3 | 3 | 0011 | 00110011 | 3 |
| 4, R | 01010 | 4 | 4 | 0100 | 10110100 | 4 |
| 5, T | 00001 | 5 | 5 | 0101 | 00110101 | 5 |
| 6, Y | 10101 | 6 | 6 | 0110 | 00110110 | 6 |
| 7, U | 11100 | 7 | 7 | 0111 | 10110111 | 7 |
| 8, I | 01100 | Halt | 8 | 1000 | 10111000 | 8 |
| 9, O | 00011 | Compute | 9 | 1001 | 00111001 | 9 |
| (+) ", Z | 10001 | Fill | + | 1010 | 00100010 | + |
| —, A | 11000 | Verify | — | 1011 | 00100001 | — |
| ., M | 00111 | Location | | 1100 | 10111101 | . |
| /, X | 10111 | Clear | Clear | 1101 | 11011101 | X |
| Car. Ret | 00010 | Enter | Enter | 1110 | 01000100 | Car. Ret |
| Blank | 00000 | Blank | Blank | 1111 | 00000000 | Blank |

*Note*: The above codes are employed in numeric input-output in the BCD and octal modes.



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TABLE III. *Alpha-Numeric-5 Input-Output Codes*

| Teletype character | Teletype code |
|---|---|
| 0, P | 01.101 |
| 1, Q | 11.101 |
| 2, W | 11.001 |
| 3, E | 10.000 |
| 4, R | 01.010 |
| 5, T | 00.001 |
| 6, Y | 10.101 |
| 7, U | 11.100 |
| 8, I | 01.100 |
| 9, O | 00.011 |
| +, Z | 10.001 |
| —, A | 11.000 |
| ., M | 00.111 |
| CARR. RETURN | 00.010 |
| /, X | 10.111 |
| ), L | 01.001 |
| :, C | 01.110 |
| ', N | 00.110 |
| !, F | 10.110 |
| s, S | 10.100 |
| H, H | 00.101 |
| $, D | 10.010 |
| &, G | 01.011 |
| (, K | 11.110 |
| FIGURE SHIFT | 11.011 |
| ;, V | 01.111 |
| ?, B | 10.011 |
| LETTERS SHIFT | 11.111 |
| LINE FEED | 01.000 |
| SPACE | 00.100 |
| BLANK | 00.000 |
| „ J | 11.010 |



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TABLE IV. *Alpha-Numeric-5 Input-Output Codes*

| Fieldata | | Lower | FADAC | Fieldata | | Lower | FADAC |
|---|---|---|---|---|---|---|---|
| Tape code | Upper case | | Binary code | Tape code | Upper case | | Binary code |
| 11000000 | Master Sp | ≀ | 000000 | 10100000 | ) | | 100000 |
| 01000001 | Upper case | ≀ | 000001 | 00100001 | — | | 100001 |
| 01000010 | Lower case | ⨯ | 000010 | 00100010 | + | | 100010 |
| 11000011 | Tab | ◇ | 000011 | 10100011 | < | | 100011 |
| 01000100 | Car. ret | < | 000100 | 00100100 | = | | 100100 |
| 11000101 | Space | ∆ | 000101 | 10100101 | > | | 100101 |
| 11000110 | A | A | 000110 | 10100110 | — | | 100110 |
| 01000111 | B | B | 000111 | 00100111 | $ | | 100111 |
| 01001000 | C | C | 001000 | 00101000 | * | | 101000 |
| 11001001 | D | D | 001001 | 10101001 | ( | | 101001 |
| 11001010 | E | E | 001010 | 10101010 | " | | 101010 |
| 01001011 | F | F | 001011 | 00101011 | : | | 101011 |
| 11001100 | G | G | 001100 | 10101100 | ? | | 101100 |
| 01001101 | H | H | 001101 | 00101101 | ! | | 101101 |
| 01001110 | I | I | 001110 | 00101110 | , | | 101110 |
| 11001111 | J | J | 001111 | 10101111 | Stop ○ | | 101111 |
| 01010000 | K | K | 010000 | 00110000 | | 0 | 110000 |
| 11010001 | L | L | 010001 | 10110001 | | 1 | 110001 |
| 11010010 | M | M | 010010 | 10110010 | | 2 | 110010 |
| 01010011 | N | N | 010011 | 00110011 | | 3 | 110011 |
| 11010100 | O | O | 010100 | 10110100 | | 4 | 110100 |
| 01010101 | P | P | 010101 | 00110101 | | 5 | 110101 |
| 01010110 | Q | Q | 010110 | 00110110 | | 6 | 110110 |
| 11010111 | R | R | 010111 | 10110111 | | 7 | 110111 |
| 11011000 | S | S | 011000 | 10111000 | | 8 | 111000 |
| 01011001 | T | T | 011001 | 00111001 | | 9 | 111001 |
| 01011010 | U | U | 011010 | 00111010 | | ' | 111010 |
| 11011011 | V | V | 011011 | 10111011 | | ! | 111011 |
| 01011100 | W | W | 011100 | 00111100 | | / | 111100 |
| 11011101 | X | X | 011101 | 10111101 | | . | 111101 |
| 11011110 | Y | Y | 011110 | 10111110 | Special □ | | 111110 |
| 01011111 | Z | Z | 011111 | 00111111 | Back space | | 111111 |



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TABLE V

| Term | Description |
|---|---|
| GPRC′ | An input to FADAC—This term, in conjunction with GRPC′ controls the run-halt mode of the computer. With an open circuit, the computer is in "Run" mode; with —10.0 ± 1.0 volts applied, the computer is in "Halt" mode. |
| GPHC′ | An input to FADAC—This term in conjunction with GPRC′ controls the "Run-Halt" mode. With an open circuit, the computer is in the "run" mode; with +6.0 ± 1.0 volts applied, the computer is in the "Halt" mode. |
| FBPR | An input to FADAC—This term supplies an "Initiate Fill" signal. Normally false (clamped to 0.0 ± 0.5 volts), FBPR is —10.0 ± 1.0 volts when true. |
| 11G<br>12G<br>13G<br>14G<br>15G<br>16G<br>17G<br>18 | Inputs to FADAC from some external input device having a Fieldata voltage swing. 0.0 ± 1.0 volt level is false; —4.0 ± 2.0 volt level is true. On paper tape, the sprocket hole reference point occurs between I3G and I4G. The input impedance of these amplifiers is 150 ohms; 0.001 microfarads to ground. |
| TG | An input to FADAC—This term is used as an alternate clock to incoming information. In conjunction with TG′, TG strobes input data on I1G, I2G, . . . I8G. To use TG′, term TGAO′ must be connected to TR′ within the input cable. The strobes supplied to TG are of two forms:<br>(1) A steady state —4.0 ± 2.0 volts (Fieldata) signal to TG and a false-going strobe (0.0 ± 0.5 volt) during the presence of information on I1G, I2G, . . . I8G to TG′. The false going strobe pulse has no contact "bounce".<br>(2) A normally closed (no information) mechanical contact supplying —4.0 ± 2.0 volts to TG′ and a normally open (information) mechanical contact supplying —4.0 ± 2.0 volts to TG. This strobe is used if TG and TG are supplied a signal which has contact "bounce". |
| TGAO′ | Refer to description of TG. |
| TG′ | Refer to description of TG. |
| FBOO | Output from FADAC—This term indicates when information is present on the output lines: D100, R200, . . . D800. |
| TFAI′<br>TFAO′<br>TFBO′<br>TFBO | Inputs to FADAC—These terms are feedback signals from some external output device. With an output device connected, lines TFAO′ and TFAI′ are connected together. Feedback signals fall into two major categories:<br>(1) Signals which, because they depend on the transfer of a switch contact, may inherently have "bounce".<br>(2) Signals which have no "bounce".<br>For feedback signals having no "bounce", a —4.0 ± 2.0 volt signal is supplied to line TFBO. Line TFBO′ toggles from —3 volts to 0 volts when information, "latched" in the output device, is no longer required on lines D100, D2,00, . . . D800. |
| D100<br>D200<br>D300<br>D400<br>D500<br>D600<br>D700<br>D800 | Output lines from FADAC transmitting either 8-level Fieldata information or 5-level Teletype information. On Paper Tape, the sprocket hole reference point occurs between D300 and D400. Outputs are in standard Fieldata voltage swing. |
| MTF | An input to FADAC—This term controls the mode of operation from another FADAC. With the MTF line open, FADAC is in the "input" mode; grounded, in the "Verify" mode. |
| IFS | An input to FADAC—This term distinguishes between Magnetic-Tape format and Paper-Tape format. An open circuit is Magnetic Tape fill; Input grounded is Paper Tape fill. |
| OFS | Same significance as IFS. |
| RDY | An output from FADAC—This term is used to signify that the computer is ready to receive information. Nominal voltage swing is ground to —3 volts. Pulse width is 3.0 ± 1.0 microseconds. |



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TABLE V—Continued

| Term | Description |
|---|---|
| STR | An input to FADAC—This term signifies information is on the input lines. Nominal voltage swing is ground to —3 volts. Pulse width is not less than 1 microsecond and not more than 2 microseconds. |
| PSIA<br>PSIB | Strobe stretching circuits used to widen the strobe pulse from 2 microseconds to approximately 6 microseconds. |
| CTTI | An input to FADAC—Distinguishes between Fieldata input and Teletype input. When true (—4.0 ± 2.0 volts), this term indicates that 5-level Teletype information is present on the information lines; when false (0.0 to ± 0.5 volts), that 8-level Fieldata information is present. |
| F14<br>F15<br>F16<br>F17<br>F18<br>F19<br>F20<br>F21<br>F22<br>F23<br>F24<br>F25<br>F26<br>F27<br>F28<br>F29<br>F30 | Inputs to FADAC—These terms can be sampled into the A register under program control. A —4.0 ± 2.0 volt input is true; a ground input is false. A one-to-one correspondence exists between the F subscripts and the corresponding bit positions of the A register. |
| FBIO | An output from FADAC to an input device—This term is normally false (0.0 to ± 0.5 volt Fieldata). FBIO goes true (—4.0 ± 2.0 volts Fieldata) when FADAC is in the mode to sample the information on the input lines. This signal returns to the false state when TEI′ goes true. Used primarily for the FADAC to FADAC tie-in. Maximum information rate to and from FADAC is dependent on the type of information transferred. Maximum rates are:<br>4, 250 characters per second in a/n6 mode<br>3, 550 characters per second in a/n5 mode<br>600 characters per second in octal or decimal mode. |
| IMTO | Output from FADAC—This term defines the period of information transfer to or from the Tape-tub. Normally false (0.0 to ± 0.5 volt), during the transfer it is true (—4.0 ± 2.0 volts). |
| AMTO | Output from FADAC—This term defines the period of address transfer to the Tape-tub. Normally false (0.0 to ± 0.5 volt), it is true (—4.0 ± 2.0 volts) during two-character output address. |
| OPL1<br>OPL2<br>OPL3<br>OPL4<br>OPL5<br>OPL6 | Outputs from FADAC—These terms are the "Discrete" output lines from FADAC. |
| OPE | An input to FADAC—This term disables the output lines. |
| FTR′ | An input to FADAC—This term indicates a FADAC to FADAC tie-in. |
| EER | An input to FADAC—This term resets the "Error" light. |
| ESU | An input to FADAC—This term selects the "Receive" button. |
| TEOP | An output from FADAC—This term is used in conjunction with information output. TEOP is the inverse of the TFBO signal. When the TFBO′ signal does not exist, TEOP is connected to TFAI′. |
| TEIP | An output from FADAC—This term is used in conjunction with information input. TEIP is the inverse of the TG signal. When the TG′ signal does not exist, TEIP is connected to TR′. |



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# APPENDIX C

# GLOSSARY OF COMPUTER TERMINOLOGY

## 10. SCOPE

**10.1** This appendix to MIL–C–60184 (MU), Computer Set, Gun Direction: 10525735, provides definitions to terms and phrases not defined in Sections 3 and 4 of the basic specification.

## 20. APPLICABLE DOCUMENTS

None.



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**30. Terminology and definitions.**

Access time—A time interval which is characteristic of a storage unit and is essentially a measure of the time required to communicate with that unit. The time interval between the instant at which information is: (1) called for from storage and the instant at which delivery is completed, i.e., the read time; or (2) ready for storage and the instant at which storage is completed, i.e., write time.

Accumulator—The main arithmetic register in which sums and other arithmetical and logical results are formed. In FADAC this is known as the A register.

Address—A label which identifies a register, location, or device in which information in stored.

Absolute address—The number which specifies an absolute storage location in memory; that is, the number assigned as an address which is the same as that used by machine when executing instructions.

Symbolic address—The description of a storage location by means of symbols, usually alphabetic letters, or alphabetic letters and numbers (e.g., A2, AG7, etc.).

Base (radix)—A quantity used explicitly to define some system of representing numbers by positional notation.

Binary—As in binary number system (base 2).

Binary-coded decimal system—A system of number representations in which each decimal digit is represented by a group of four binary digits.

Bit—A binary digit.

Block—A group of words considered or handled as a unit: a record.

Check—A process of partial or complete testing of (1) the correctness of machine operations or (2) the correctness of results produced by a routine.

Built-in-check—An automatic indication arising from the failure of the computer to verify a test built into the hardware of the computer (e.g., parity check, overflow check, etc.).

Clear—To fill a storage location or a register with zeroes and, when not specifically stated otherwise, a plus sign. To replace information in a storage device by zero.

Clock pulse—A synchronizing pulse used to gate information into all flip-flops except Read flip-flops.

Code—Symbolic or numerical equivalents of the instructions intelligible to a computer which are used in solving a problem.

Operation code—The part of an instruction that designates the operation to be performed.

Computer—A device capable of accepting information, applying prescribed processes to the information, and supplying the results of these processes.

Counter—A device, register, or storage location for storing integers, permitting these integers to be increased or decreased by unity or by an arbitrary integer, and capable of being reset to zero or to an arbitrary integer.

Error—The amount of loss of precision in a quantity; the difference between an accurate quantity and its calculated approximation.

Extract—To erase or clear all bits within a word which do not compare with the contents of a specified storage location. Thus, the content of each bit position



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of the accumulator is compared with the content of the corresponding bit position of the storage location. If both bits are "1", the accumulator is left unaffected; otherwise, a "0" replaces the contents of the corresponding bit position in the accumulator. See also "mask."

Fixed point number—A number in which the point is explicitly located at some predetermined position (radix point).

Flag—Any device (usually a word) which conveys information from within a stored program computer to the routine being executed; this device may be set automatically by the computer or from the routine being executed: a storage device having a capacity of one binary digit.

Halt—

Manual Halt—A computer mode under external control.

Program Halt—A computer mode under program control initiated by the "HALT" command or when the external control is removed.

Indicator—Any device which present information from within a stored programmed computer on the console of the computer. This device may be set automatically by the computer as the result of certain conditions or from the routine being executed.

Input (noun)—Information transferred from a secondary or external source (e.g., paper tape) into the computer.

Instruction—A set of characters which defines an operation together with one or more addresses (or no address) and which, as a unit, causes the computer to operate accordingly on the indicated quantities.

Location—A position in memory which may be used to store information: a storage register, storage cell, or memory cell.

Magnitude—The absolute value of a number, or the size of a quantity.

Mask—A control or screening word used with the extract instruction; an extractor.

Mode—A specific computer state (i.e., input, output, compute, etc.).

Operand—The contents of the memory location specified by the address part of an instruction; one of the quantities involved in an arithmetic operation.

Output—Information transferred from the computer to an external medium.

Overflow (fixed point)—The generation of a quantity beyond the capacity of the register which is to receive the result.

Overflow—(floating point)—The generation, in an arithmetic operation, of an exponent part (characteristic) in which the magnitude exceeds the maximum allowable value.

Overflow indicator—An indicator which is turned on when an overflow is produced as the result of the execution of a command.

Record—To write information for the purpose of preserving for use at a later time.

Register—The hardware for storing one or more computer words.

Restore—(reset)—To return a device to zero or to an initial condition.

Routine—A set of coded instructions arranged in proper sequence to direct the computer to solve a given problem or perform certain calculations.

Select—Initiate the fill mode for a device such as the Keyboard or Mechanical Reader.



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Storage—The memory of a computer.

Volatile storage—A type of storage which does not retain information when the supply of electrical power to the computer is interrupted.

Nonvolatile storage—A type of storage which preserves information when the supply of electrical power to the computer is interrupted.

Storage location—See location.

Store—To record information into a storage location.

Strobing pulse—A synchronizing pulse used to gate memory information into the Read flip-flop.

Switch—Any manually set device which conveys information from the console of a computer to the routine stored within the computer.

Tape tub—An external magnetic tape storage device.

Transfer (jump)—An instruction or signal which, conditionally or unconditionally, specifies the location of the next instruction and directs the computer to that instruction.

Word—A set of characters occupying one memory location. This is usually a number or an instruction.

Write—To record information in a register, location, or other storage device.



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"to detach this form, cut along this line"

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**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION: Frame, Tent, Maintenance, Medium, Light, Metal, T 58-10, 20 Feet By 64 Feet, Complete - MIL-F-40132B

ORGANIZATION

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT: $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

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☐ YES ☐ NO IF "YES", IN WHAT WAY?

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